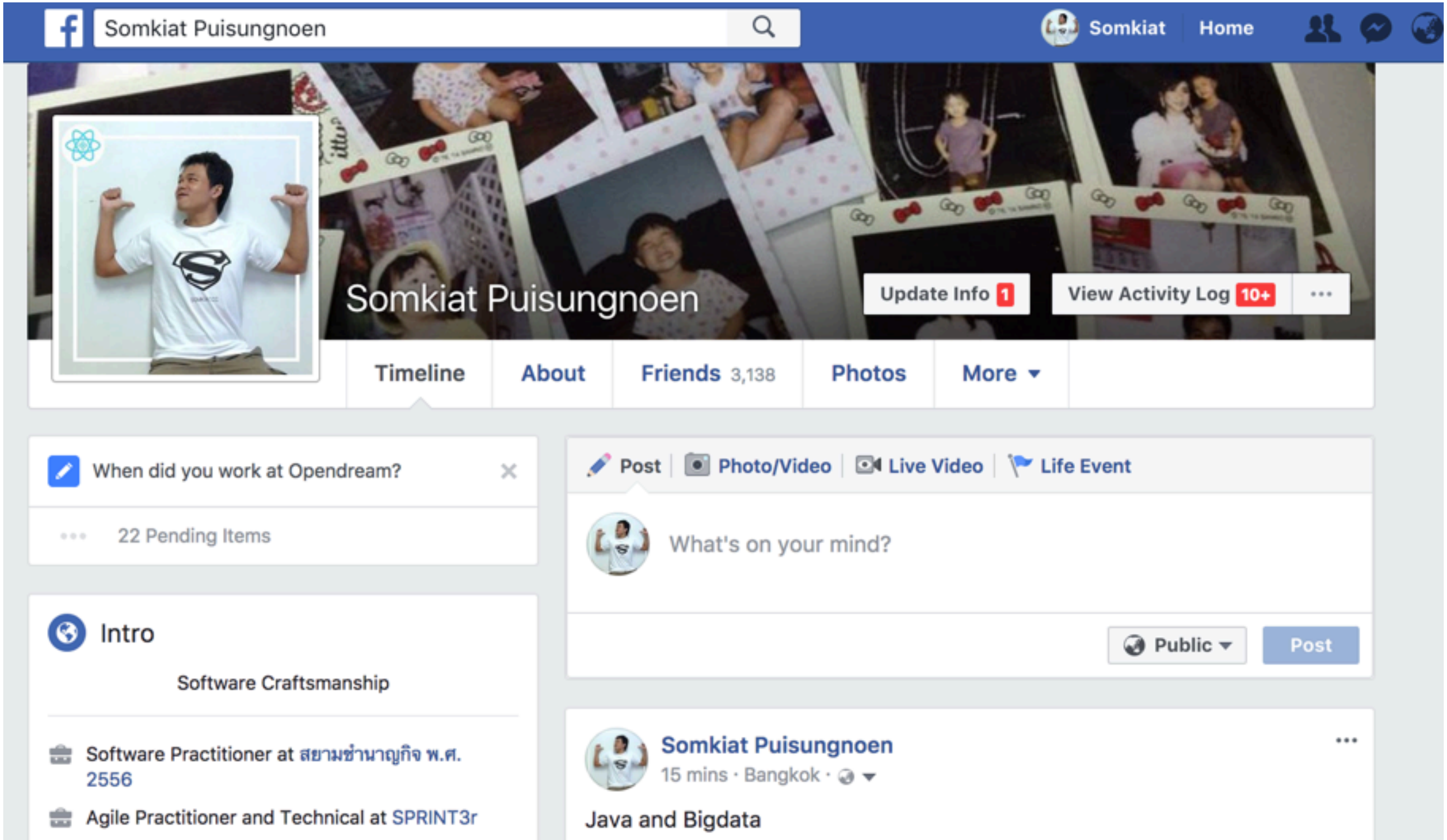
Java Programming
Spring Boot


Somkiat Puisungnoen
Somkiat
Home
Somkiat Puisungnoen
Update Info 1
View Activity Log 10+
Timeline
About
Friends 3,138
Photos
More
When did you work at Opendream?
22 Pending Items
Intro
Software Craftsmanship
Software Practitioner at สยามชำนาญกิจ พ.ศ. 2556
Agile Practitioner and Technical at SPRINT3r
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Life Event
What's on your mind?
Public
Post
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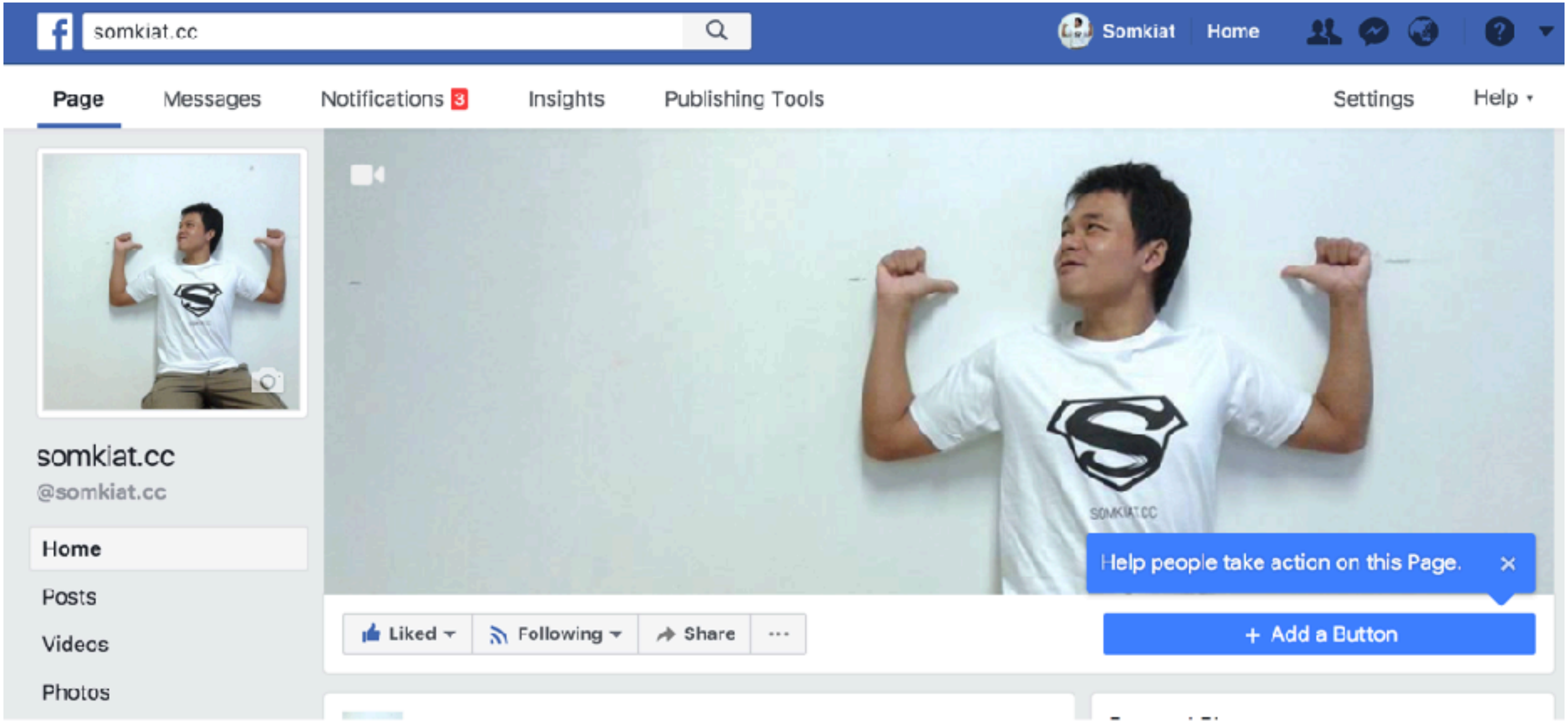
15 mins · Bangkok ·
Java and Bigdata


somkiat.cc
Somkiat
Home
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@somkiat.cc
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Videos
Photos
Help people take action on this Page.
Liked
Following
Share
+ Add a Button

https://github.com/up1/course-basic-java-5-days

Topics (1)

- Introduction to Java Programming
- Installation and configuration
- IDE
- Basic of Java Programming
- Mistake from Java programmer
- Write automated tests with JUnit
- SOLID :: How to design testable application

Topics (2)

- Introduction to Spring Boot
- Create starter project
- Structure of project
- Develop RESTful APIs
- Working with Database with JPA
- Testing with Spring Boot

Java Programming

TIOBE Index 2025

Sep 2025	Sep 2024	Change	Programming Language	Ratings	Change
1	1		Python	25.98%	+5.81%
2	2		C++	8.80%	-1.94%
3	4	^	C	8.65%	-0.24%
4	3	v	Java	8.35%	-1.09%
5	5		C#	6.38%	+0.30%
6	6		JavaScript	3.22%	-0.70%
7	7		Visual Basic	2.84%	+0.14%
8	8		Go	2.32%	-0.03%
9	11	^	Delphi/Object Pascal	2.26%	+0.49%
10	27	^^	Perl	2.03%	+1.33%
11	9	v	SQL	1.86%	-0.08%

https://www.tiobe.com/tiobe-index/

Software Requirements

Java Development Kit
IntelliJ IDEA (community edition)

Download JDK 17/21/25

Java 25, Java 21, and earlier versions available now

JDK 25 is the latest *Long-Term Support (LTS)* release of the Java SE Platform.

Learn about Java

JDK 21 is the previous Long-Term Support (LTS) release of the Java SE Platform.

Earlier JDK versions are available below.

JDK 25 **JDK 21**

Java SE Development Kit 25 downloads

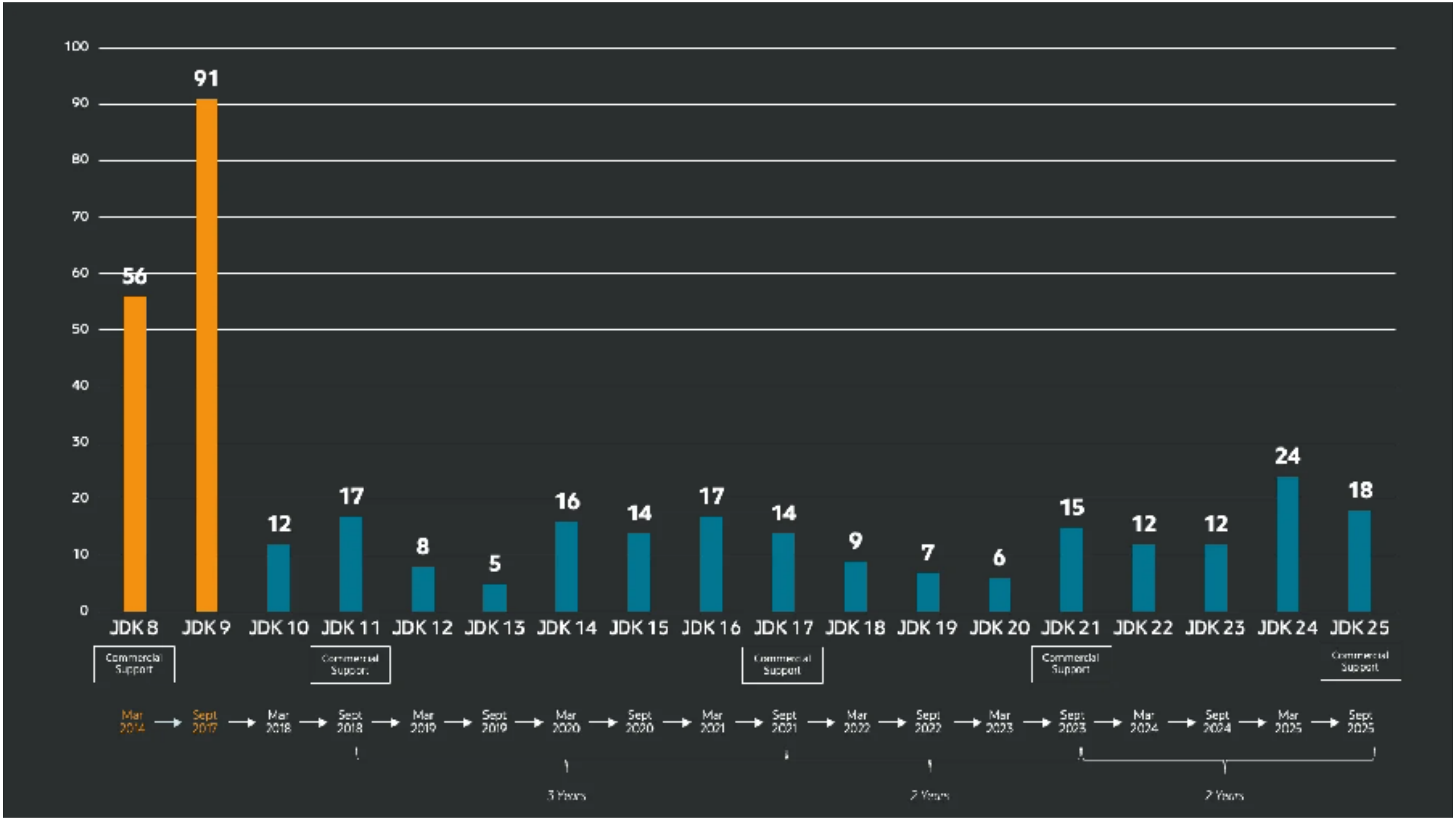
JDK 25 binaries are free to use in production and free to redistribute, at no cost, under the Oracle No-Fee Terms and Conditions (NFTC).

JDK 25 will receive updates under the NFTC, until September 2028, a year after the release of the next LTS. Subsequent JDK 25 updates will be
(OTN) and production use beyond the limited free grants of the OTN license will require a fee.

https://www.oracle.com/eg/java/technologies/downloads/

Java 25

History of Java from 8 to 25



https://www.infoq.com/news/2025/09/java25-released/

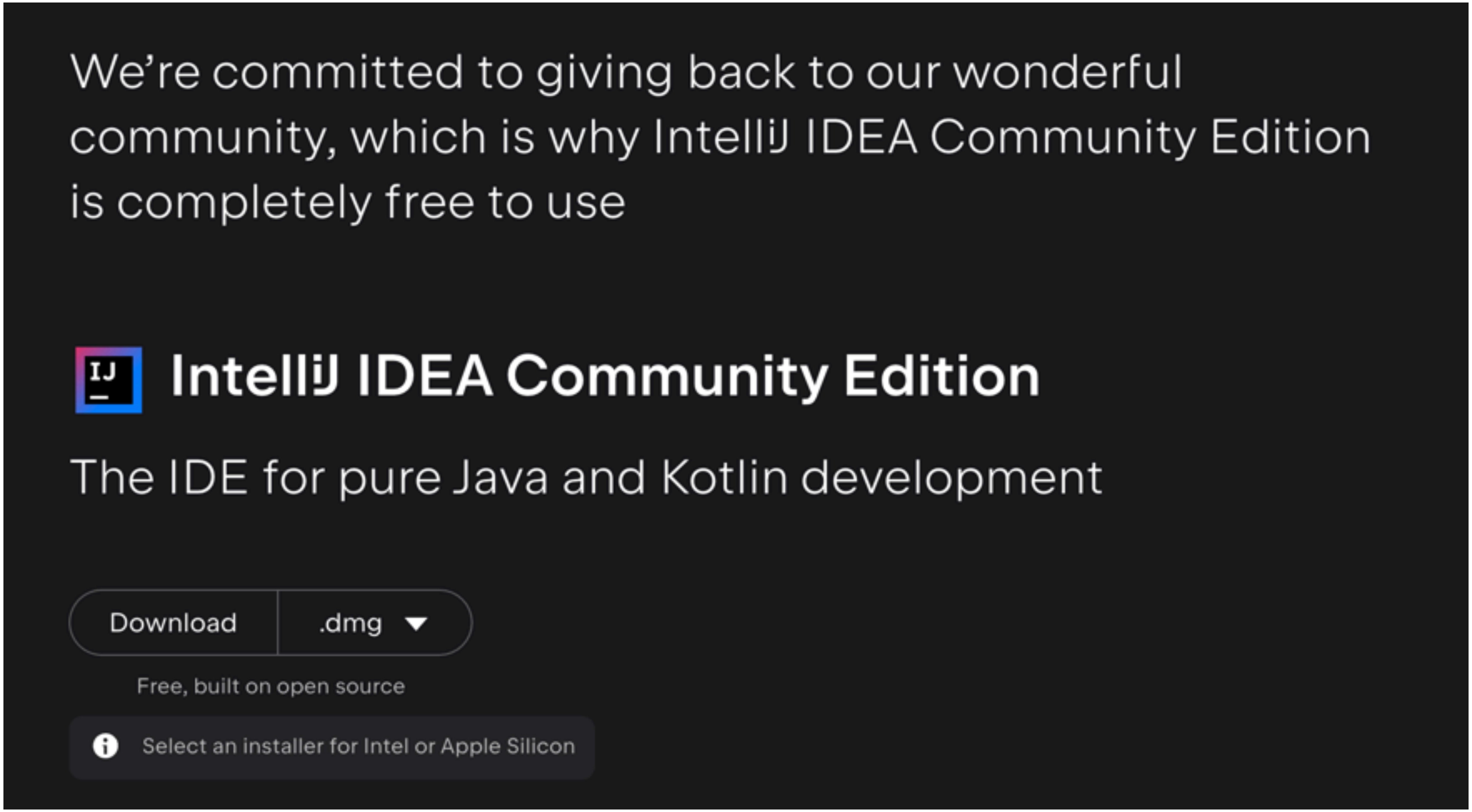
Config JAVA_HOME and PATH

set JAVA_HOME=<path of JDK>
set PATH=.;%JAVA_HOME%\bin;%PATH%

Testing configuration

$javac -version
$java -version

Download IntelliJ IDEA



https://www.jetbrains.com/idea/download/

Programming Language

Programming Language

Program is a set of instruction that tell a computer

We **execute** the instruction lists from program

Instructions are described using **Programming Language**

Categories of language

- High-level Language
- Assembly Language
- Machine Language
- Hardware

Categories of language

Java

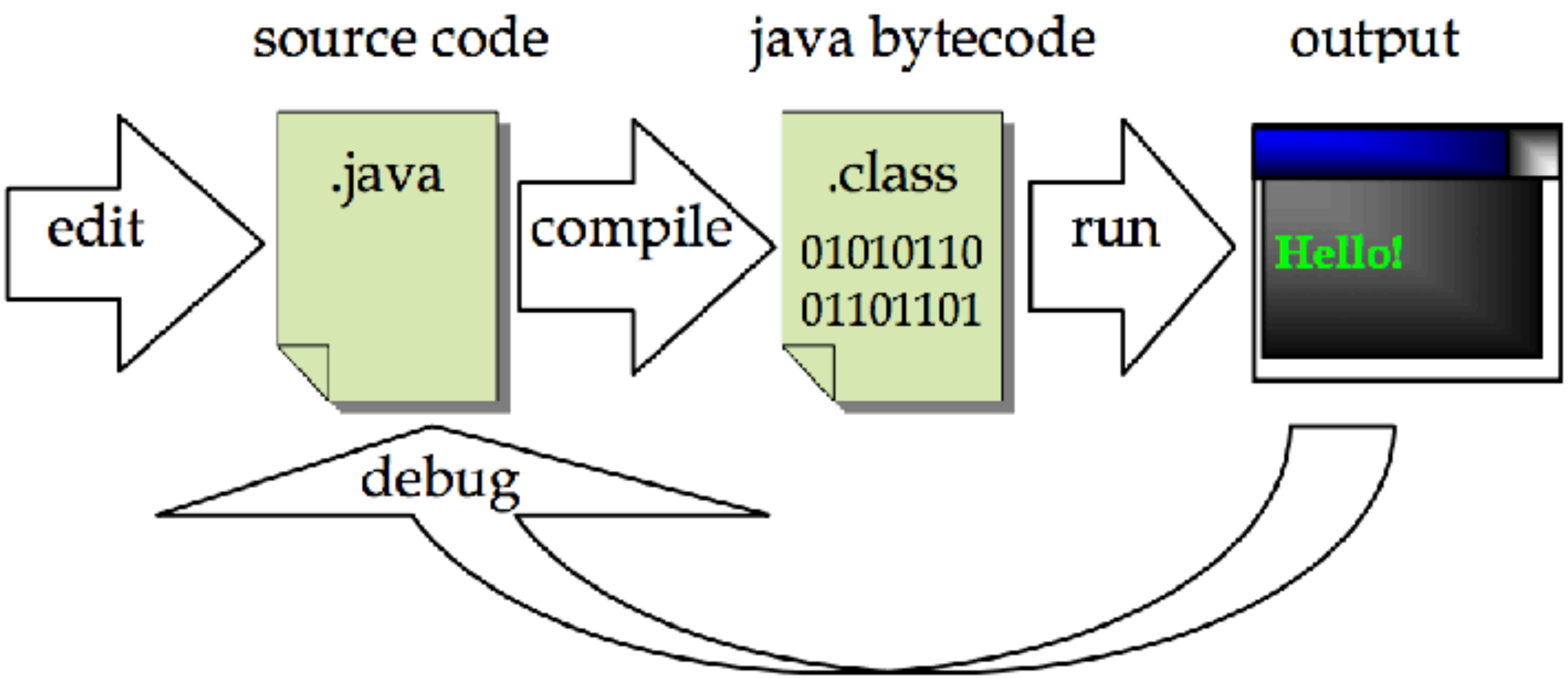
High-level Language

Assembly Language

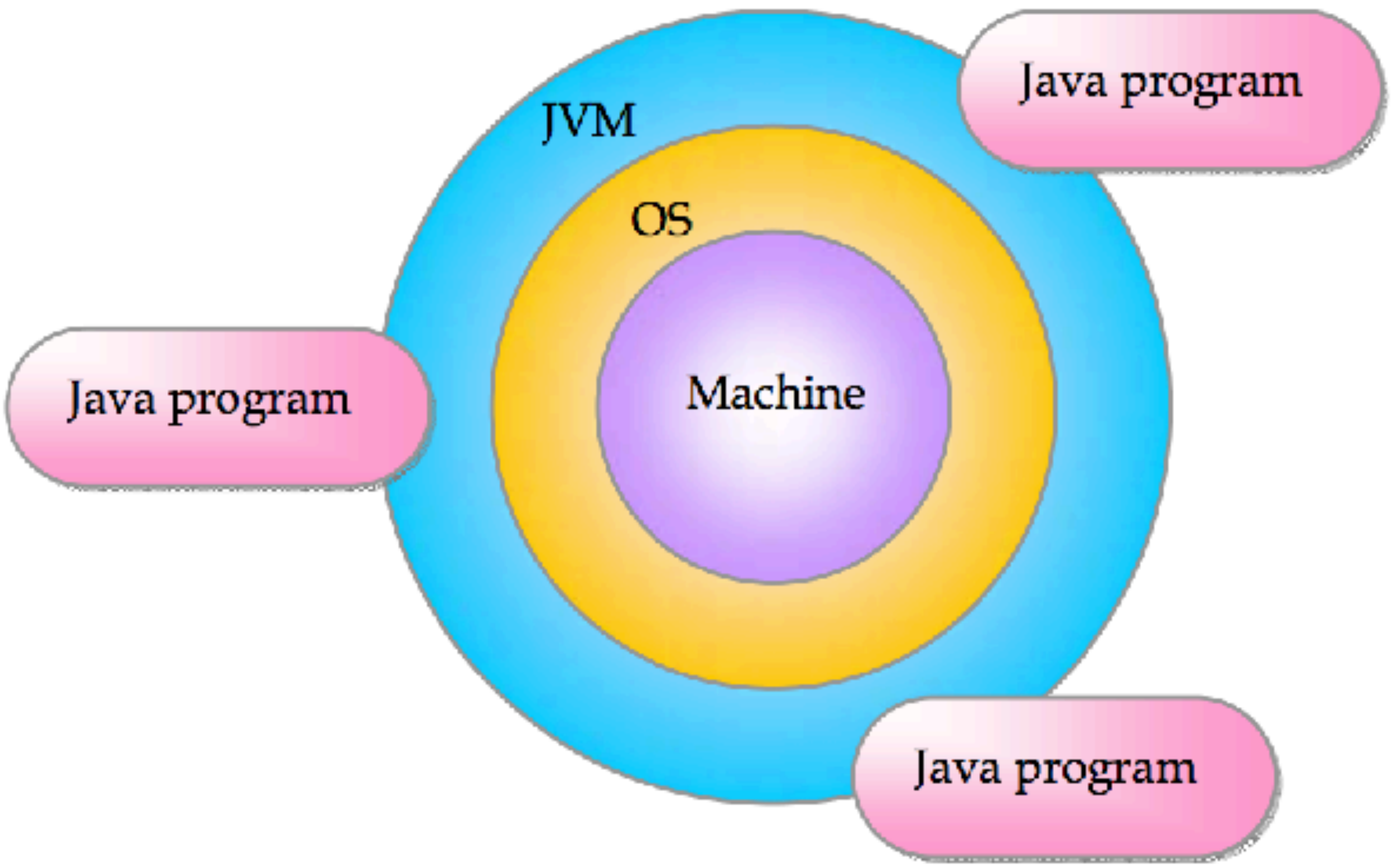
Machine Language

Hardware

Java workflow



Write once run anywhere ?



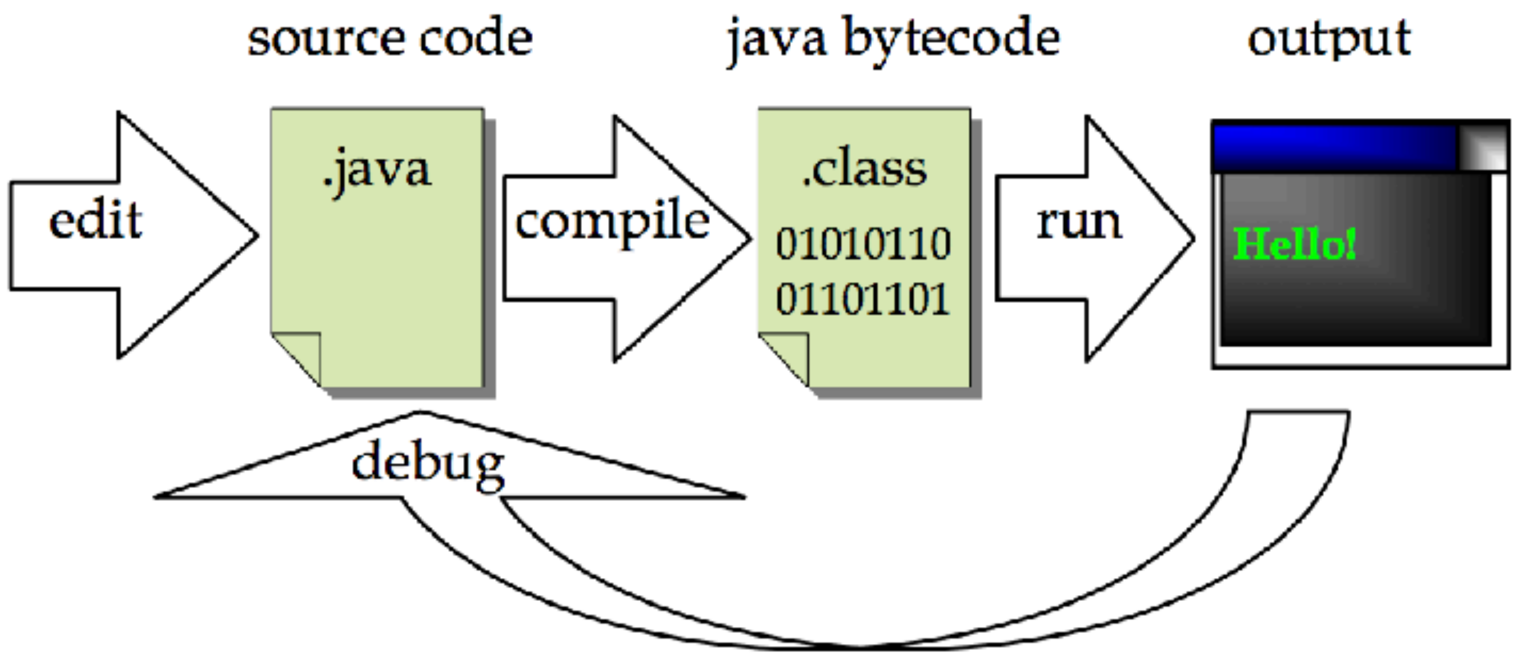
Introduction to Java

Structure of Java program

Create file HelloWorld.java

```java
public class HelloWorld {

    // Show message in a console
    public static void main(String[] args) {
        System.out.println("Hello World");

    }

}
```

Compile and Run program



$javac HelloWorld.java

$java HelloWorld

Hello World

Keywords

abstract	continue	for	new	switch
assert	default	goto	package	synchronized
boolean	do	if	private	this
break	double	implements	protected	throw
byte	else	import	public	throws
case	enum	instanceof	return	transient
catch	extends	int	short	try
char	final	interface	static	void
class	finally	long	strictfp	volatile
const	float	native	super	while

https://docs.oracle.com/javase/tutorial/java/nutsandbolts/_keywords.html

Comments

Documentation comment
Single line comment (C++ style)
Multi-line comment (C style)

Comments

```java
/**
 * 1. Documentation comment
 *
 * @author somkiat
 */
public class HelloWorld {

    // 2. Single line comment (C++ style)
    public static void main(String[] args) {
        /*
         3. Multi-line comment (C style)
         */
        System.out.println("Hello World");

    }

}
```

Formatting/ Readability

Whitespace characters in program are ignored during the compilation

```java
public class HelloWorld {
public static void main(String[] args) {
System.out.println("Hello World");}}
```

```java
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello World");
    }
}
```

Any fool can write code that a computer can understand. Good programmers write code that humans can understand.

Martin Fowler

QuoteAddicts.com

Java Code convention

Oracle (1997)

https://goo.gl/pfPc1S

Google

https://github.com/google/styleguide
https://google.github.io/styleguide/javaguide.html

Spring Framework

https://github.com/spring-projects/spring-framework/wiki/Code-Style
https://github.com/47deg/coding-guidelines/tree/master/java/spring

Basic of Java

Basic of Java Part 1

Variables name
Expressions
Statements
Data types
Operators

Variables and assign value

Symbolic names of memory location
A variable must have name and data type (static type)
Must be declared before using it

```
int number;
int counter = 1;
String name;
double averageScore = 3.5;
```

Naming rule for Java identifier

Can’t be a Java keyword

Case-sensitive

Unlimited length of Unicode letters and digits

Must begin with a letter, underscore(_) and dollar sign ($)

White space not allowed

White space ?

Such as spacebar

isWhitespace

`public static boolean isWhitespace(char ch)`

Determines if the specified character is white space according to Java. A character is a Java whitespa

- It is a Unicode space character (SPACE_SEPARATOR, LINE_SEPARATOR, or PARAGRAPH_SEPARATOR) '\u202F').
- It is '\t', U+0009 HORIZONTAL TABULATION.
- It is '\n', U+000A LINE FEED.
- It is '\u000B', U+000B VERTICAL TABULATION.
- It is '\f', U+000C FORM FEED.
- It is '\r', U+000D CARRIAGE RETURN.
- It is '\u001C', U+001C FILE SEPARATOR.
- It is '\u001D', U+001D GROUP SEPARATOR.
- It is '\u001E', U+001E RECORD SEPARATOR.
- It is '\u001F', U+001F UNIT SEPARATOR.

https://docs.oracle.com/javase/8/docs/api/java/lang/Character.html#isWhitespace-char-

Naming convention

Use meaningful name

For compound words use **camelCase**

Class name begin with an **Uppercase** letter

Variable and **method** name begin with a **Lowercase** letter

For a **constant use all uppercase letter** and underscore(_) to separate word

Meaningful name

```java
int d = 10;
int days = 10;
int daysSinceCreation = 10;
int daysSinceLastModification = 10;
```

Expressions

Expression is a value, variable, method or one of their combination that can be evaluated to a value

```
3.14
a + b
a + c -10
a > 10
Math.sqrt(4)
“Hi, all”
(a + 3 > b) && ( a - 3 < c)
```

Statements

Complete sentence that causes some action to occur
End with semicolon (;)

```
int a;
int b = 10;
c = a + b;
boolean isPass =  a > b;
System.out.println("Hello");
```

Data types (Statically typed)

1. Primitive data types
2. Classes

1. Primitive data types

Basic of Java data types have 8 types
Can’t add and change

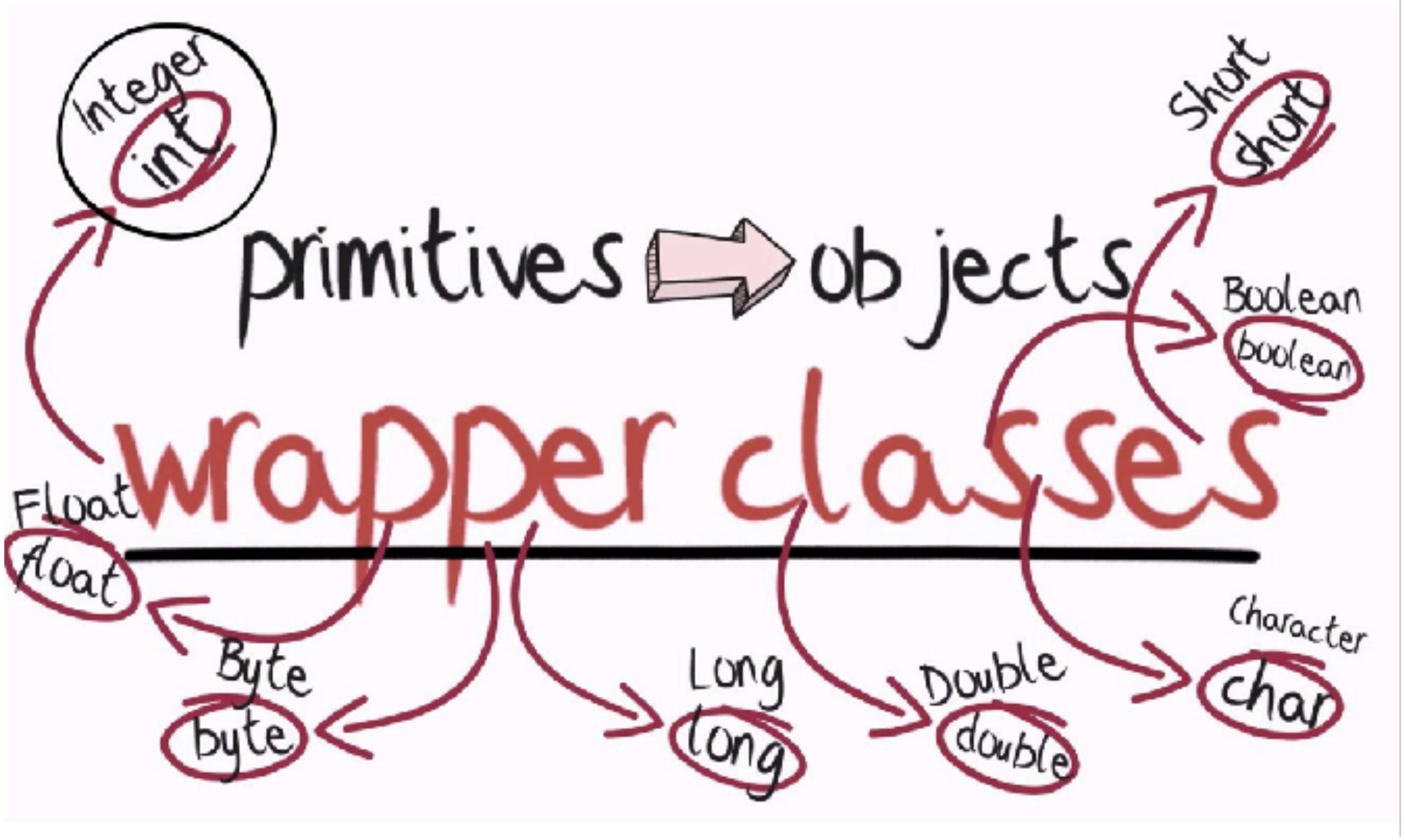
Value type	Primitive data types
Integer (-10, 10, 0)	byte, short, int, long
Floating-point (1.0, -1.5, 3.144)	float, double
Character (‘a’, ‘ก’)	char
Logical (true, false)	boolean

https://docs.oracle.com/javase/tutorial/java/nutsandbolts/datatypes.html

Primitive data types

Primitive data types		Min/Max value
byte	8-bit signed two complement integer	-128 / 127
short	16-bit signed two complement integer	-32,768 / 32,767
int	32-bit signed two complement integer	-2^31 / 2^31 - 1
long	64-bit signed two complement integer	-2^63 / 2^63 - 1
float	32-bit IEEE 754 floating point	
double	64-bit IEEE 754 floating point	
boolean		true, false
char	16-bit Unicode character	\u0000 / \uffff

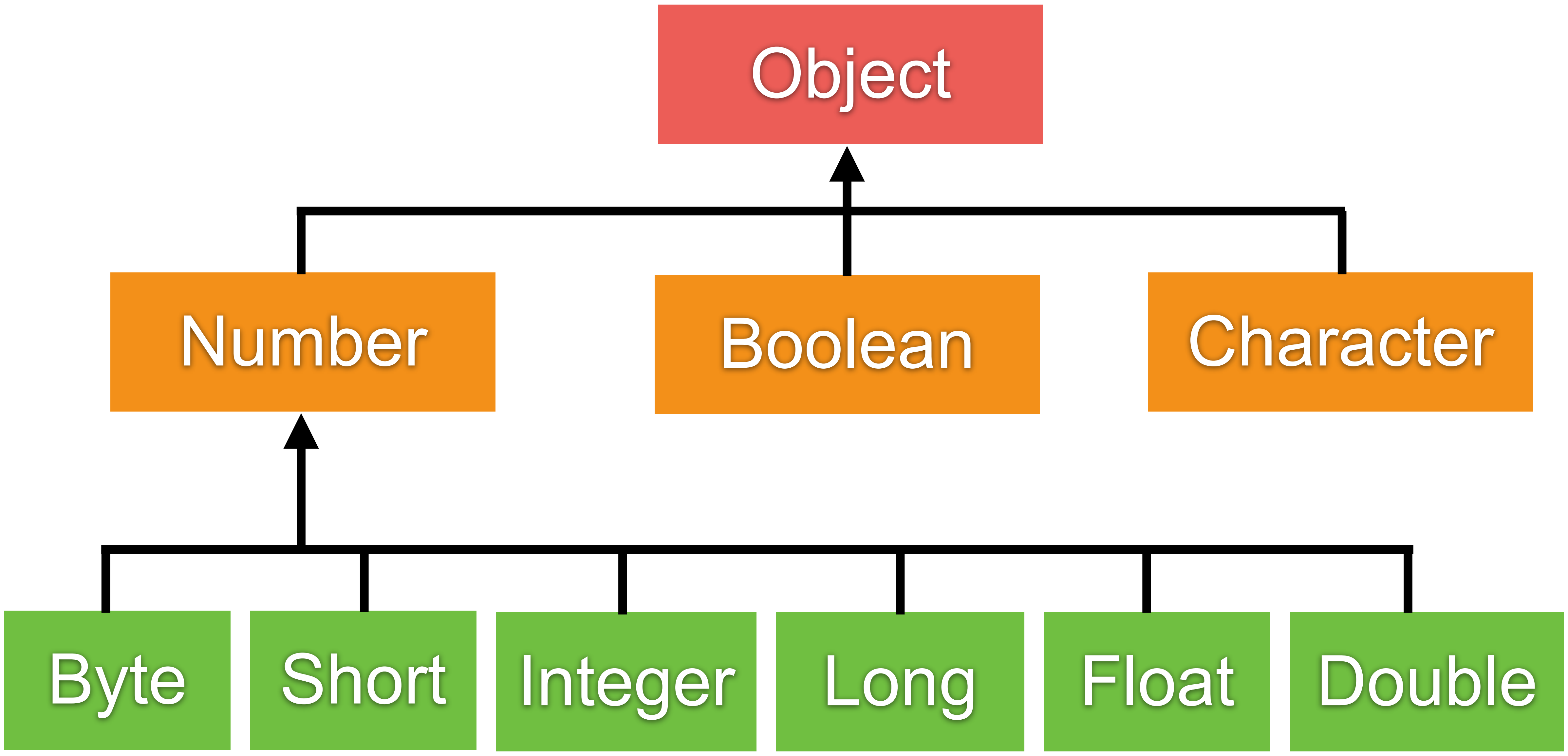
Wrapper class



Wrapper class

Primitive data types	Wrapper class
byte	Byte
short	Short
int	Integer
long	Long
float	Float
double	Double
boolean	Boolean
char	Character

Wrapper class hierarchy



Min/Max value of primitive type

Primitive data types	Minimum value	Maximum value
byte	Byte.MIN_VALUE	Byte.MAX_VALUE
short	Short.MIN_VALUE	Short.MAX_VALUE
int	Integer.MIN_VALUE	Integer.MAX_VALUE
long	Long.MIN_VALUE	Long.MAX_VALUE
float	Float.MIN_VALUE	Float.MAX_VALUE
double	Double.MIN_VALUE	Double.MAX_VALUE
char	Character.MIN_VALUE	Character.MAX_VALUE

Working with wrapper class (1)

Working with wrapper class (2)

```
// Wrapping
int number = 10;
Integer number2 = new Integer(number);
Integer number3 = Integer.valueOf(number);

// Unwrapping
int valueBack2 = number2.intValue();
int valueBack3 = number3.intValue();
System.out.println(valueBack2);
System.out.println(valueBack3);
```

Wrapper class conversion

Autoboxing and Autounboxing

```
// Autoboxing
int number = 10;
Integer number2 = number;

// Autounboxing
int valueBack2 = number2;
System.out.println(valueBack2);
```

Final variables

Can’t change the value of variables (immutable)
Use keyword **final**

```
// Constant variable
private static final int NUMBER_ONE = 1;

private void init() {
    // Final variable
    final int number = 10;
    number = 200;  // Can't modify value
}
```

2. Classes

Complex data types

Classes are **standard** from Java such as String, Integer

We can create new classes as we need

Working with String

Not primitive data type

Standard class in Java

Represent a sequence of one or more characters

```
String shortName = "S";
String name = "Somkiat Puisungngoen";
```

https://docs.oracle.com/javase/8/docs/api/java/lang/String.html

Working with String operation

```java
String name = "Somkiat";
System.out.println(name.concat("  Pui"));
System.out.println(name.substring(1, 2));
System.out.println(name.charAt(0));
System.out.println(name.startsWith("S"));
System.out.println(name.endsWith("t"));
```

Operators

Operators that require 2 operands are called **Binary operator**

Operators that require only 1 operand are called **Unary operator**

Parentheses, () are called **Grouping operator**

Operators

Arithmetic operators
Unary operators
Equality and Relational operators
Conditional operators
Bitwise and bit shift operators

Arithmetic operators

Operator	Description
+	Addition operator (can use for String concatenation)
-	Subtraction operator
*	Multiplication operator
/	Division operator
%	Remainder operator

String with additional operator

```
String firstName = "Somkiat";
String lastname = "Puisungnoen";
int age = 30;

String result = firstName + lastname + age;

System.out.println(result);
```

Formatting result

```java
String firstName = "Somkiat";
String lastname = "Puisungnoen";
int age = 30;
```

```java
System.out.printf("%s %s %d"
        , firstName, lastname, age);
```

Formatting result with String

```
String firstName = "Somkiat";
String lastname = "Puisungnoen";
int age = 30;

String result = String.format("%s %s %d",
        firstName, lastname, age);

System.out.println(result);
```

Unary operators

Required only one operand

Operator	Description
+	Unary plus operator
-	Unary minus operator
++	Increment operator
--	Decrement operator
!	Logical complement operator

Equality and relational operator

Operator	Description
==	Equal to
!=	Not equal to
>	Greater than
>=	Greater than or equal to
<	Less than
<=	Less than or equal to

Conditional operator

Operator	Description
&&	Conditional-AND
||	Conditional-OR
?:	Ternary (shorthand for if-then-else statement)

Type comparison with instanceof

```java
String object1 = new String();

System.out.println(object1 instanceof String);
System.out.println(object1 instanceof Object);
```

Bitwise and bit shift operator

Operator	Description
&	Bitwise AND operator
|	Bitwise inclusive OR operator
^	Bitwise exclusive OR operator
~	Unary bitwise complement
>>	Signed left shift
<<	Signed right shift
>>>	Unsigned right shift

Useful methods from Math (1)

Method	Description
abs()	returns the absolute value of the input value
round()	returns the integer nearest to the input value
ceil()	returns the smallest integer that is bigger than or equal to the input value
floor()	returns the biggest integer that is smaller than or equal to the input value
exp()	returns the exponential of the input value
max()	returns the bigger between the two input values
min()	returns the smaller between the two input values

https://docs.oracle.com/javase/8/docs/api/java/lang/Math.html

Useful methods from Math (2)

Method	Description
pow()	returns the value of the first value raised to the power of the second value.
sqrt()	returns the square root of the input value.
sin()	returns the trigonometric sine value of the input value
cos()	returns the trigonometric cosine value of the input value.
tan()	returns the trigonometric tangent value of the input value.

https://docs.oracle.com/javase/8/docs/api/java/lang/Math.html

Workshop with Math

The distance, d, between two points in the three-dimensional space (x1,y1,z1) and (x2,y2,z2) can be computed from:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

The following program computes and shows the distance between (2,1,3) and (0,0,6)

Compound assignment

Original	Shorthand
x = x + 1	x += 1
x = x - 1	x -= 1
x = x * 1	x *= 1
x = x / 1	x /= 1

Caution for complex expression

Original	Shorthand
x = x / y +1	x /= y + 1 (Wrong !!)
x = x / y +1	x /= (y + 1)

Operator precedence (1)

No.	Operator	Precedence
1	Grouping	()
2	Postfix	expr++, expr--
3	Unary/Prefix	++expr, --expr, +expr, -expr, ~, !
4	Multiplicative	*, /, %
5	Addtitive	+, -
6	Shift	>>, <<, >>>
7	Relational	<, >, <=, >=, instance

https://docs.oracle.com/javase/tutorial/java/nutsandbolts/operators.html

Operator precedence (2)

No.	Operator	Precedence
8	Equality	==, !=
9	Bitwise AND	&
10	Bitwise exclusive OR	^
11	Bitwise inclusive OR	|
12	Logical AND	&&
13	Logical OR	||
14	Ternary	?:
15	Assignment	=, +=, <<=, >>= …

Increment/Decrement operator

Prefix

the value inside the variable is incremented (or decremented) and then use to evaluate the expression

Postfix

the value inside the variable is used to evaluate the expression and then the value is incremented (or decremented)

Workshop

```java
int a, b, c = 0;
a = c++;
b = ++a;
c++;
b = ++c + a++;
a = --b + c++;

System.out.println(a); // ?
System.out.println(b); // ?
System.out.println(c); // ?
```

a	b	c
0	0	0

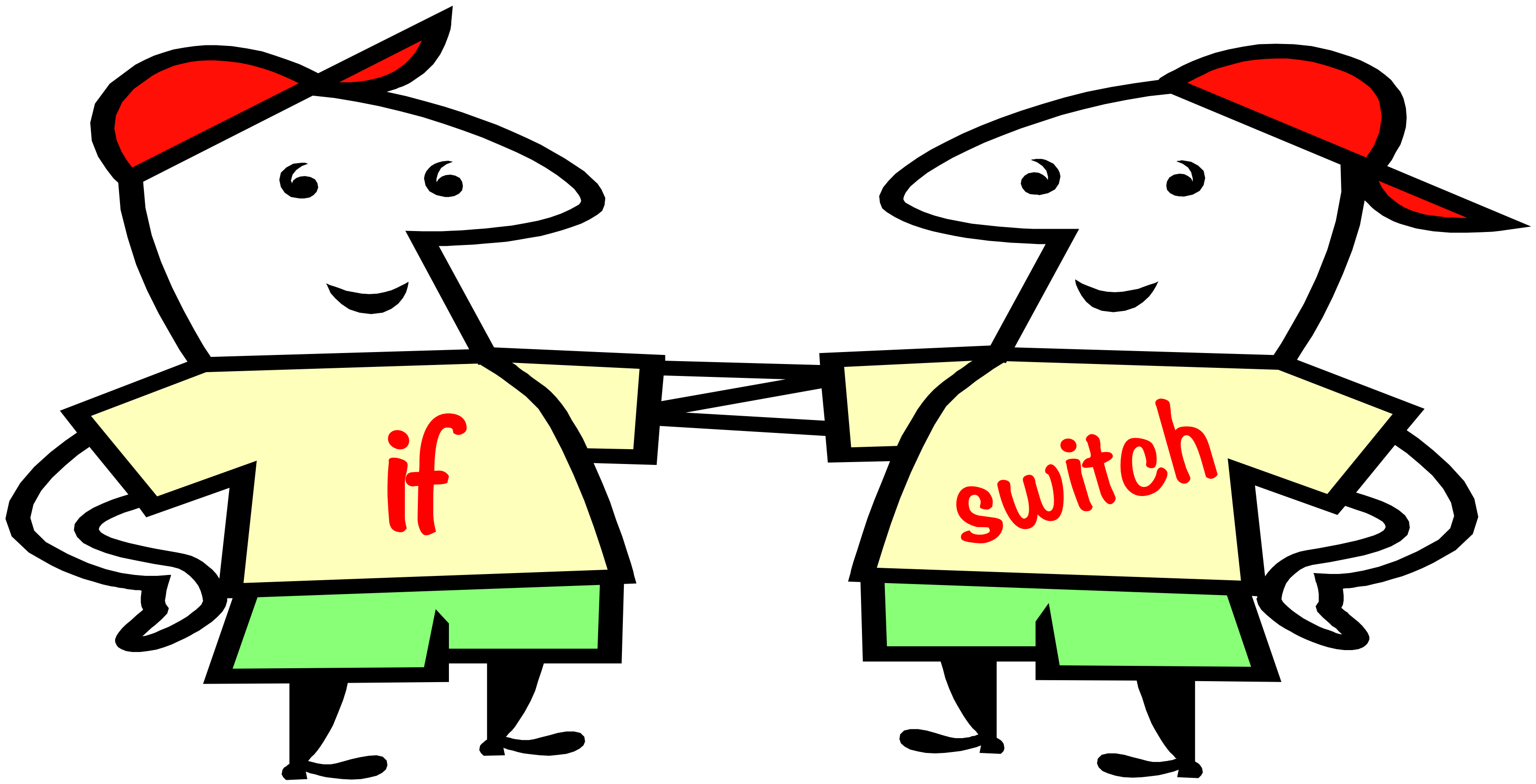
Basic of Java Part 2

Basic of Java Part 2

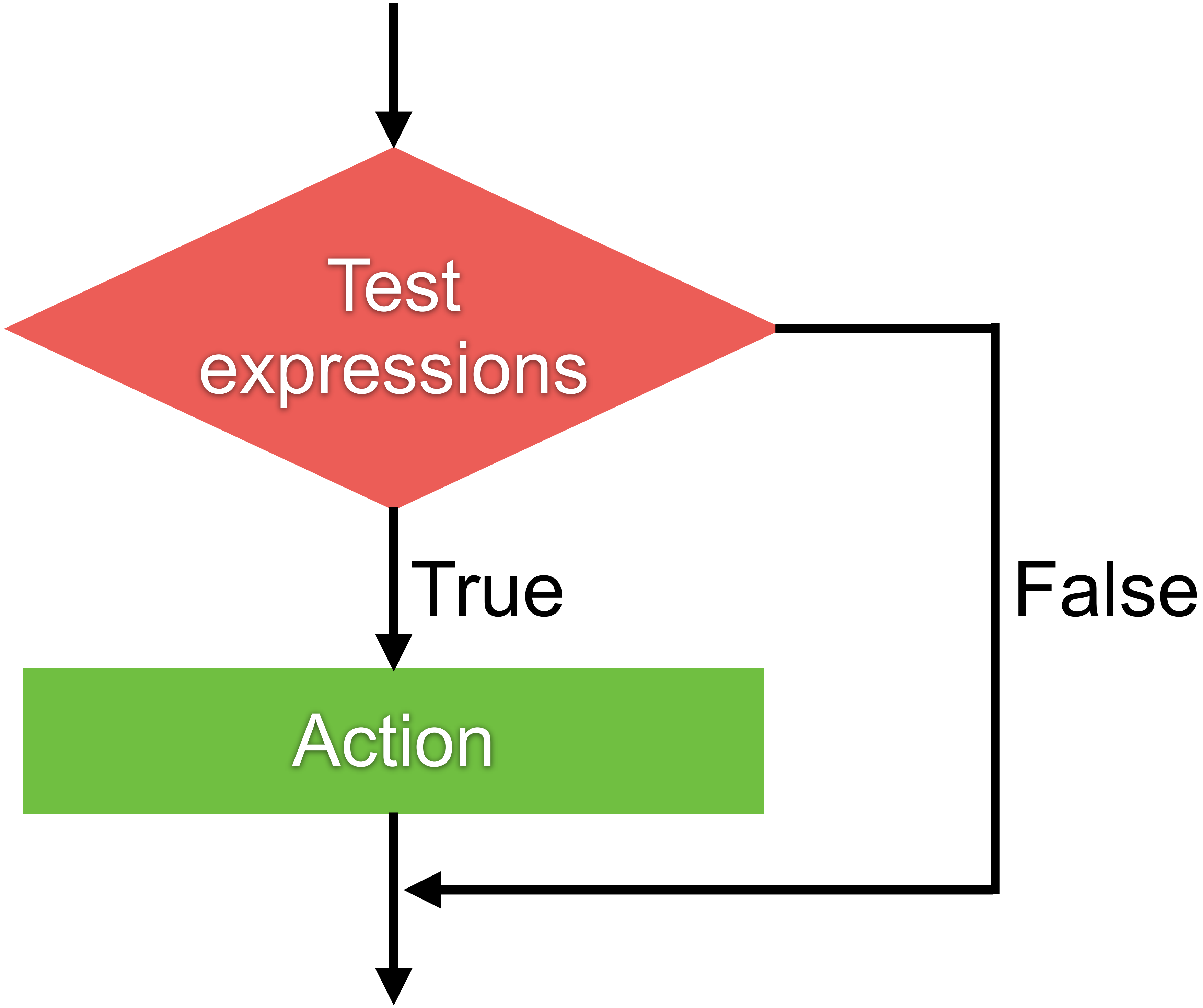
Control flow statements
Conditional statements (if-else/switch case)
For loop
While loop
Do-while loop

Conditional statements

if-else if - else
switch-case



IF - ELSE IF - ELSE



IF - ELSE IF - ELSE

```java
int score = 50;

if (score >= 80 && score <= 100) {
    System.out.println("Grade A");
} else if (score >= 70 && score < 80) {
    System.out.println("Grade B");
} else if (score >= 60 && score < 70) {
    System.out.println("Grade C");
} else if (score >= 50 && score < 60) {
    System.out.println("Grade D");
} else {
    System.out.println("Grade F");
}
```

Ternary operator (short if)

```java
int score = 50;
String grade = score > 80 ? "A" : score > 70 ? "B" : "C";
System.out.println(grade);
```

Easy to read and understand ?

Workshop
String comparison

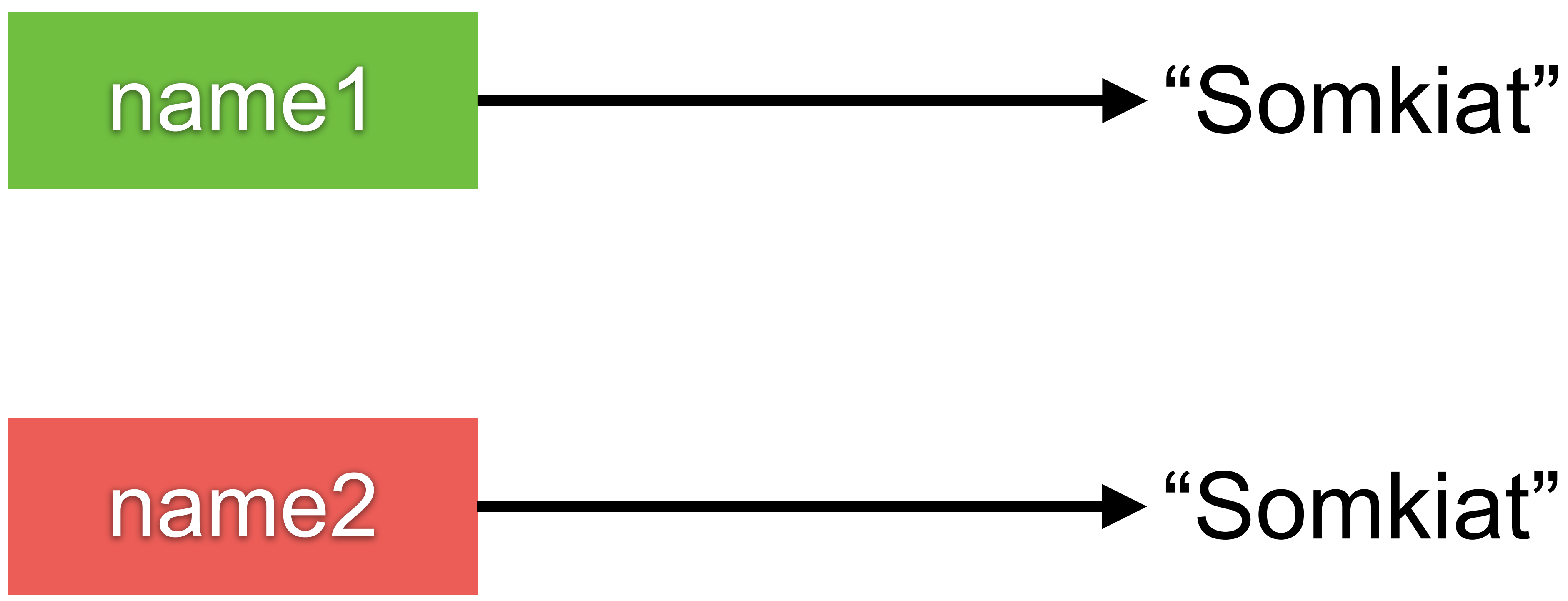
String comparison (1)

```java
String name1 = new String("Somkiat");
String name2 = new String("Somkiat");

if(name1 == name2) {
    System.out.println("Equal");
} else {
    System.out.println("Not equal");
}
```

String comparison (2)

```java
String name1 = new String("Somkiat");
String name2 = new String("Somkiat");

if(name1.equals(name2)) {
    System.out.println("Equal");
} else {
    System.out.println("Not equal");
}
```

String Equality



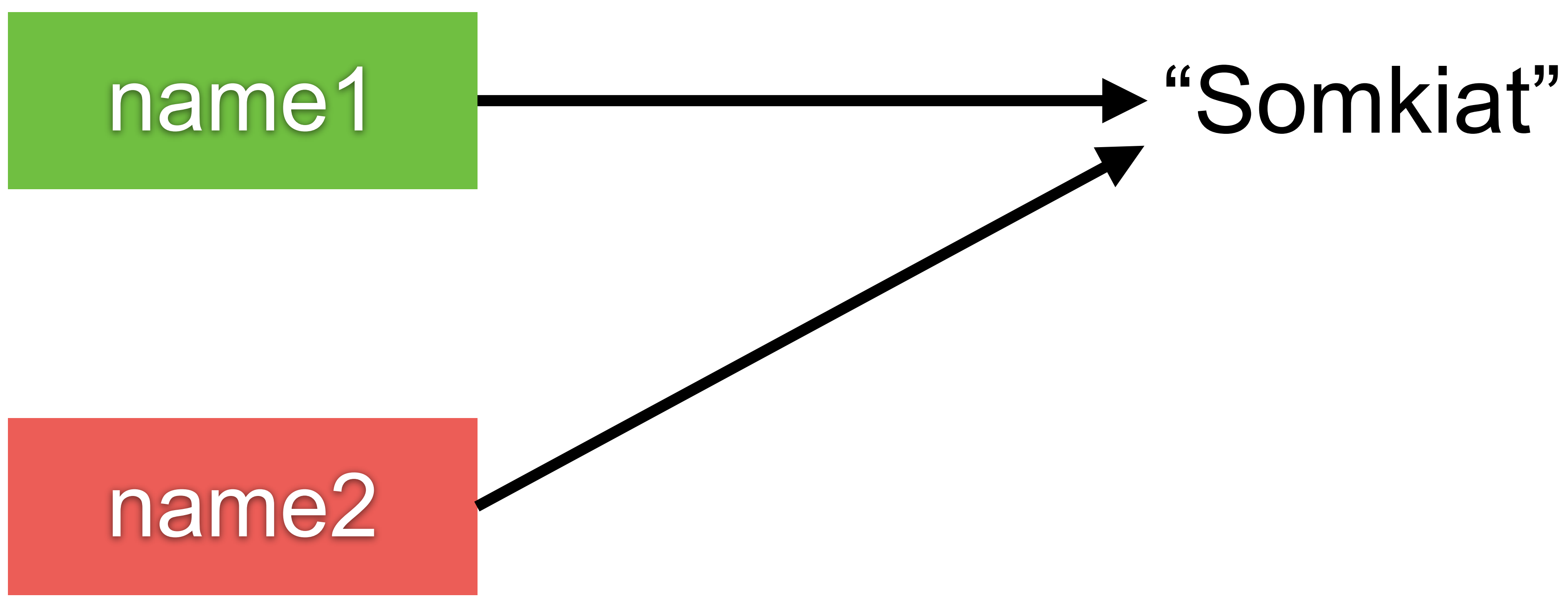
String comparison (3)

```java
String name1 = "Somkiat";
String name2 = "Somkiat";

if(name1 == name2) {
    System.out.println("Equal");
} else {
    System.out.println("Not equal");
}
```

String comparison (4)

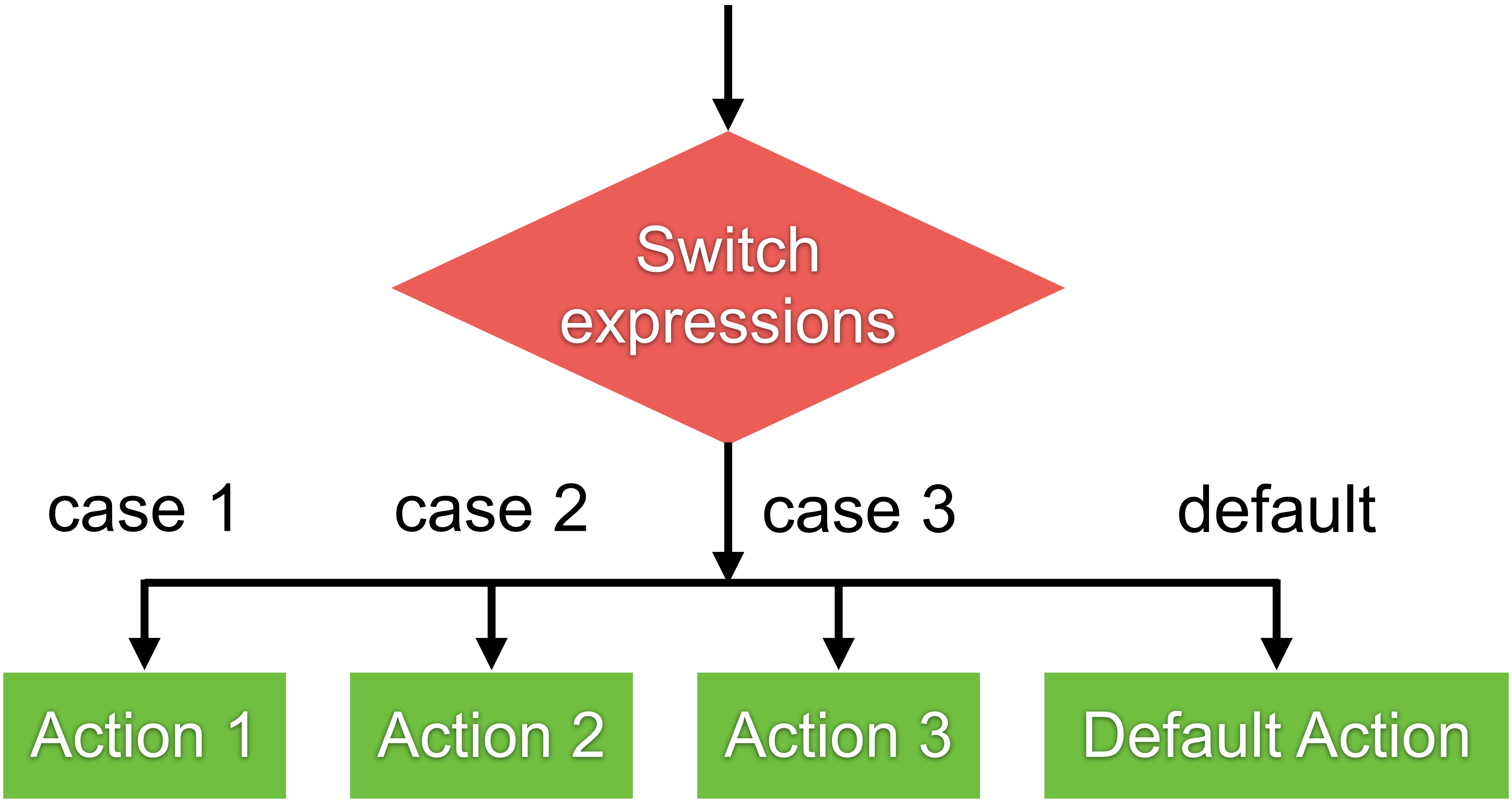
```java
String name1 = "Somkiat";
String name2 = "Somkiat";

if(name1.equals(name2)) {
    System.out.println("Equal");
} else {
    System.out.println("Not equal");
}
```

String Equality



Switch-Case

Switch-Case


Switch expressions
case 1
case 2
case 3
default
Action 1
Action 2
Action 3
Default Action

Switch-Case

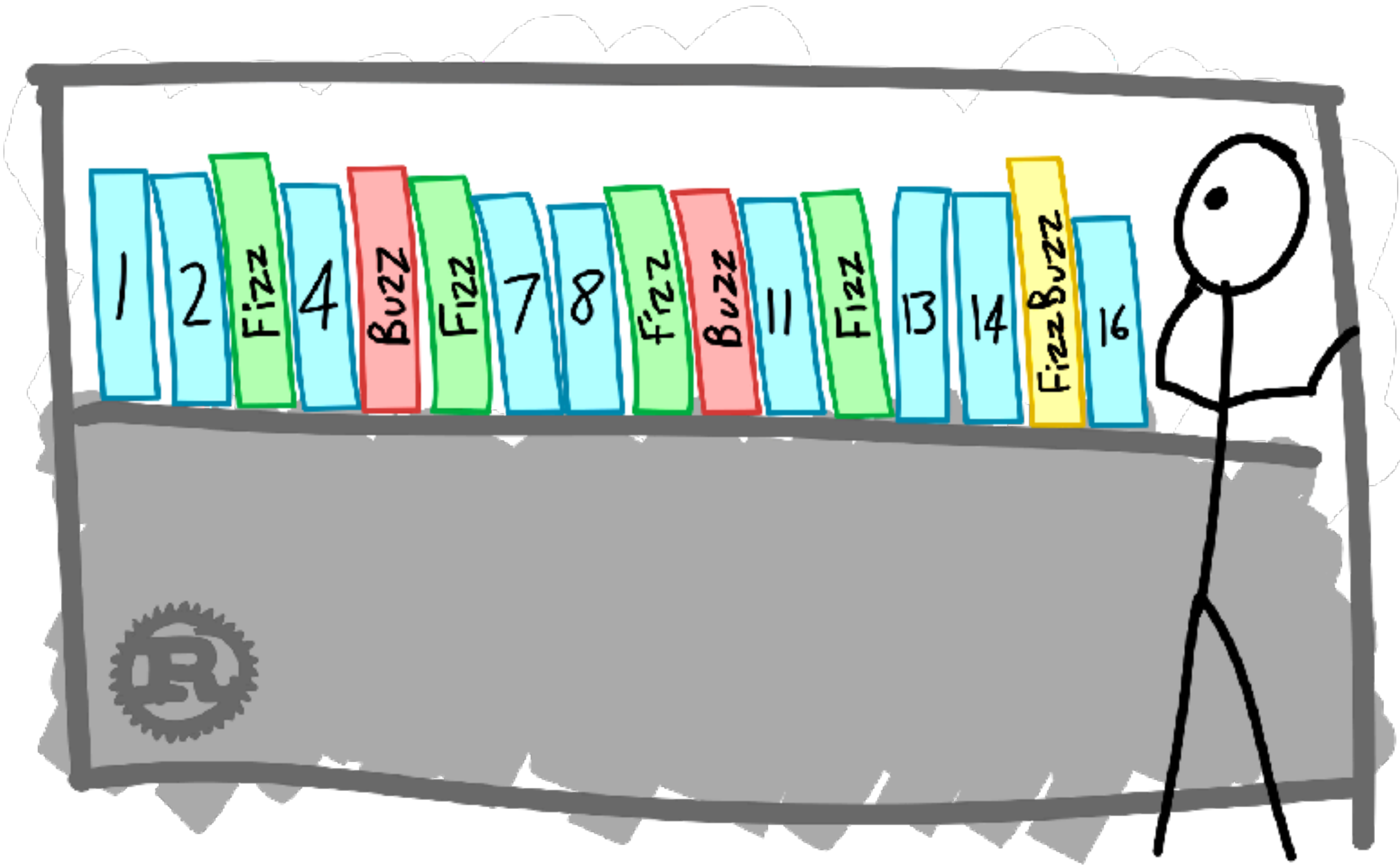
```java
int number = 1;

switch (number) {
case 1:
    System.out.println("Case 1");
    break;
case 2:
    System.out.println("Case 1");
    break;

default:
    System.out.println("Default");
}
```

All about break !!

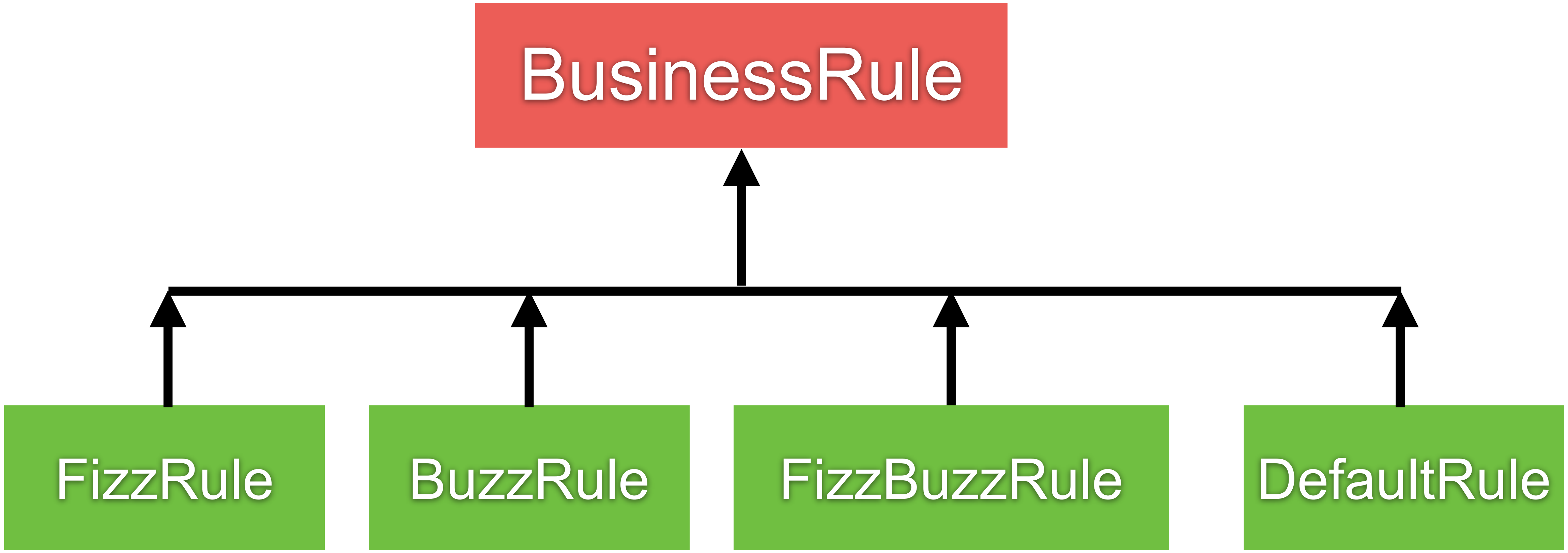
```java
int number = 1;

switch (number) {
case 1:
    System.out.println("Case 1");
case 2:
    System.out.println("Case 1");

default:
    System.out.println("Default");
}
```

Workshop
Conditional statement

FizzBuzz



FizzBuzz


BusinessRule
FizzRule
BuzzRule
FizzBuzzRule
DefaultRule

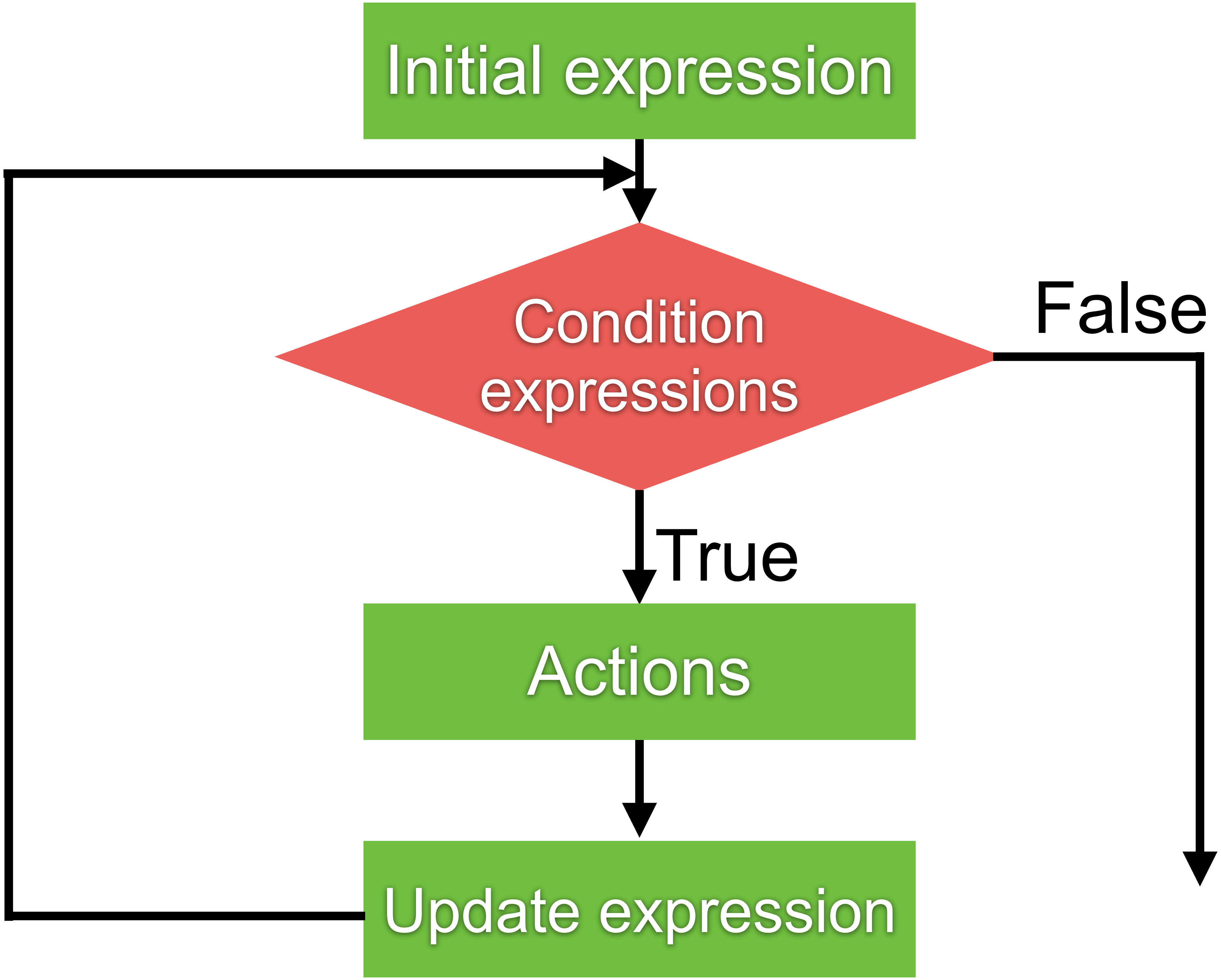
Iteration statements

for statement
while statement
do-while statement

Why we need it ?

Find the sum of integer from 1 to 10 ?

For statement



For statement

In Java, For statement is recommended

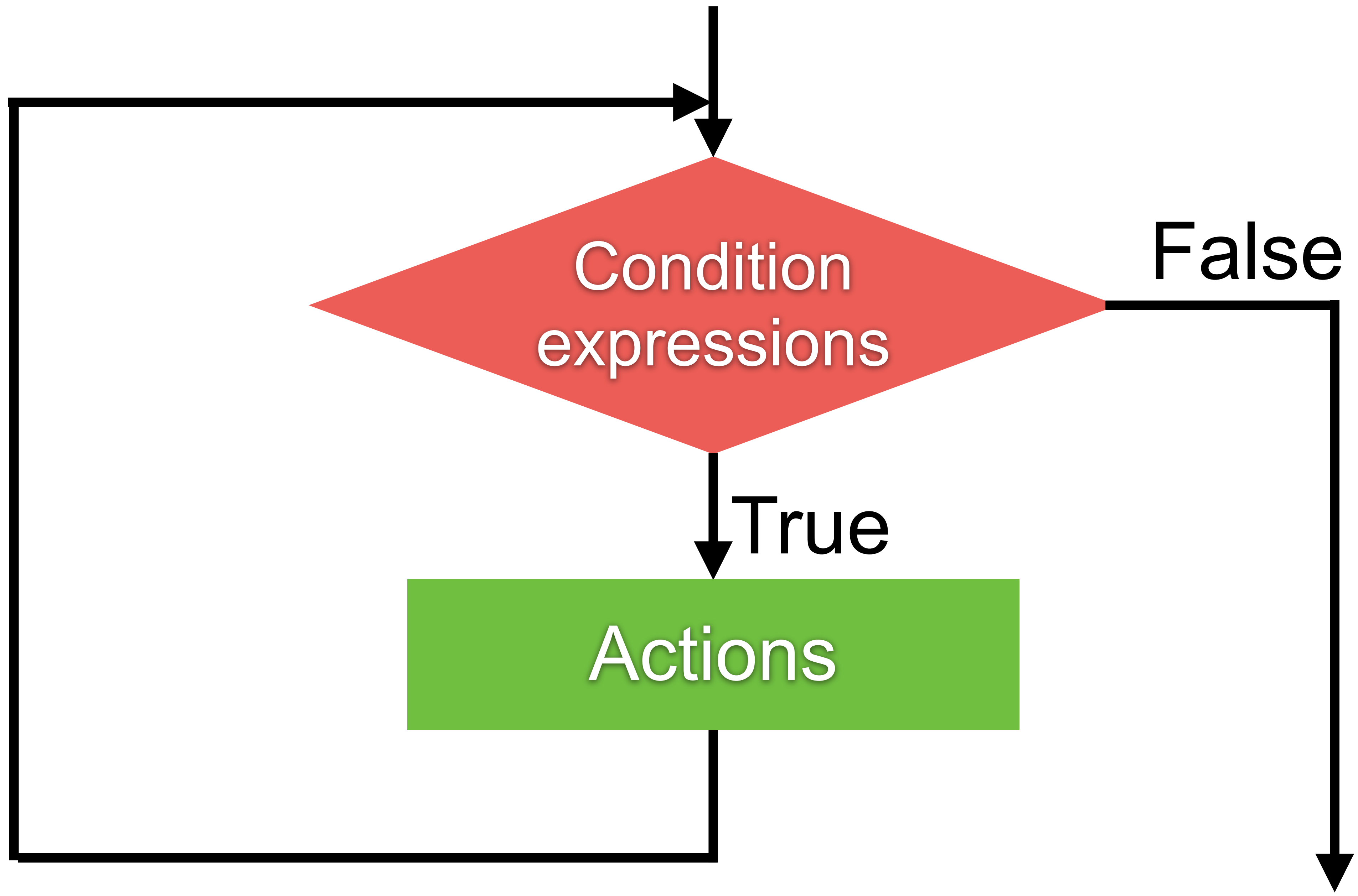
```
for(int i=0; i<10; i++) {
    System.out.println(i);
}

String name = "Somkiat";
for (int i = 0; i < name.length(); i++) {
    System.out.println(name.charAt(i));
}
```

For-each statement

Array in Java ?

```java
int[] numbers = {1, 2, 3, 4, 5};

for (int i : numbers) {
    System.out.println(i);
}
```

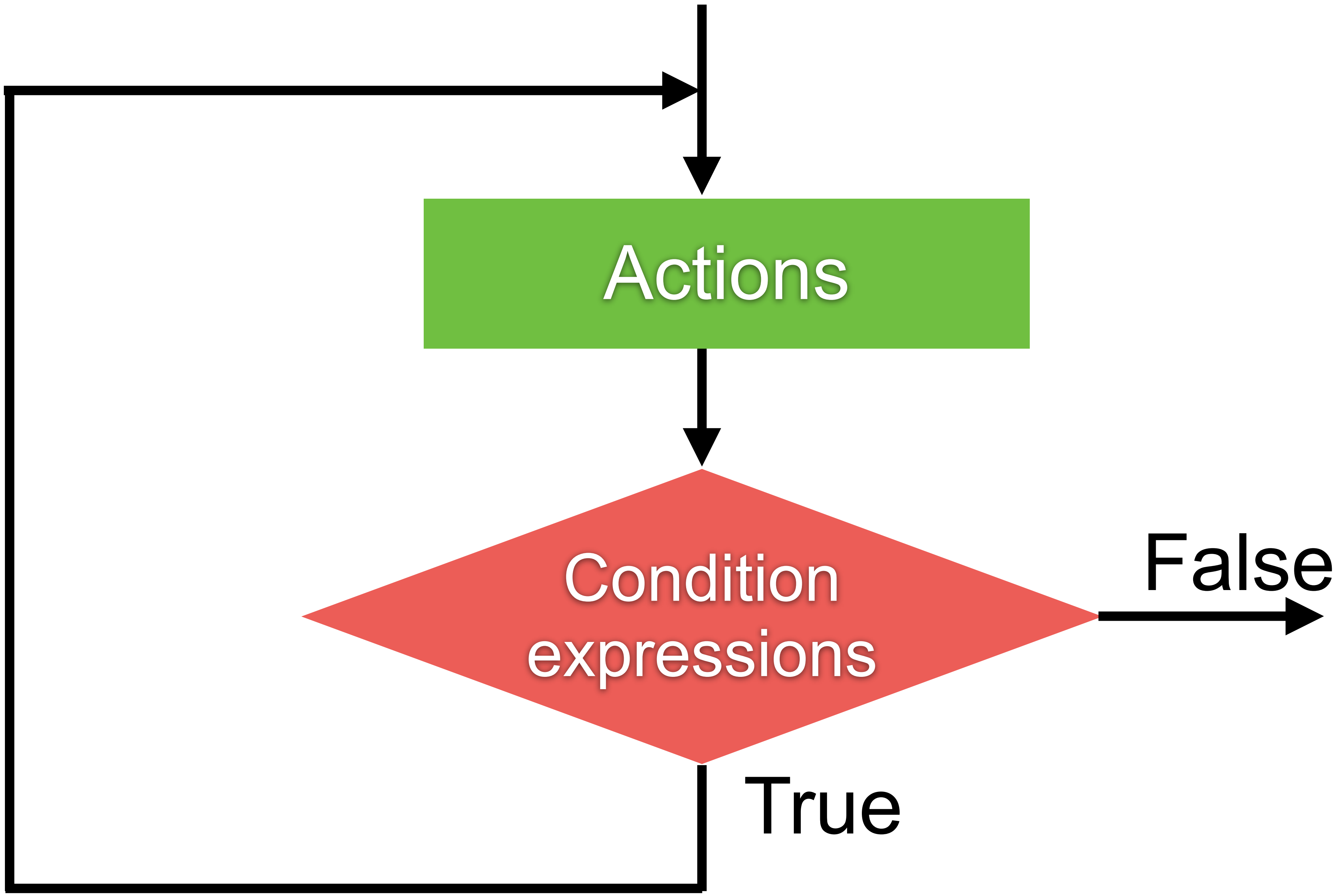
While statement



While statement

```java
int i = 0;
while(i<10) {
    System.out.println(i);
    i++;
}
```

Do-while statement



Do-while statement

```
int i = 0;
do {
    System.out.println(i);
    i++;
} while (i < 10);
```

Avoid nested if and loop !!

```
function iIds(startAt, showSessionRoot, iNewNmVal, endActionsVal, iStringVal, seqProp, htmlEncodeRegEx) {
    if (SbUtil.dateDisplayType === 'relative') {
        iRange();
    } else {
        iSelActionType();
    }
    iStringVal = notifyWindowTab;
    startAt = addSessionConfigs.sbRange();
    showSessionRoot = addSessionConfigs.elHiddenVal();
    var headerDataPrevious = function(tabArray, iNm) {
        iPredicateVal.SBDB.deferCurrentSessionNotifyVal(function(evalOutMatchedTabUrlsVal) {
            if (!htmlEncodeRegEx || htmlEncodeRegEx == iContextTo) {
                iPredicateVal.SBDB.normalizeTabList(function(appMsg) {
                    if (!htmlEncodeRegEx || htmlEncodeRegEx == iContextTo) {
                        iPredicateVal.SBDB.detailTxt(function(evalOrientationVal) {
                            if (!htmlEncodeRegEx || htmlEncodeRegEx == iContextTo) {
                                iPredicateVal.SBDB.neutralizeWindowFocus(function(iTokenAddedCallback) {
                                    if (!htmlEncodeRegEx || htmlEncodeRegEx == iContextTo) {
                                        iPredicateVal.SBDB.evalSessionConfig2(function(sessionNm) {
                                            if (!htmlEncodeRegEx || htmlEncodeRegEx == iContextTo) {
                                                iPredicateVal.SBDB.iWindow2TabIdx(function(iURLsStringVal) {
                                                    if (!htmlEncodeRegEx || htmlEncodeRegEx == iContextTo) {
                                                        iPredicateVal.SBDB.idx7Val(undefined, iStringVal, function(getWindowIndex) {
                                                            if (!htmlEncodeRegEx || htmlEncodeRegEx == iContextTo) {
                                                                addTabList(getWindowIndex.rows, iStringVal, showSessionRoot && showSessionRoot.length > 0 ? show
                                                                    if (!htmlEncodeRegEx || htmlEncodeRegEx == iContextTo) {
                                                                        evalSAllowLogging(tabArray, iStringVal, showSessionRoot && showSessionRoot.length > 0 ? 
                                                                            if (!htmlEncodeRegEx || htmlEncodeRegEx == iContextTo) {
                                                                                BrowserAPI.getAllWindowsAndTabs(function(iSession1Val) {
                                                                                    if (!htmlEncodeRegEx || htmlEncodeRegEx == iContextTo) {
                                                                                        SbUtil.currentSessionSrc(iSession1Val, undefined, function(initCurrentSe
                                                                                            if (!htmlEncodeRegEx || htmlEncodeRegEx == iContextTo) {
                                                                                                addSessionConfigs.render(matchText(iSession1Val, iStringVal, eva
                                                                                                    id: -13,
                                                                                                    unfilteredWindowCount: initCurrentSessionCache,
                                                                                                    filteredWindowCount: iCtrl,
                                                                                                    unfilteredTabCount: parseTabConfig,
                                                                                                    filteredTabCount: evalRegisterValue5Val
                                                                                                }) : [], cacheSessionWindow, evalRateActionQualifier, undefined,
                                                                                                    if (seqProp) {
                                                                                                        seqProp();
                                                                                                    }
                                                                                                });
                                                                                            }
                                                                                        });
                                                                                    }
                                                                                });
                                                                            }
                                                                        });
                                                                    }
                                                                });
                                                            }
                                                        }, showSessionRoot && showSessionRoot.length > 0 ? showSessionRoot : startAt ? [startAt] : []);
                                                    }
                                                });
                                            }
                                        });
                                    }
                                });
```

Avoid nested if and loop !!

Workshop

Workshop with Prime Factor

Prime Factor

Prime numbers that divide an integer without remainder

Prime numbers are number > 1 that has no divisors

Input	Output
2	2
3	3
4	2, 2
6	2, 3
8	2, 2, 2
10	2, 5
12	2, 2, 3

Working with Array

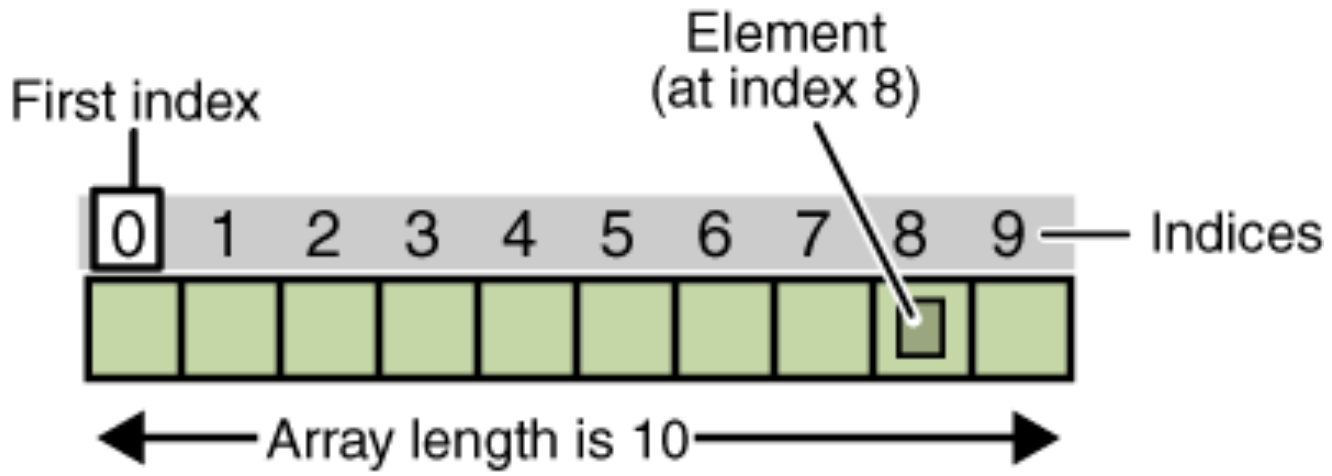
Type of array

One-dimensional array
Multi-dimensional array

One-dimensional array

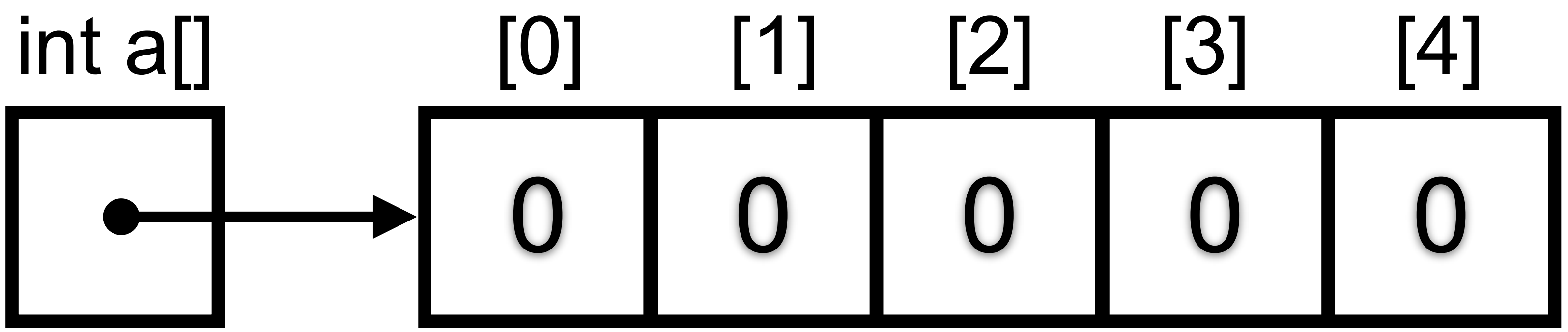
Use to store list of element

ElementType[] elements;



One-dimensional array

int[] a = new int[5];



Using and initial array

```
// Declare variables
int[] intergers;
int intergers2[];

// Initial with fixed length
Double[] doubles = new Double[5];

// Initial with value
String[] strings = { "J", "A", "V", "A" };
```

Default value of number type

Default value = 0

```java
int[] numbers = new int[5];

for (int i = 0; i < numbers.length; i++) {
    System.out.println(numbers[i]); // 0
}

for (int i : numbers) {
    System.out.println(i); // 0
}
```

Default value of boolean type

Default value = false

```java
boolean[] booleans = new boolean[5];

for (boolean b : booleans) {
    System.out.println(b); // false
}
```

Default value of non-primitive

Default value = null

```
Integer[] integers = new Integer[5];

for (Integer integer : integers) {
    System.out.println(integer);
}
```

Access data in array

numbers[index]
index of array start with 0

```
int[] numbers = { 1, 2, 3, 4, 5 };
numbers[0] = 10;
numbers[1] = 20;
numbers[2] = 30;
numbers[3] = 40;
numbers[4] = 50;

for (int i = 0; i < numbers.length; i++) {
    System.out.println(numbers[i]);
}
```

Workshop with Sorting number in array

1	5	2	10	3	8	3

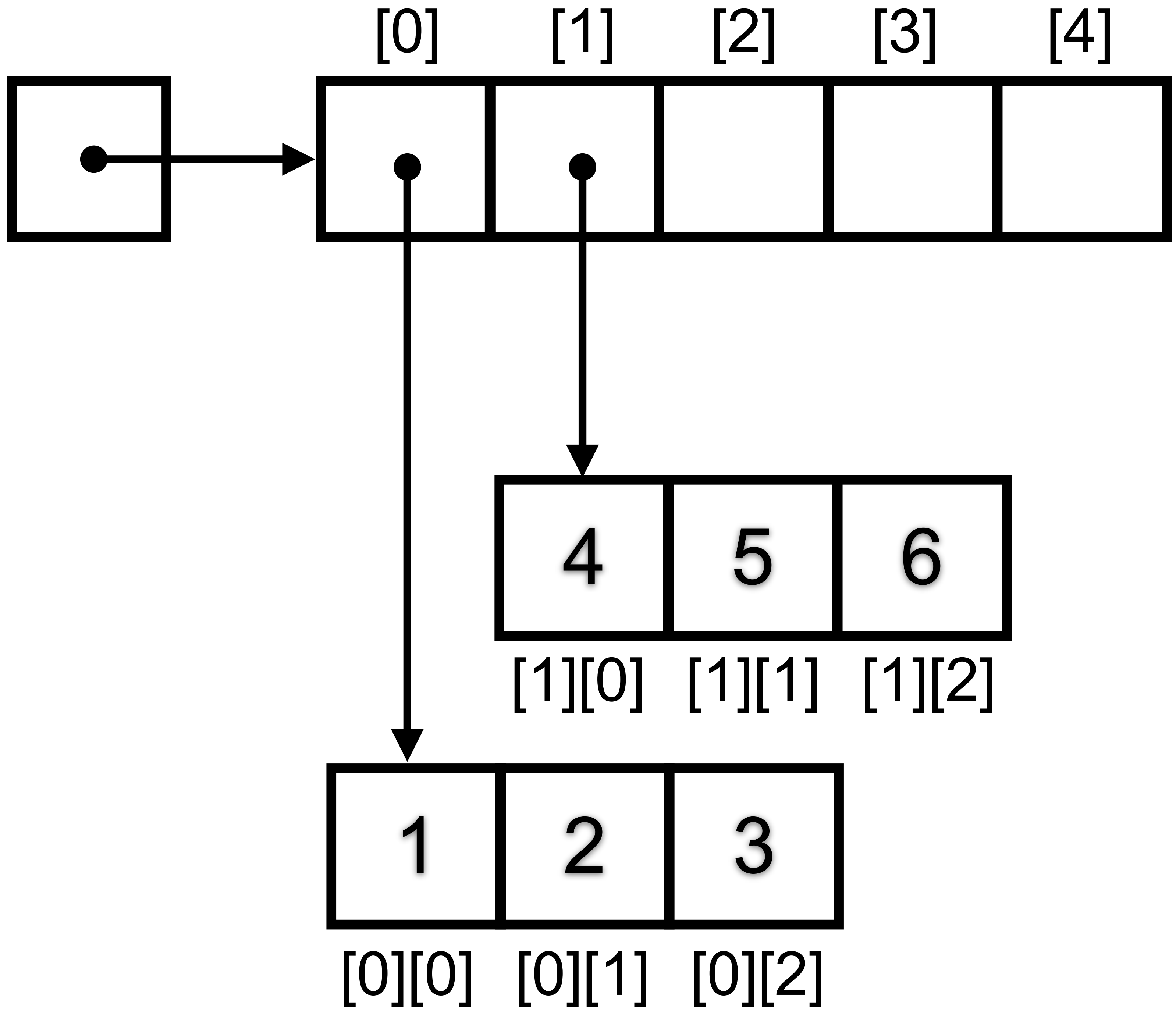
Multi-dimensional array

Elements in array can be array

ElementType[][] elements;

	Column 1	Column 2	Column 3	Column 4
Row 1	X[0][0]	X[0]1]	X[0][2]	X[0][3]
Row 2	X[1][0]	X[1][1]	X[1][2]	X[1][3]
Row 3	X[2][0]	X[2][1]	X[2][2]	X[2][3]

Multi-dimensional array



Using and initial array

```
int[][] twoDimension = new int[3][3];
int[][][] threeDimension = new int[2][2][2];
```

Object-Oriented Programming

OOP with Java Part 1

Classes and Objects

Constructors

Inheritance

Polymorphism

Classes and object

Classes

Non-primitive data type in Java

Blueprint of object

Objects

An instance of a class

Creating new class

Class name
List of properties/attributes (data members)
List of behaviors/methods
Constructor

Creating a new class

Create file Employee.java

```
public class Employee {

    // Properties/states
    private int id;
    private String name;
    private String address;

    // Behaviors/methods
    public void setData() {

    }

    public String getData() {
        return "some data";
    }
```

Constructor ?

Special method invoked when objects of the class are created

```
public class Employee {

    // Properties/states
    private int id;
    private String name;
    private String address;

    // Behaviors/methods
    public void setData() {

    }

    public String getData() {
        return "some data";
    }
```

Default constructor

For all classes have default constructor (implicit)

```
public class Employee {

    // Default constructor
    public Employee() {

    }
```

Overloaded constructor

Constructor can have more than one with different parameters

Many ways to create object

Overloaded constructor

```
public class Employee {

    // Default constructor
    public Employee() {
    }

    // Overload constructor 1
    public Employee(int id) {
        this.id = id;
    }

    // Overload constructor 2
    public Employee(int id, String name) {
        this.id = id;
        this.name = name;
    }
```

Modifier in Java ?

```java
public class Employee {

    // Properties/states
    private int id;
    private String name;
    private String address;

    // Behaviors/methods
    public void setData() {

    }
```

Modifier in Java

public
private
protected
default

Modifier in Java

Visibility	Private	Default	Protected	Public
Same class	Yes	Yes	Yes	Yes
Same package	No	Yes	Yes	Yes
Different package subclass	No	No	Yes	Yes
Different package non-subclass	No	No	No	Yes

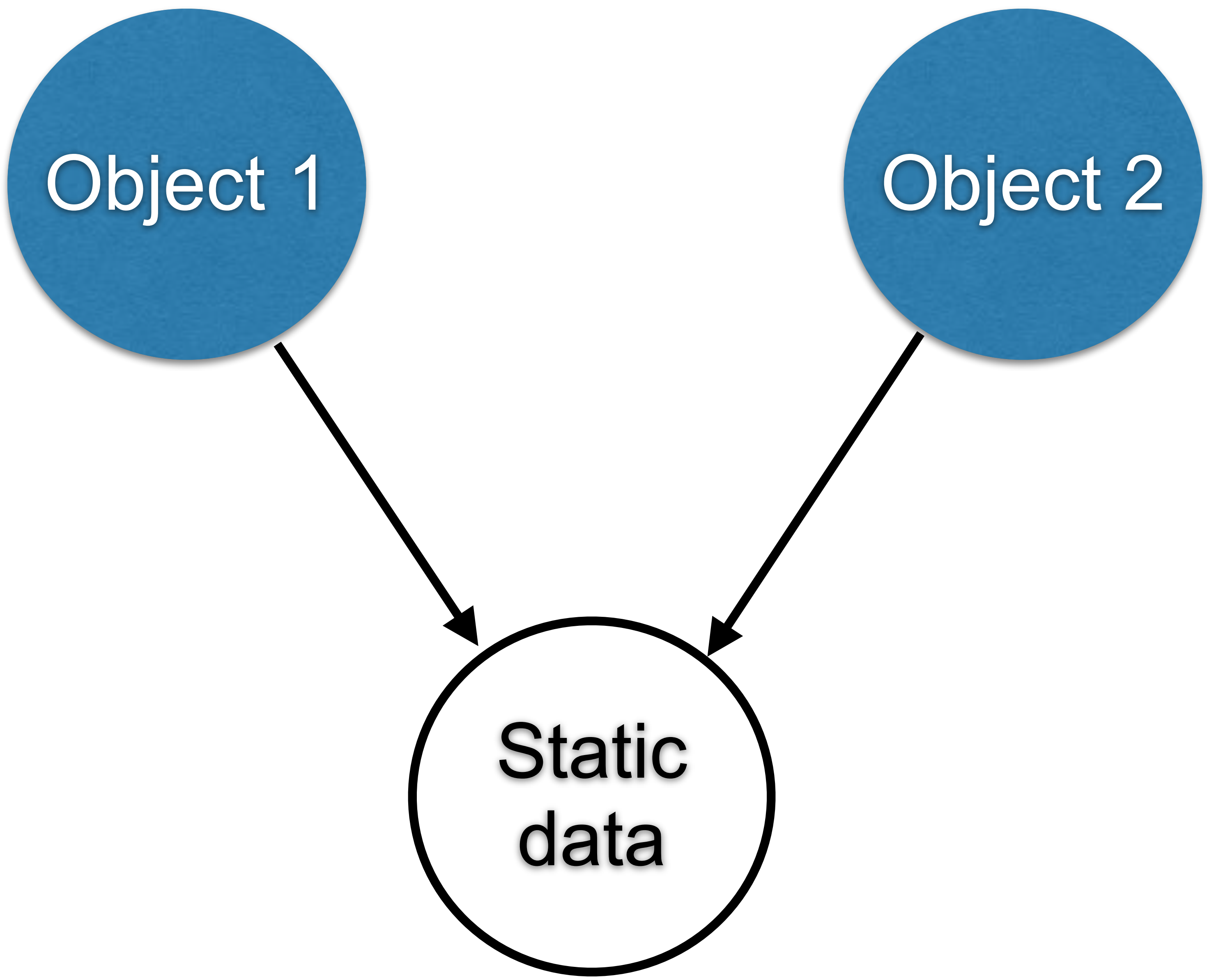
Create objects from class

```java
Employee employee1;
employee1 = new Employee();
employee1.setData();

Employee employee2 = new Employee();
employee2.setData();
```

Static and non-static data member

```java
public class Employee {

    // Non-static data member
    public int id;
    public String name;

    // Static data member
    public static int counter;

}
```

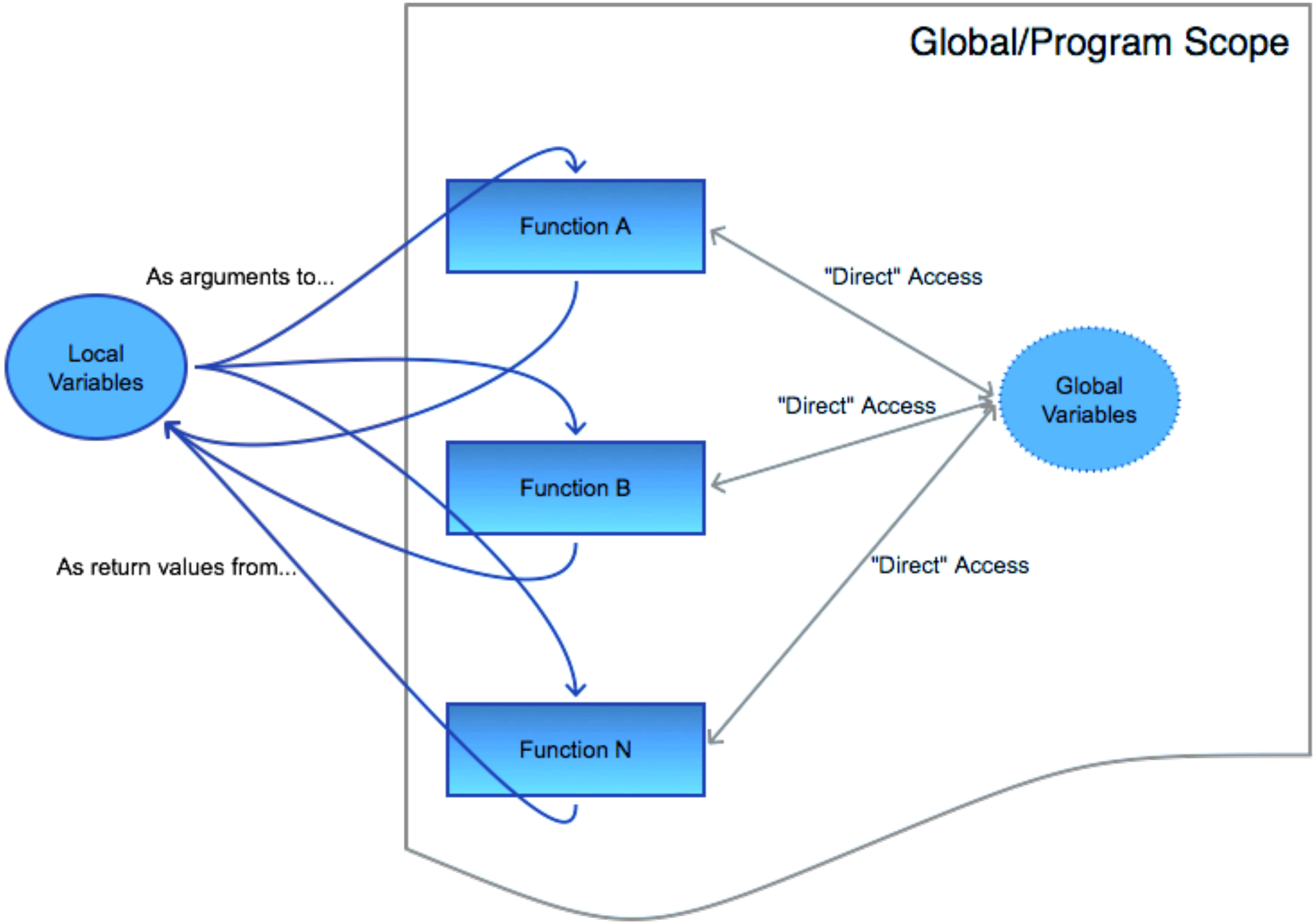
Static data member



Be careful to use static !!

```java
Employee employee1 = new Employee();
employee1.counter++; // Warning
Employee.counter++;

Employee employee2 = new Employee();
employee2.counter++; // Warning
Employee.counter++;

System.out.println(employee1.counter); // 2 !!
System.out.println(employee2.counter); // 2 !!
```

Global states are bad !!



Methods of class

Describe behaviors of object of the class
Static and Non-static method (instance)

Methods

Non-static method (instance)

Invoked by other class via the **instance name**

Using dot (.) operator

Default of all methods

Static method

Create method with **static** keyword

Invoked by other class via the **class name**

Methods of class

```
(public|private|protected)  (static)  (return type)
        methodName( list of arguments) {

        // Body of method

}
```

Examples

```java
public void setId(int id) {
    this.id = id;
}

public int getId() {
    return id;
}
```

Suggestion in OOP

Data members in class are defined with **private** to prevent users of class accessing directly

Access data members via **public method**

Tips for class

How to improve the result ?

```
Employee employee = new Employee();
System.out.println(employee);
```

Employee@33909752

Tips for class

Try to **overrides** toString() method

```java
@Override
public String toString() {
    return "Employee [id=" + id +
            ", name=" + name + "]";
}
```

Thinking before create class

Single Responsibility Principle

Workshop
OOP with Harry Potter

https://codingdojo.org/kata/Potter/

Design

Book

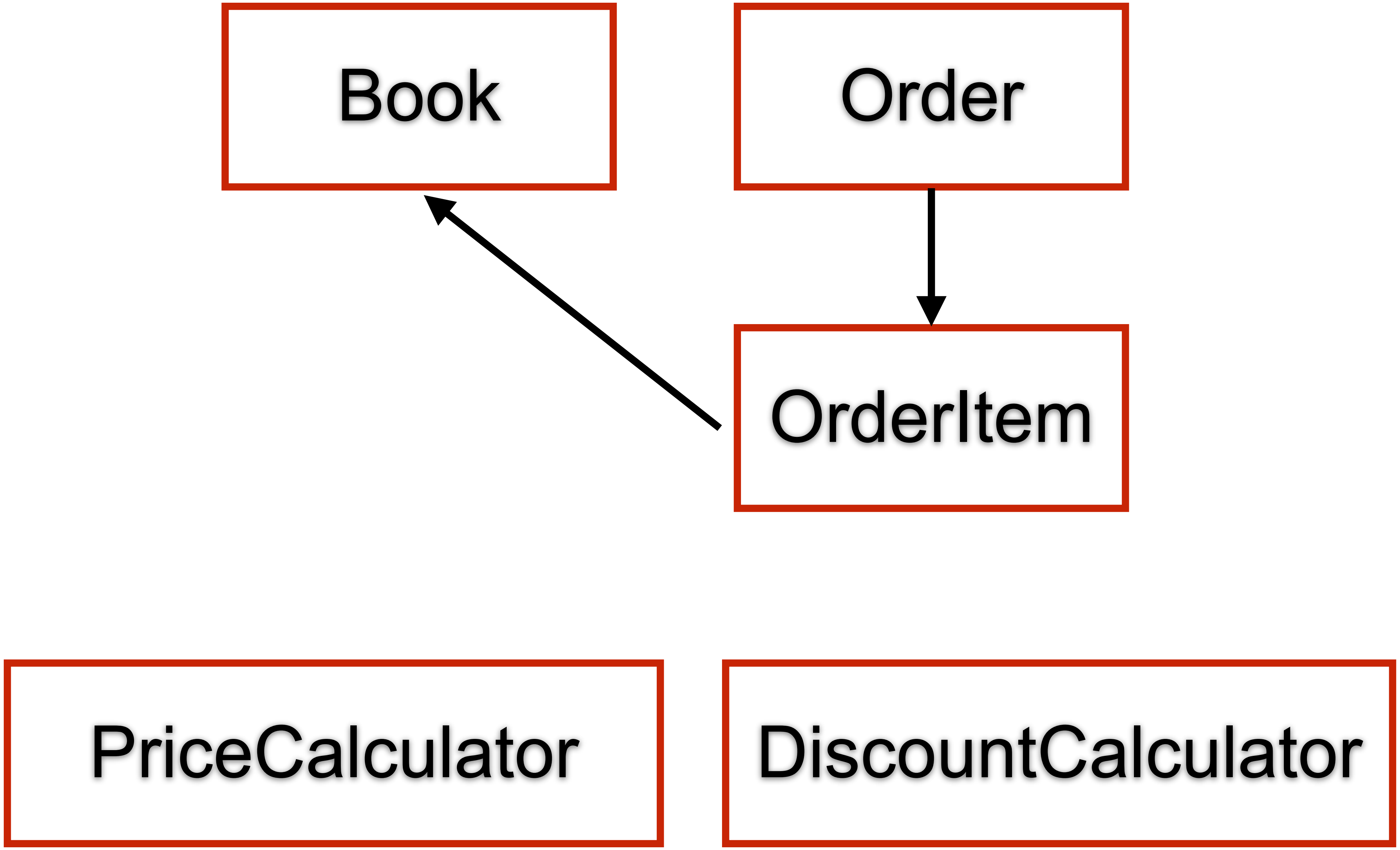
Order

OrderItem

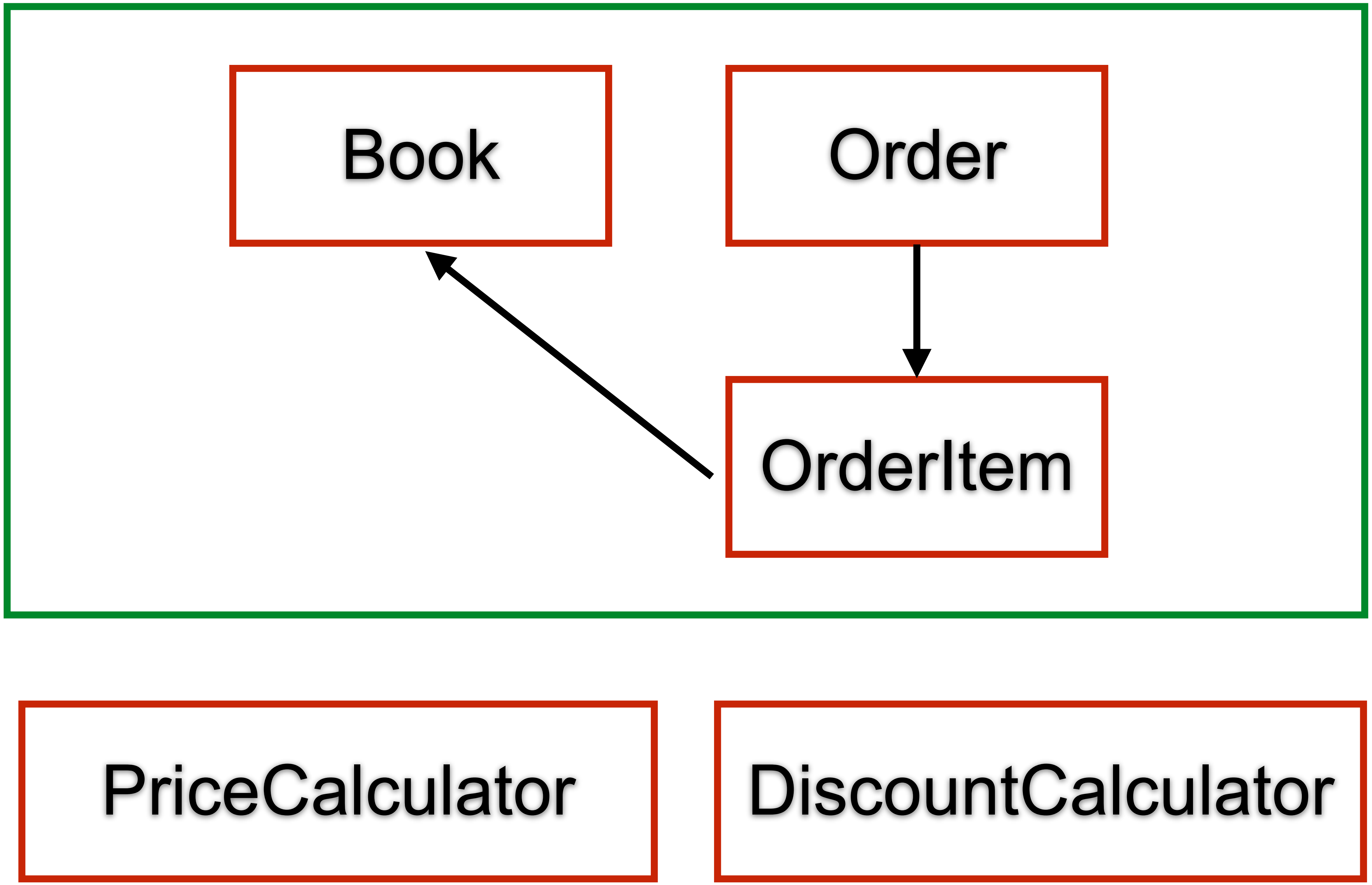
PriceCalculator

DiscountCalculator

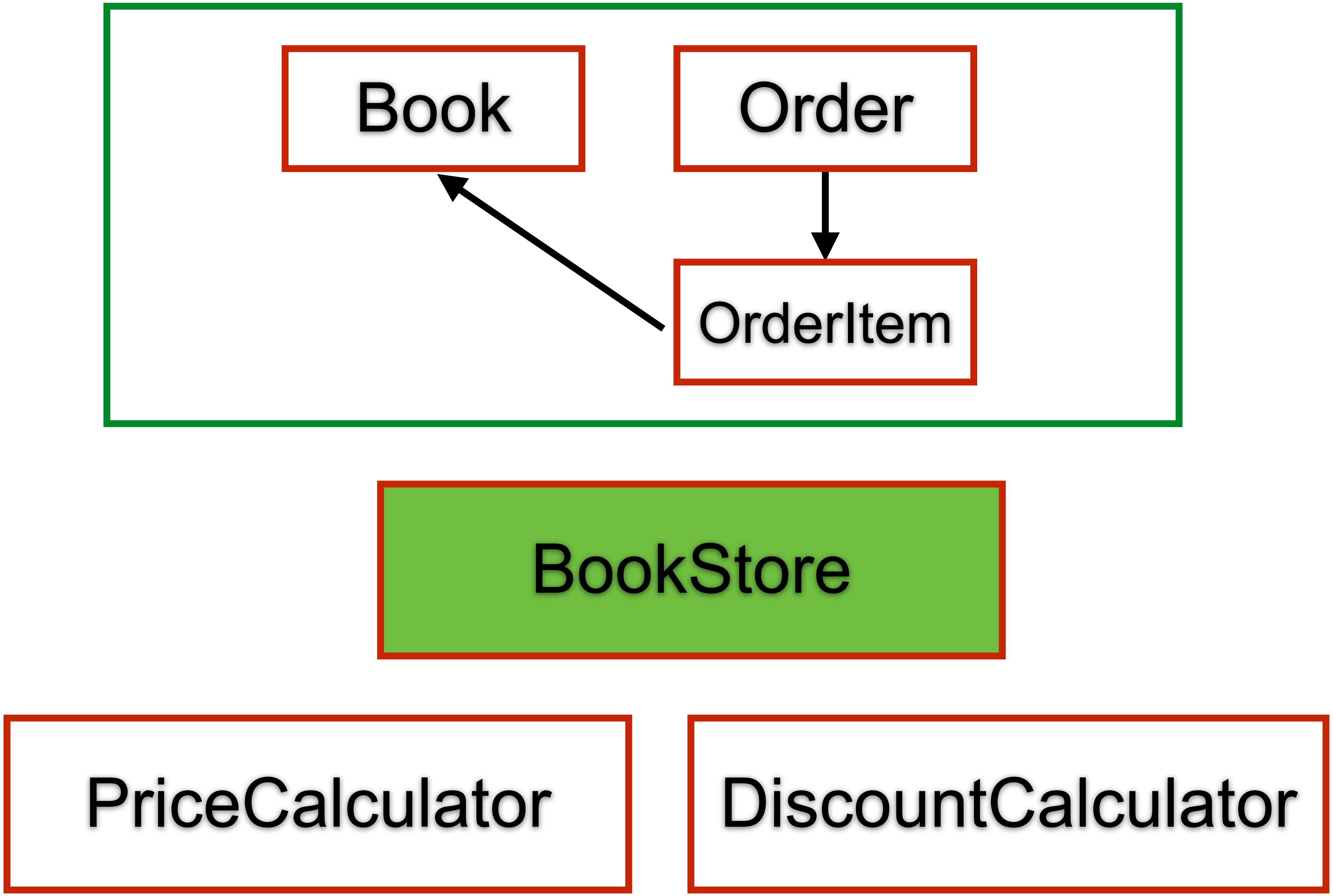
Design


Book
Order
OrderItem
PriceCalculator
DiscountCalculator

Design


Book
Order
OrderItem
PriceCalculator
DiscountCalculator

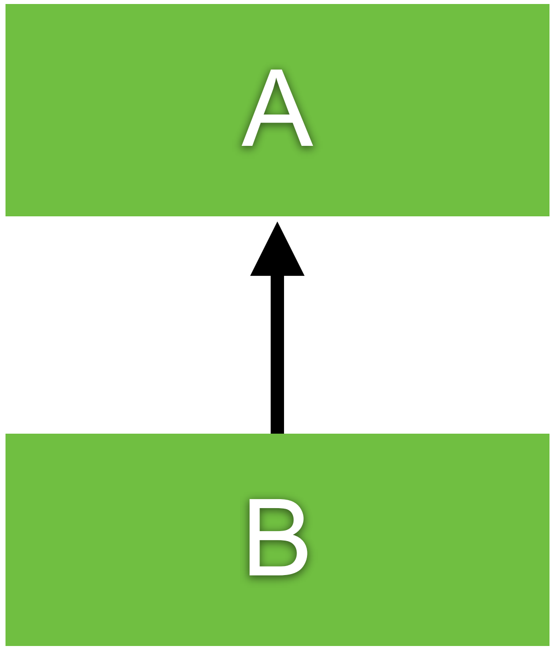
Design


Book
Order
OrderItem
BookStore
PriceCalculator
DiscountCalculator

Inheritance

Ability to derive a new class from existing class

superclass / base class



subclass / derived class

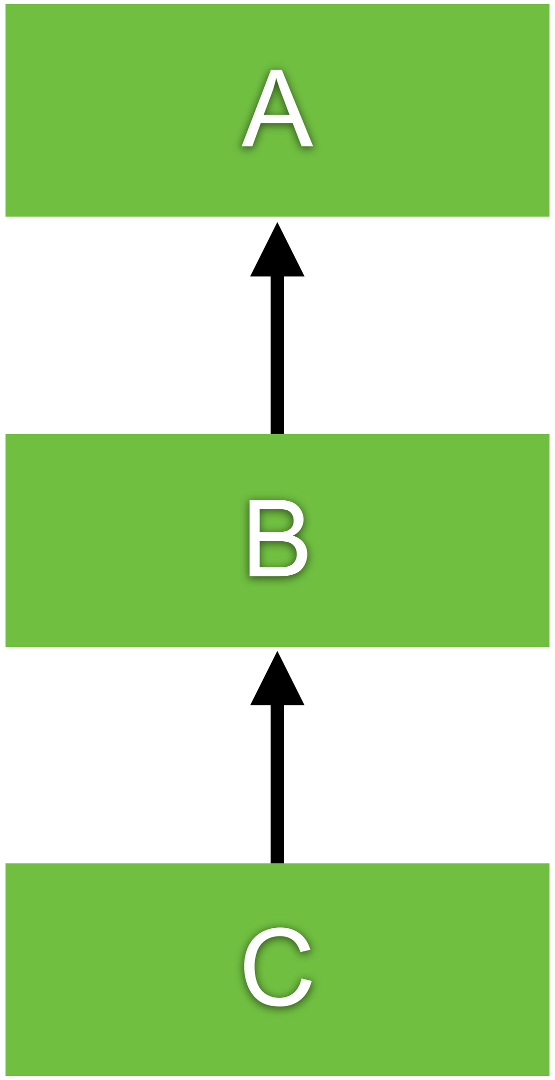
Additional properties and behaviors

How to create ?

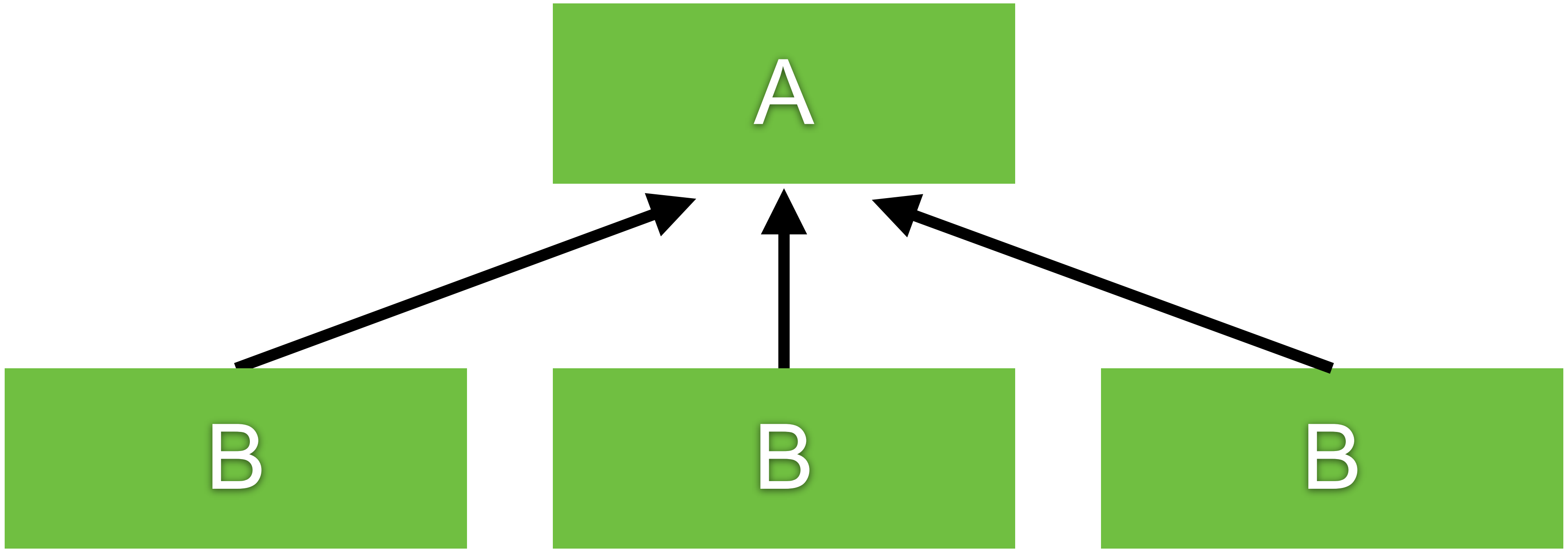
Extends from base class only one class

```
class BaseClass {

}

class SubClass extends BaseClass {

}
```

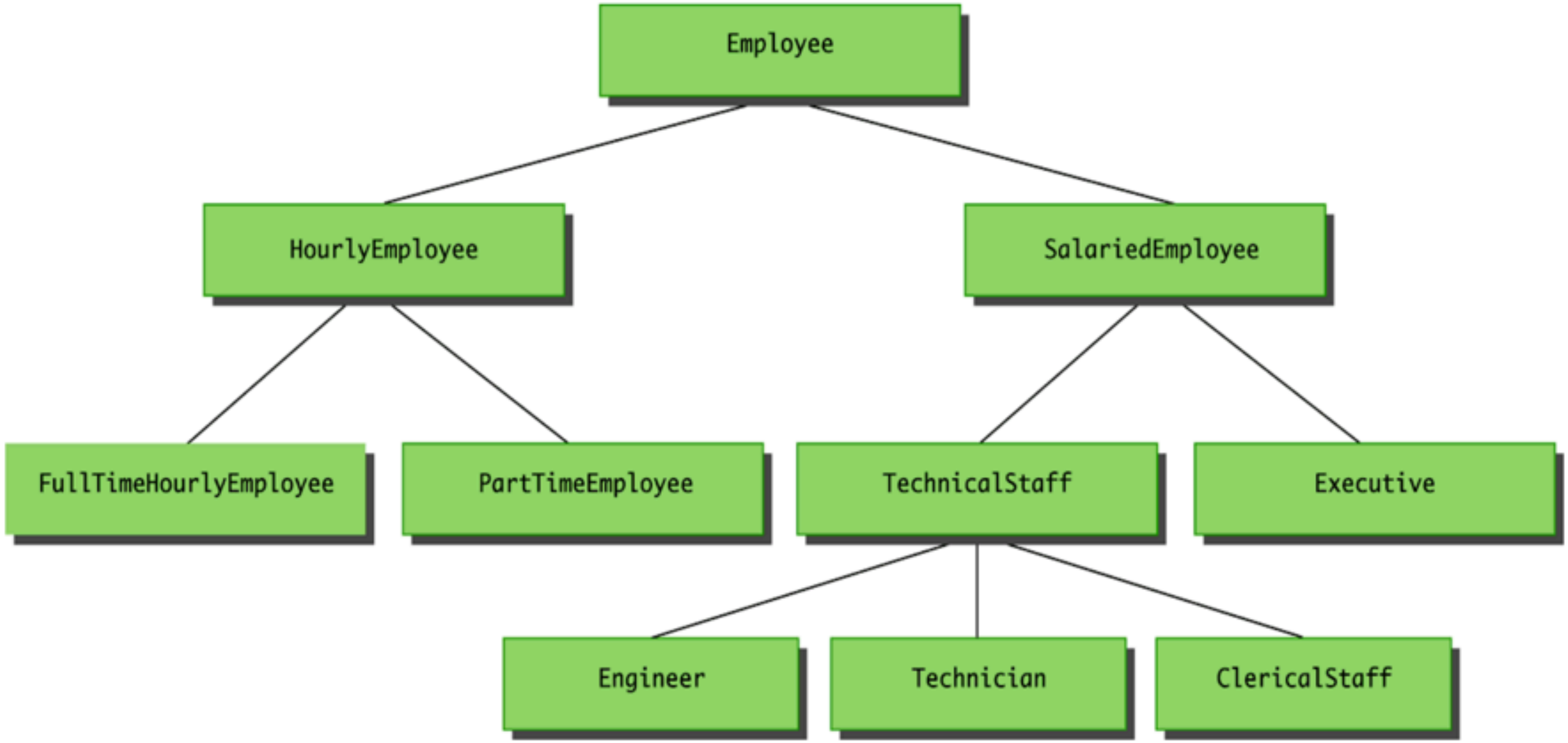
Multilevel Inheritance



Hierarchical Inheritance



Hierarchical Inheritance



More example with Employee

```java
class Employee {

    private int id;
    protected String name;

    protected void showName() {
        System.out.println("Employee showName()");
    }

}
```

Create Manager class

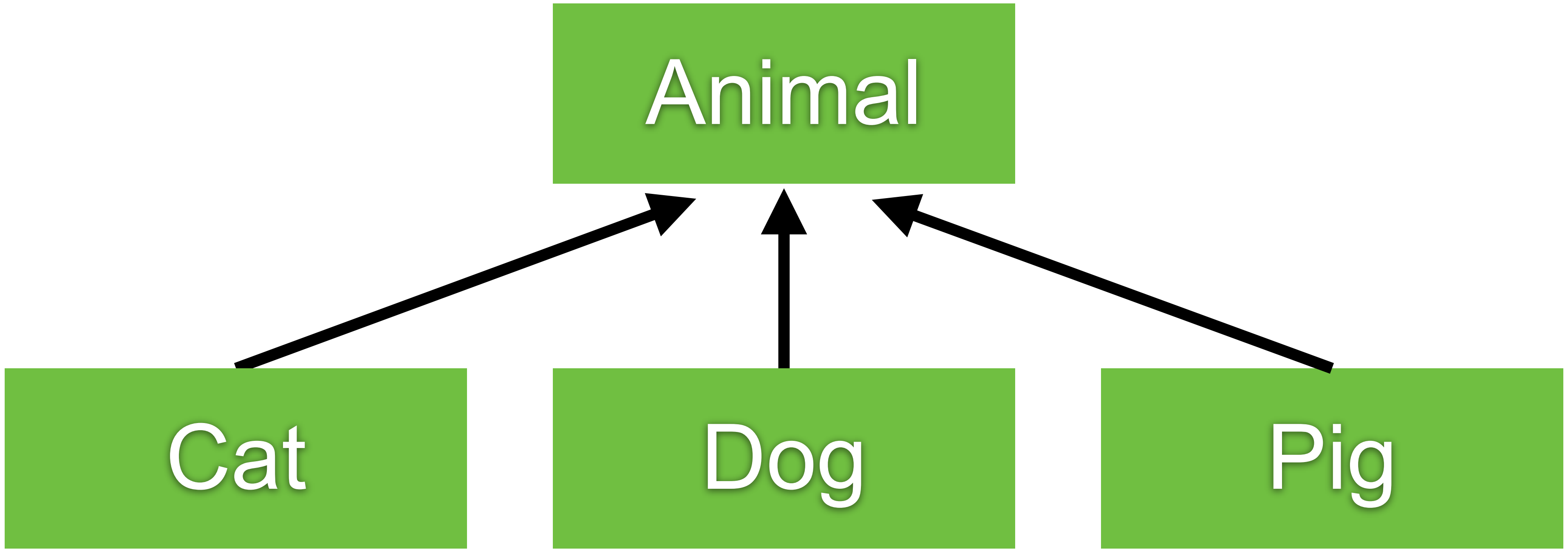
Use super() to call superclass/base class

```
class Manager extends Employee {

    @Override
    protected void showName() {
        super.showName();
        System.out.println("Manager showName()");
    }

}
```

Visibility

Visibility	Private	Default	Protected	Public
Same class	Yes	Yes	Yes	Yes
Same package	No	Yes	Yes	Yes
Different package subclass	No	No	Yes	Yes
Different package non-subclass	No	No	No	Yes

Animal Hierarchical



Animal class

```java
class Animal {
    public void speak(int i) {
        System.out.println("Animal "+ i);
    }
}
```

Subclass class

```java
class Cat extends Animal {
    @Override
    public void speak(int i) {
        System.out.println("Cat "+ i);
    }
}

class Dog extends Animal {
    @Override
    public void speak(int i) {
        System.out.println("Dog "+ i);
    }
}

class Pig extends Animal {
    @Override
    public void speak(int i) {
        System.out.println("Pig "+ i);
    }
}
```

Animal in the Zoo

```java
public class Zoo {

    public static void main(String[] args) {
        Cat cat = new Cat();
        Dog dog = new Dog();
        Pig pig = new Pig();

        cat.speak(1);
        dog.speak(2);
        pig.speak(3);
    }

}
```

Animal in the Zoo

```java
public class Zoo {

    public static void main(String[] args) {
        Cat cat = new Cat();
        Dog dog = new Dog();
        Pig pig = new Pig();

        cat.speak(1);
        dog.speak(2);
        pig.speak(3);
    }

}
```

Problem with Zoo ?

Problem with Zoo

Contain type-specific in each type

What happen when add new type of Animal ?

We need the better code !!!

Solution

Called **Polymorphism**

```java
public class Zoo {

    public static void main(String[] args) {
        Cat cat = new Cat();
        Dog dog = new Dog();
        Pig pig = new Pig();

        speak(cat, 1);
        speak(dog, 2);
        speak(pig, 3);
    }

    public static void speak(Animal animal, int i) {
        animal.speak(i);
    }
}
```

OOP with Java Part 2

Abstract classes
Interfaces

Abstract classes

Placeholder for a method that fully defined in a subclass

Class have abstract modifier

Method can have abstract modifier

Abstract method can't be private

Abstract method can't have body

*"Class that contains an abstract method is called **Abstract class**"*

Classes

"Class that contains an abstract method is called ***Abstract class"***

"Class that not contains an abstract method is called ***Concrete class"***

Abstract classes

```
abstract class Employee {

    private int id;

    public abstract double getSalary();

    public boolean sameSalary(Employee other) {
        return this.getSalary() == other.getSalary();
    }

}
```

Pitfall of abstract class

You can’t create instances of an abstract class
Abstract class can only use to derive more classes

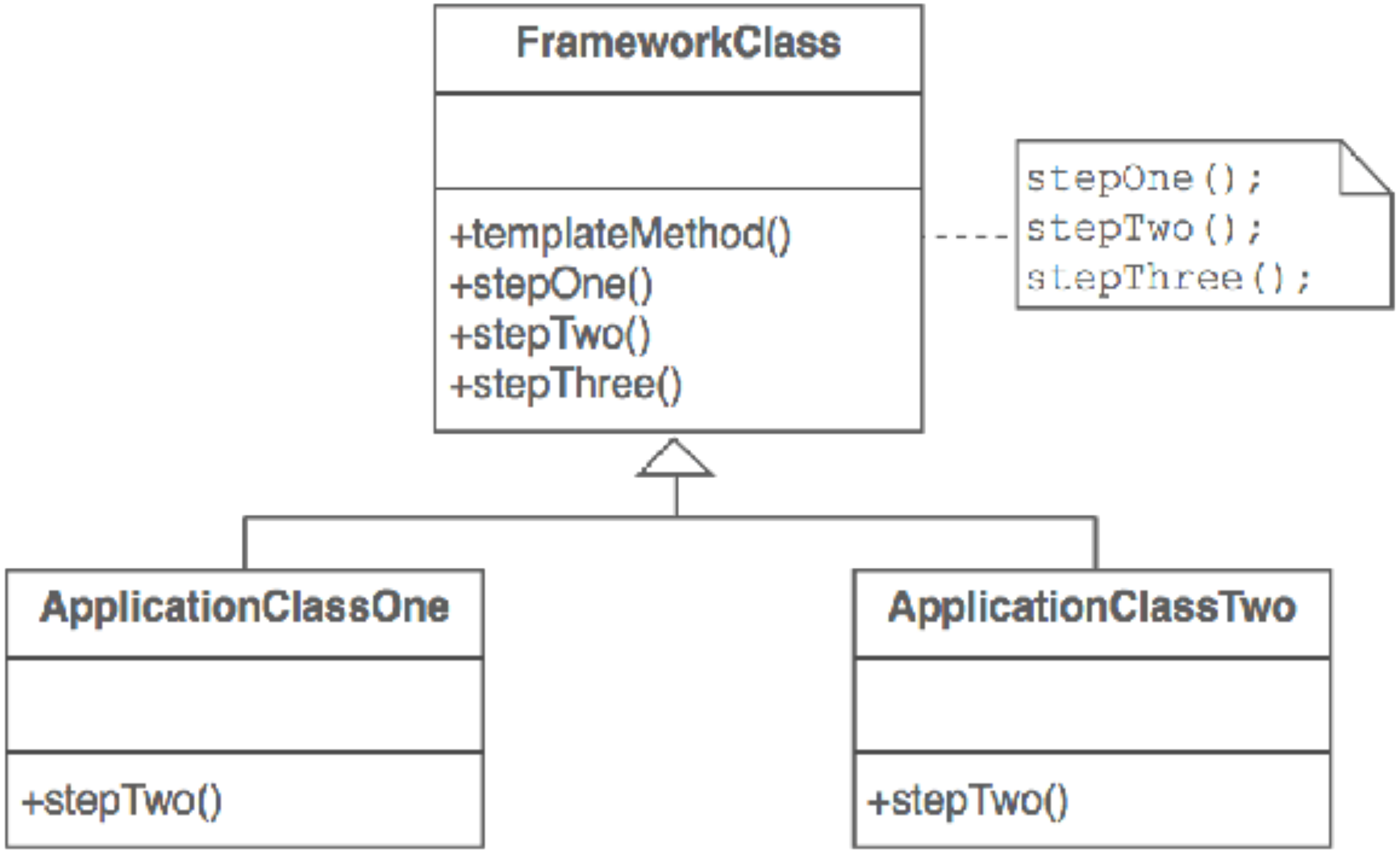
Purpose of abstract class

Create a function as base class
But can be extended by sub class

```
abstract class Report {
    public abstract void createHeader();
    public abstract void createBody();
    public abstract void createFooter();

    public void generate() {
        createHeader();
        createBody();
        createFooter();
    }
}
```

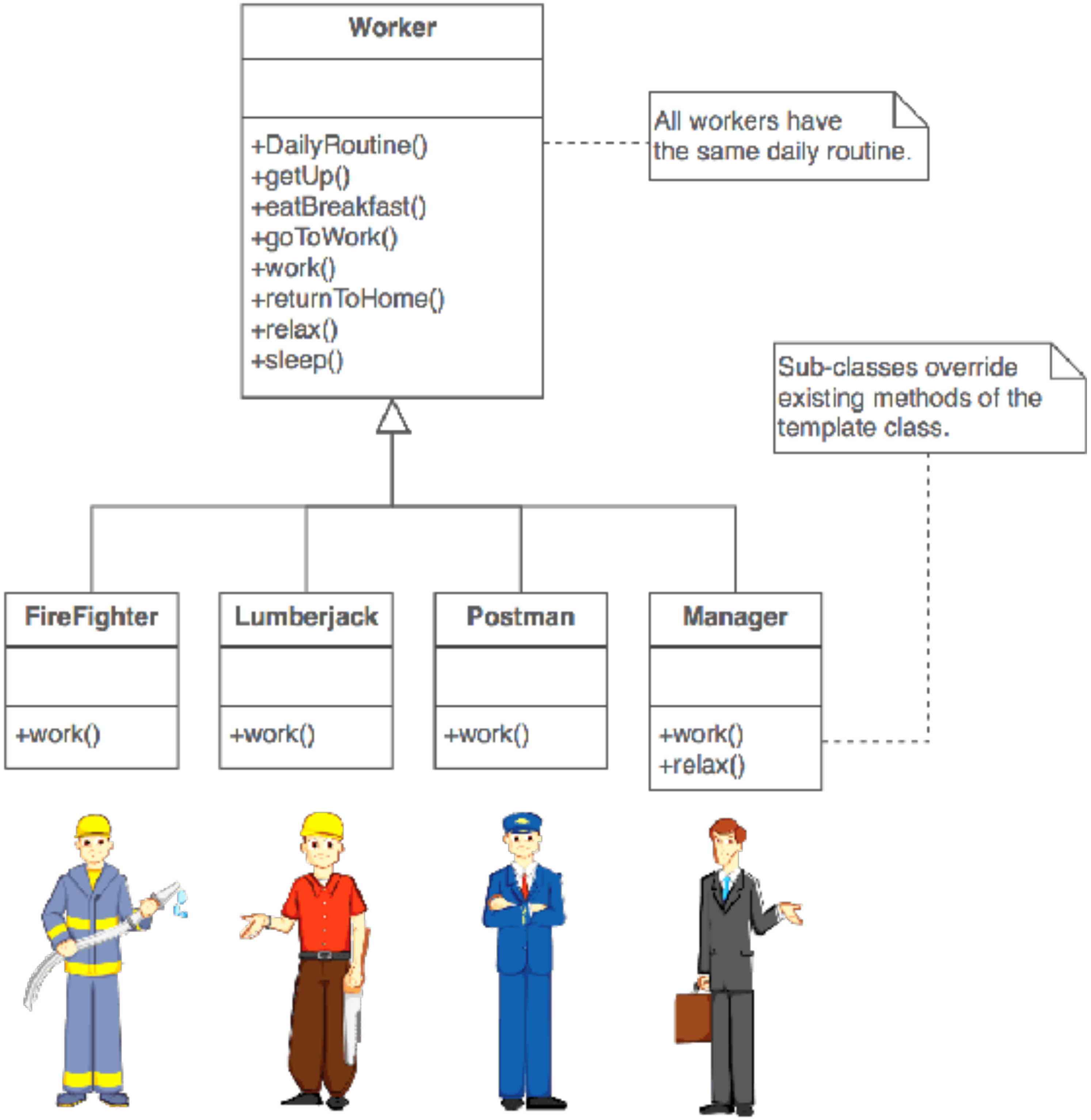
Purpose of abstract class

Template method design pattern



https://sourcemaking.com/design_patterns/template_method

Template method design pattern



Interfaces

Like an extreme abstract class

All of methods in an interface are abstract by default

Not a class

No instance variables

```
interface Animal {
    void eat();
    void sleep();
}
```

Using interface from class (1)

```java
class Cat implements Animal {

    @Override
    public void eat() {
    }

    @Override
    public void sleep() {
    }

}
```

Using interface from class (2)

```java
class Cat implements Animal {

    @Override
    public void eat() {
    }

    @Override
    public void sleep() {
    }

}
```

Need to implements all method from interface

Multiple implements interfaces

Class can be implement multiple interfaces

```
interface Animal {
    void eat();
    void sleep();
}

interface Animal2018 {
    void die();
}
```

Multiple implements interfaces

```java
class BlackPanther implements Animal, Animal2018 {

    @Override
    public void die() {
    }

    @Override
    public void eat() {
    }

    @Override
    public void sleep() {
    }

}
```

Extending an interface

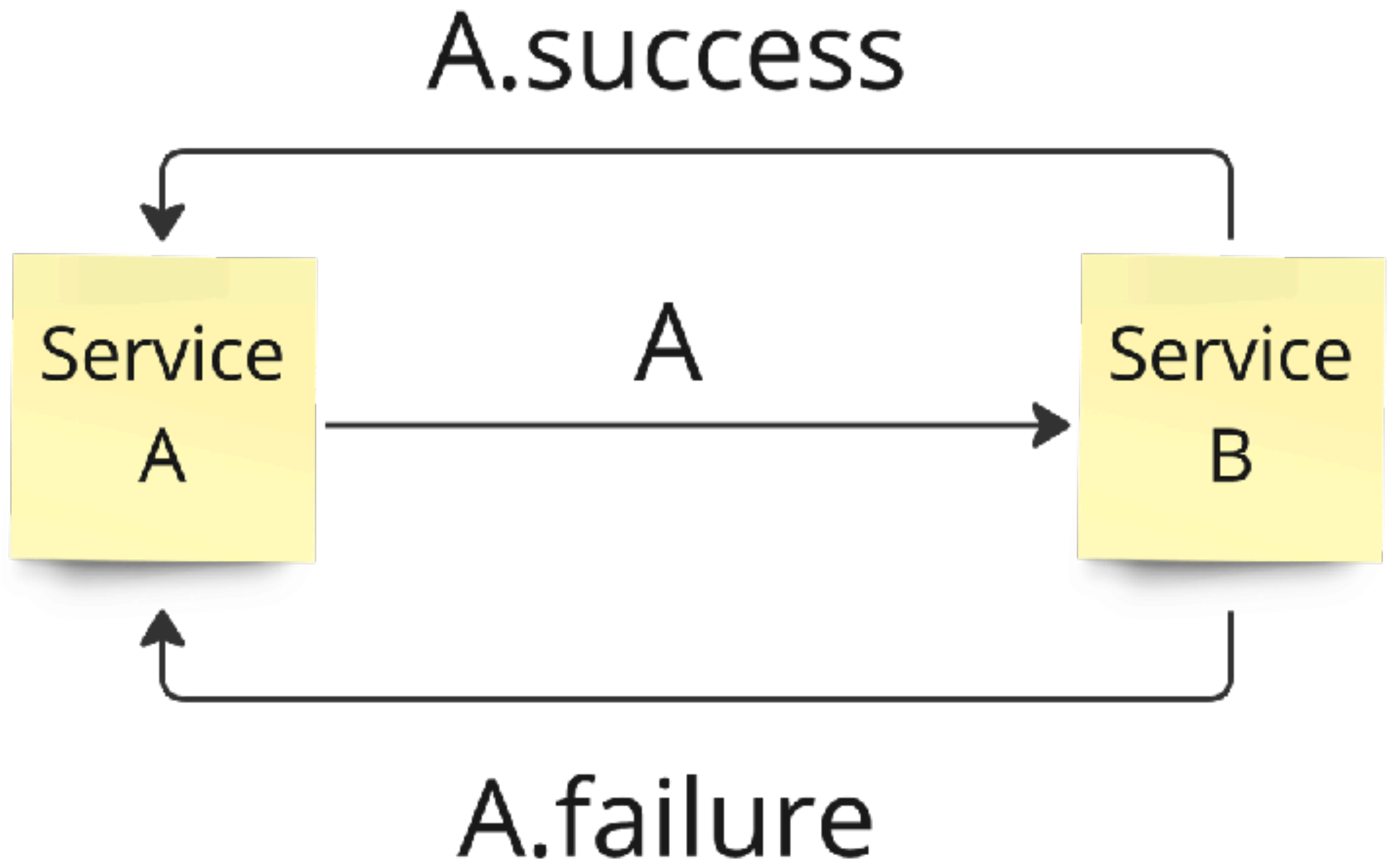
An new interface can add method definitions to existing interface by **extending** the old

```
interface Animal {
    void eat();
    void sleep();
}
```

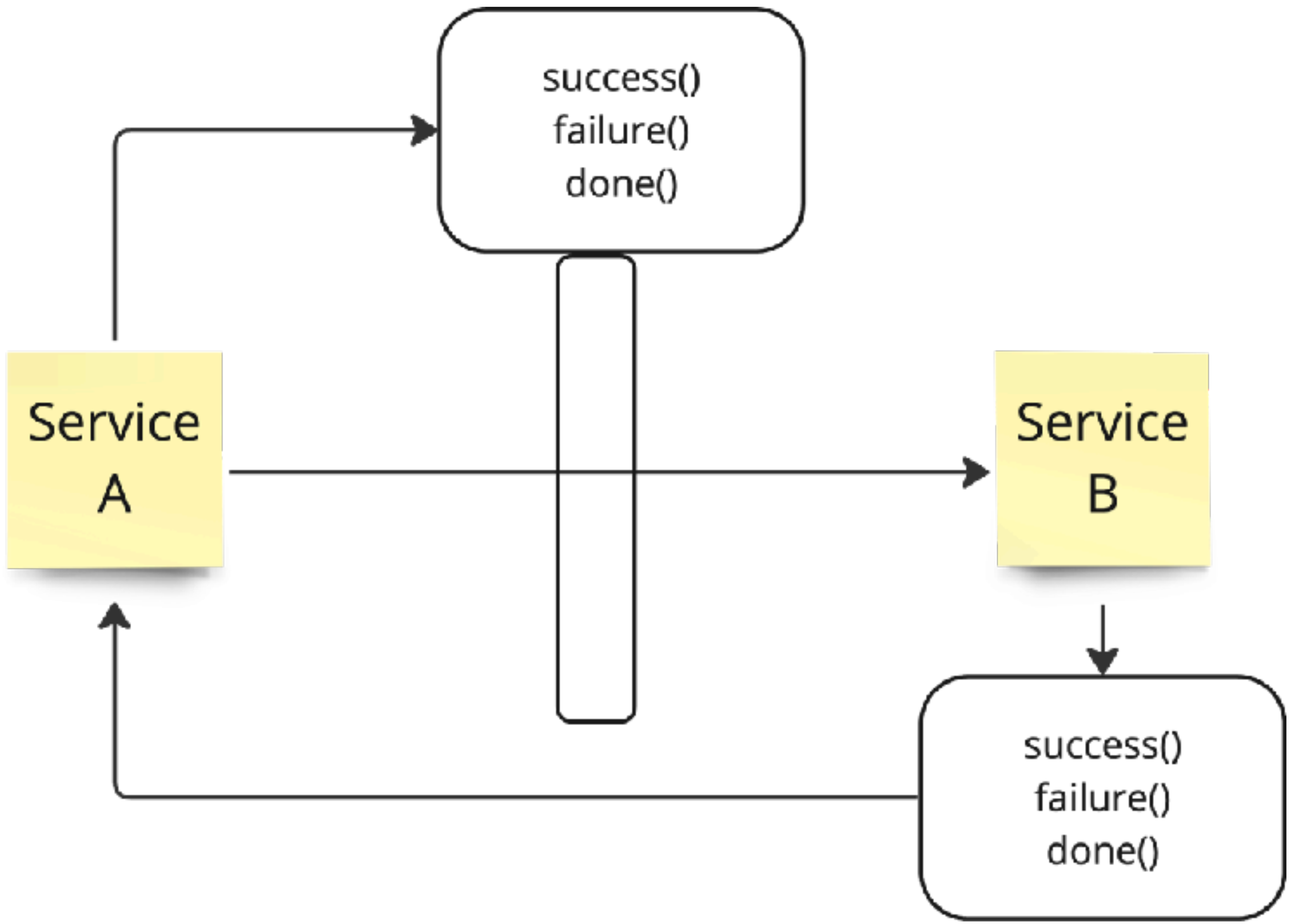
```
interface Animal2018 extends Animal {
    void die();
}
```

Callback with Interface

Call back with class



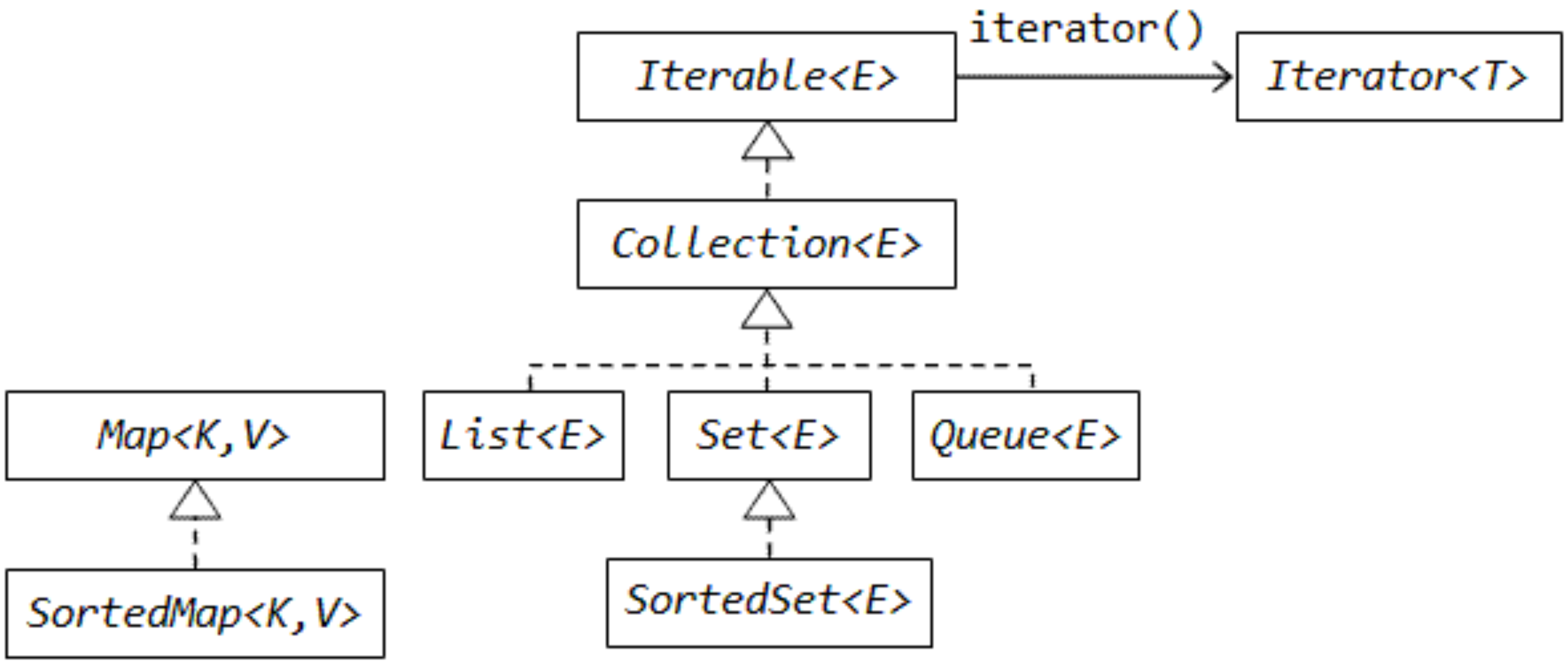
Call back with interface



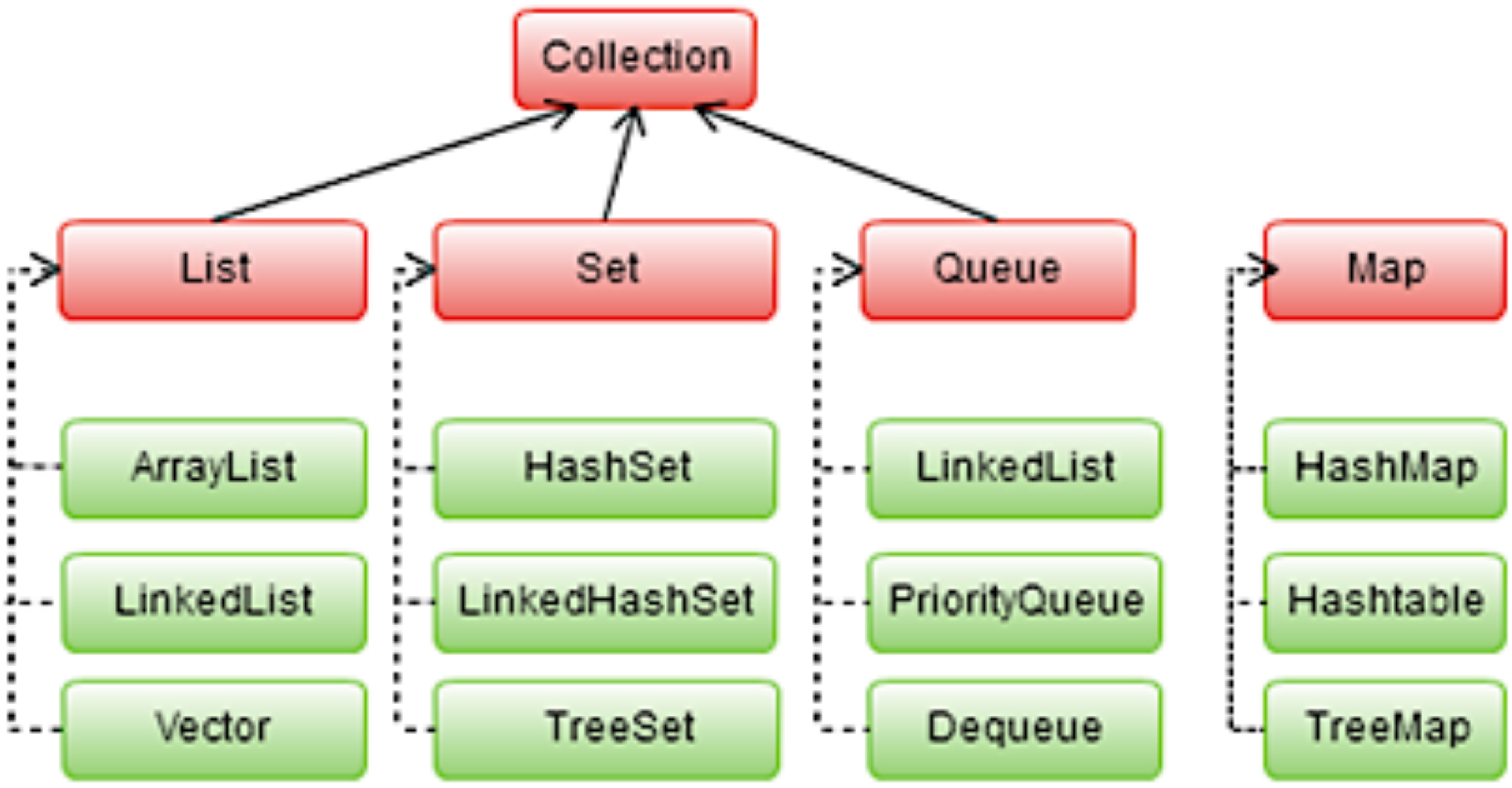
Java Collection Framework

https://docs.oracle.com/javase/8/docs/technotes/guides/collections/overview.html

Java Collection Framework



Java Collection Framework



Java Collections Cheat Sheet

REBELLABS by ZEROTURNAROUND

Notable Java collections libraries

Fastutil
http://fastutil.di.unimi.it/
Fast & compact type-specific collections for Java
Great default choice for collections of primitive types, like int or long. Also handles big collections with more than 2^{31} elements well.

Guava
https://github.com/google/guava
Google Core Libraries for Java 6+
Perhaps the default collection library for Java projects. Contains a magnitude of convenient methods for creating collection, like fluent builders, as well as advanced collection types.

Eclipse Collections
https://www.eclipse.org/collections/
Features you want with the collections you need
Previously known as gs-collections, this library includes almost any collection you might need: primitive type collections, multimaps, bidirectional maps and so on.

JCTools
https://github.com/JCTools/JCTools
Java Concurrency Tools for the JVM.
If you work on high throughput concurrent applications and need a way to increase your performance, check out JCTools.

What can your collection do for you?

Collection class	Thread-safe alternative	Your data				Operations on your collections						
		Individual elements	Key-value pairs	Duplicate element support	Primitive support	Order of iteration			Performant 'contains' check	Random access		
						FIFO	Sorted	LIFO		By key	By value	By index
HashMap	ConcurrentHashMap	✗	✓	✗	✗	✗	✗	✗	✓	✓	✗	✗
HashBiMap (Guava)	Maps.synchronizedBiMap (new HashBiMap())	✗	✓	✗	✗	✗	✗	✗	✓	✓	✓	✗
ArrayListMultimap (Guava)	Maps.synchronizedMultiMap (new ArrayListMultimap())	✗	✓	✓	✗	✗	✗	✗	✓	✓	✗	✗
LinkedHashMap	Collections.synchronizedMap (new LinkedHashMap())	✗	✓	✗	✗	✓	✗	✗	✓	✓	✗	✗
TreeMap	ConcurrentSkipListMap	✗	✓	✗	✗	✗	✓	✗	✓*	✓*	✗	✗
Int2IntMap (Fastutil)		✗	✓	✗	✓	✗	✗	✗	✓	✓	✗	✓
ArrayList	CopyOnWriteArrayList	✓	✗	✓	✗	✓	✗	✓	✗	✗	✗	✓
HashSet	Collections.newSetFromMap (new ConcurrentHashMap<>())	✓	✗	✗	✗	✗	✗	✗	✓	✗	✓	✗
IntArrayList (Fastutil)		✓	✗	✓	✓	✓	✗	✓	✗	✗	✗	✓
PriorityQueue	PriorityBlockingQueue	✓	✗	✓	✗	✗	✓**	✗	✗	✗	✗	✗
ArrayDeque	ArrayBlockingQueue	✓	✗	✓	✗	✓**	✗	✓**	✗	✗	✗	✗

* O(log(n)) complexity, while all others are O(1) - constant time

** when using Queue interface methods: offer() / poll()

How fast are your collections?

Collection class	Random access by index / key	Search / Contains	Insert
ArrayList	O(1)	O(n)	O(n)
HashSet	O(1)	O(1)	O(1)
HashMap	O(1)	O(1)	O(1)
TreeMap	O(log(n))	O(log(n))	O(log(n))

Remember, not all operations are equally fast. Here's a reminder of how to treat the Big-O complexity notation:

O(1) - constant time, really fast, doesn't depend on the size of your collection

O(log(n)) - pretty fast, your collection size has to be extreme to notice a performance impact

O(n) - linear to your collection size: the larger your collection is, the slower your operations will be

BROUGHT TO YOU BY JRebel

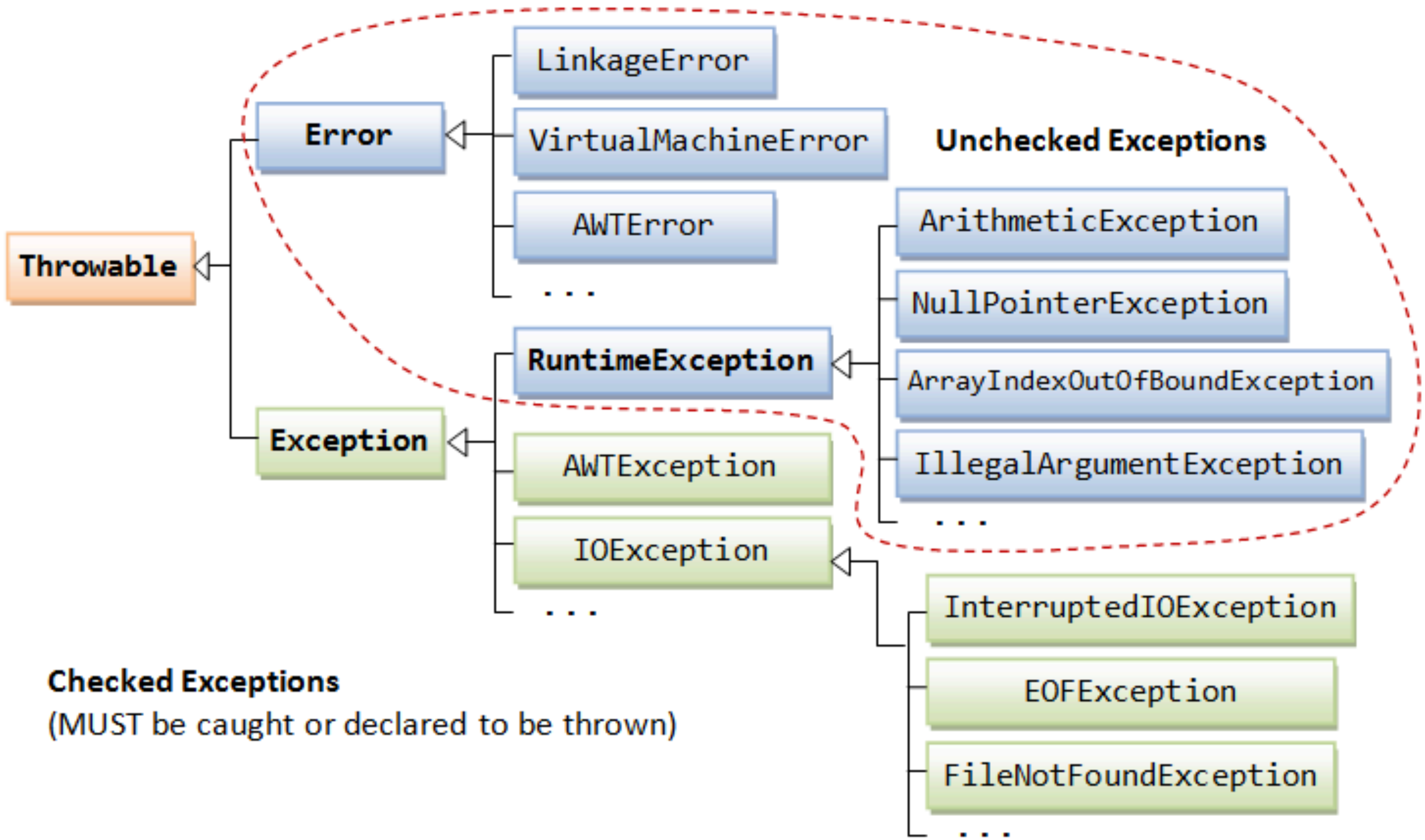
http://files.zeroturnaround.com/pdf/zt_java_collections_cheat_sheet.pdf

Types of Error in Java

Types of Errors

- Compile-time errors
- Runtime errors
- Logical errors

Exception classes



Lambda expression

Java 8 Best Practices Cheat Sheet

REBELLABS by ZEROTURNAROUND

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Default methods

Evolve interfaces & create traits

```java
//Default methods in interfaces
@FunctionalInterface
interface Utilities {
  default Consumer<Runnable> m() {
    return (r) -> r.run();
  }
  // default methods, still functional
  Object function(Object o);
}
class A implements Utilities { // implement
  public Object function(Object o) {
    return new Object();
  }
  {
    // call a default method
    Consumer<Runnable> n = new A().m();
  }
}
```

Lambdas

Syntax:
(parameters) -> expression
(parameters) -> { statements; }

```java
// takes a Long, returns a String
Function<Long, String> f = (l) -> l.toString();
// takes nothing gives you Threads
Supplier<Thread> s = Thread::currentThread;
// takes a string as the parameter
Consumer<String> c = System.out::println;

// use them with streams
new ArrayList<String>().stream().
// peek: debug streams without changes
peek(e -> System.out.println(e)).
// map: convert every element into something
map(e -> e.hashCode()).
// filter: pass some elements through
filter(e -> ((e.hashCode() % 2) == 0)).
// collect all values from the stream
collect(Collectors.toCollection(TreeSet::new))
```

java.util.Optional

A container for possible null values

```java
// Create an optional
Optional<String> optional =
Optional.ofNullable(a);
// process the optional
optional.map(s -> "RebelLabs:" + s);
// map a function that returns Optional
optional.flatMap(s -> Optional.ofNullable(s));

// run if the value is there
optional.ifPresent(System.out::println);
// get the value or throw an exception
optional.get();

// return the value or the given value
optional.orElse("Hello world!");
// return empty Optional if not satisfied
optional.filter(s -> s.startsWith("RebelLabs"));
```

Rules of Thumb

Traits: 1 default method per interface
Don't enhance functional interfaces
Only conservative implementations

Expressions over statements
Refactor to use method references
Chain lambdas rather than growing them

Fields - use plain objects
Method parameters, use plain objects
Return values - use Optional
Use *orElse()* instead of *get()*

http://files.zeroturnaround.com/pdf/zt_java8_best_practices.pdf

Lambda expression

Functionality as a method arguments

Code as data

Help code more compact

Demo with Sorting data

Sorting data in List with Comparator

Employee id=1 name=First

Employee id=2 name=Second

Employee id=3 name=Third

Create class Employee

```
class Employee {
    private int id;
    private String name;
    private double salary;

     //Setter and Getter methods
```

Sorting with java.util.Comparator

```
Comparator<Employee> sortByName = new Comparator<Employee>() {
            @Override
            public int compare(Employee e1, Employee e2) {
                return e1.getName().compareTo(e2.getName());
            }
        };
```

Using with Collections.sort()

```
List<Employee> list = new ArrayList<>();
list.add(new Employee(500, "First", 150000));
list.add(new Employee(504, "Second", 120000));
list.add(new Employee(503, "Third", 100000));
list.add(new Employee(730, "Forth", 45000));

Collections.sort(list, sortByName);
list.forEach(System.out::println);
```

Run and see output

```
demo.java8.lambda.Employee@e9e54c2
demo.java8.lambda.Employee@65ab7765
demo.java8.lambda.Employee@1b28cdfa
demo.java8.lambda.Employee@eed1f14
```

Try to improve format of output !!
more readable

Overide toString()

```
class Employee {
    private int id;
    private String name;
    private double salary;

    @Override
    public String toString() {
        return "Employee [id=" + id + ", name=" + name + ", salary=" +
salary + "]";
    }
```

```
Employee [id=500, name=First, salary=150000.0]
Employee [id=504, name=Second, salary=120000.0]
Employee [id=503, name=Third, salary=100000.0]
Employee [id=730, name=Forth, salary=45000.0]
```

Sorting with Lambda expression

```
Comparator<Employee> sortByNameWithLambda =
                (Employee e1, Employee e2) ->
                           e1.getName().compareTo(e2.getName());

Collections.sort(list, sortByName);
list.forEach(System.out::println);
```

Workshop

Sort by name (Z to A)
Sort by ID (0 to 9)

Java Stream API

Java Stream API

A new package **java.util.stream**

New functionality which contains classes for processing sequences of elements

Java Stream API

Working with List and Array

```
String[] datas = new String[] {"P", "U", "I"};
Stream<String> streams = Arrays.stream(datas);
streams.forEach(System.out::println);

Stream<String> streams2 = Stream.of("P", "U", "I");
streams2.forEach(System.out::println);
```

Java Stream API

Sequential and Parallel

```
List<String> myList = Arrays.asList("P", "U", “I");

myList.stream().forEach(System.out::println);

myList.parallelStream().forEach(System.out::println);
```

Operations

Iterating
Filtering
Mapping
Matching
Collecting

Iterating

```
myList.stream().forEach(System.out::println);
```

Filtering

```
myList.stream()
            .filter(element -> element.contains("P"))
            .forEach(System.out::println);
```

Matching

Try to validate all elements of sequence

```
boolean isValid = myList.stream()
                        .anyMatch(element -> element.contains("P"));

out.println(isValid);
```

Mapping

Try to convert elements by apply a special function

```
List<String> results =
    myList.stream().map(element -> element.toLowerCase())
                    .collect(Collectors.toList());

out.println(results);
```

Collecting

Try to convert stream to collection or map

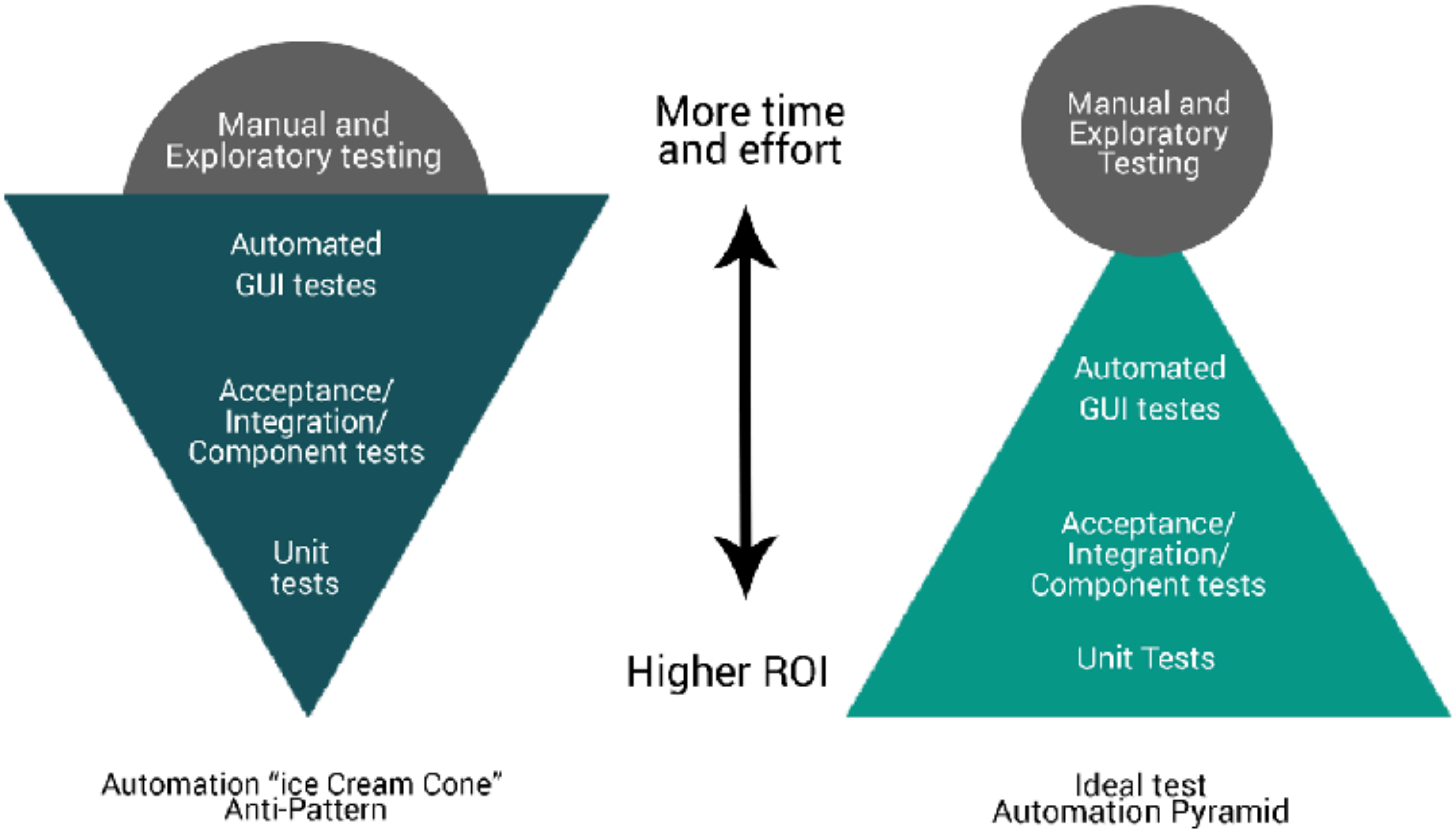
```
List<String> results =
    myList.stream().map(element -> element.toLowerCase())
                    .collect(Collectors.toList());

out.println(results);
```

Workshop

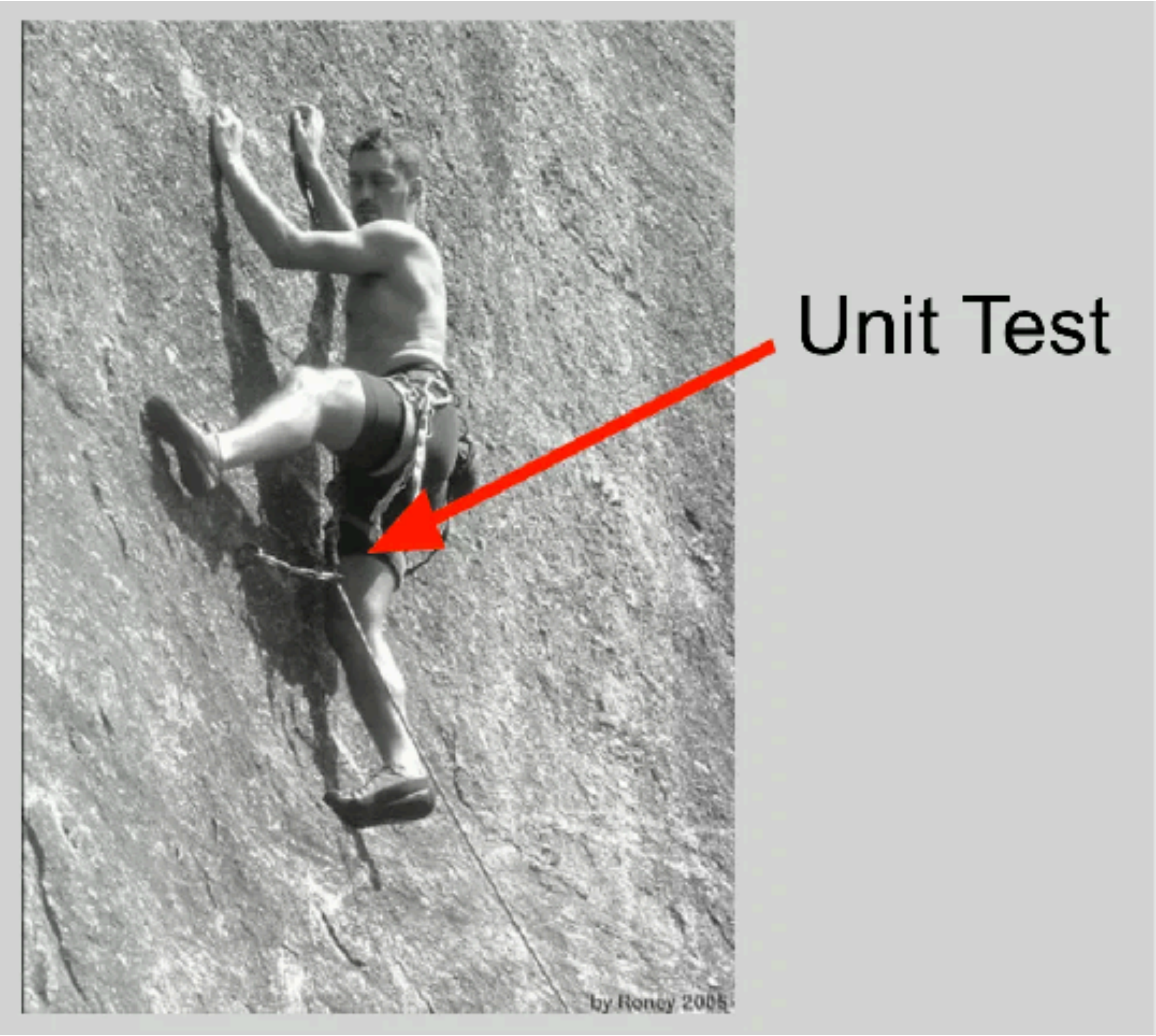
United States of America   => **USA**
united states of america   => **USA**
Light Amplification by Stimulation of Emitted Radiation   => **LASER**
Jordan Of the World   => **JTW**

Automated testing with Java

Automation test



Unit test



https://less.works/less/technical-excellence/unit-testing.html

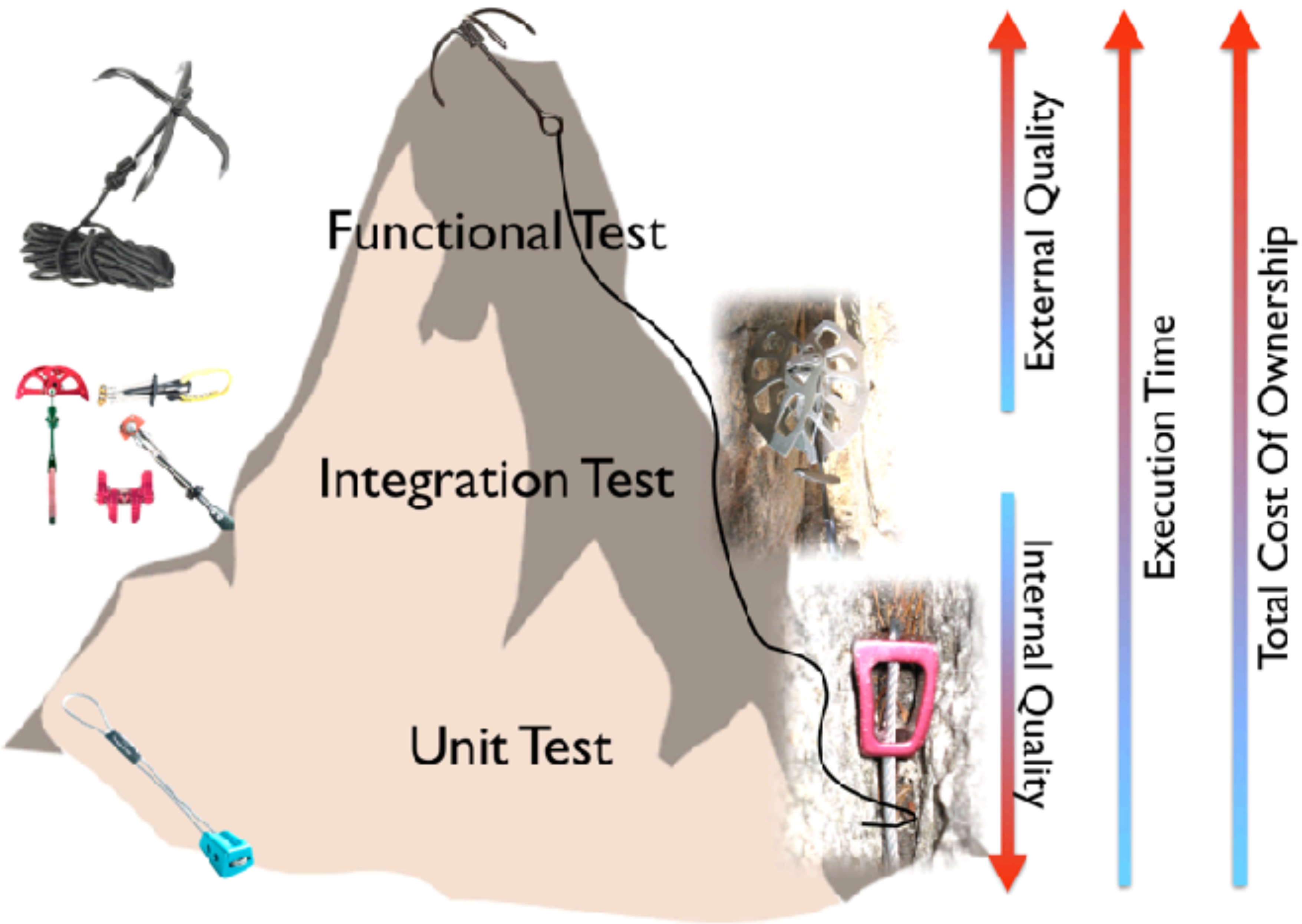
Purpose of Unit test

Facilitates changes
Simplified integration
Documentation
Design tool

Why Unit test ?

Total cost of ownership
Internal vs External quality
Quality of feedback

Quality of feedback



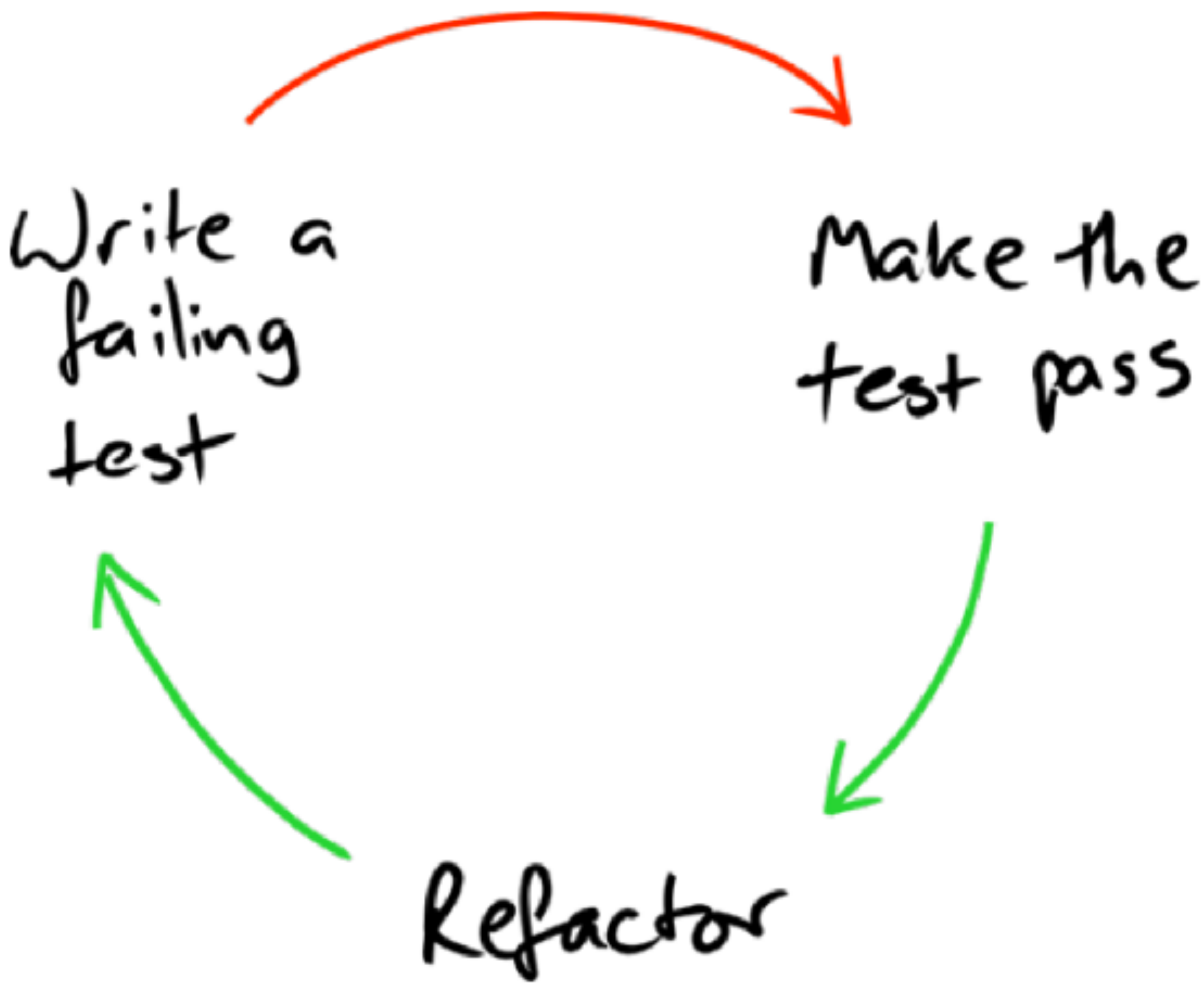
https://less.works/less/technical-excellence/unit-testing.html

Golden rule of Unit test

“Each unit test case
should be **very limited** in scope”

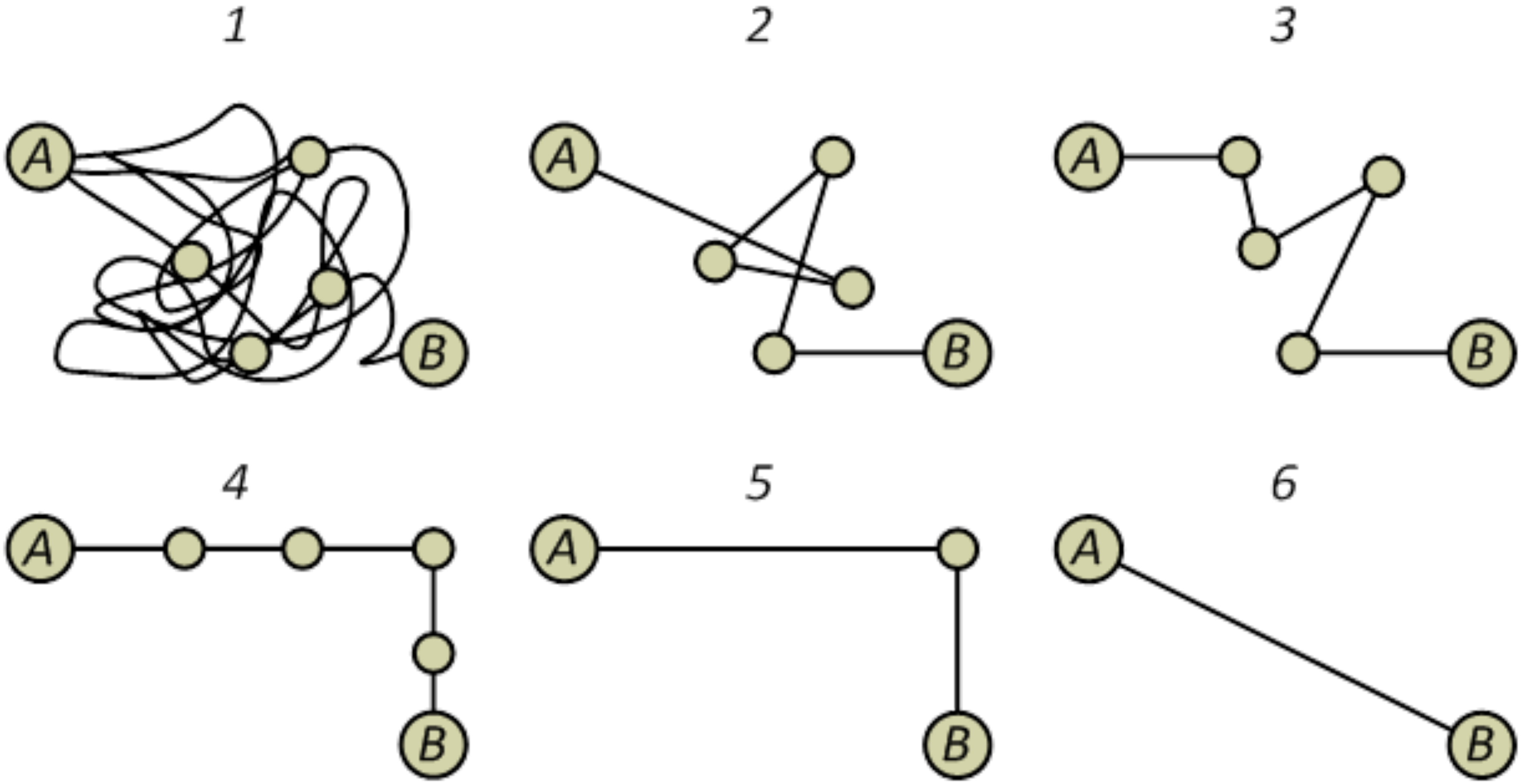
Test-Driven Development (TDD)

TDD Cycle



Refactor

Change the structure of code without change the behaviour of code

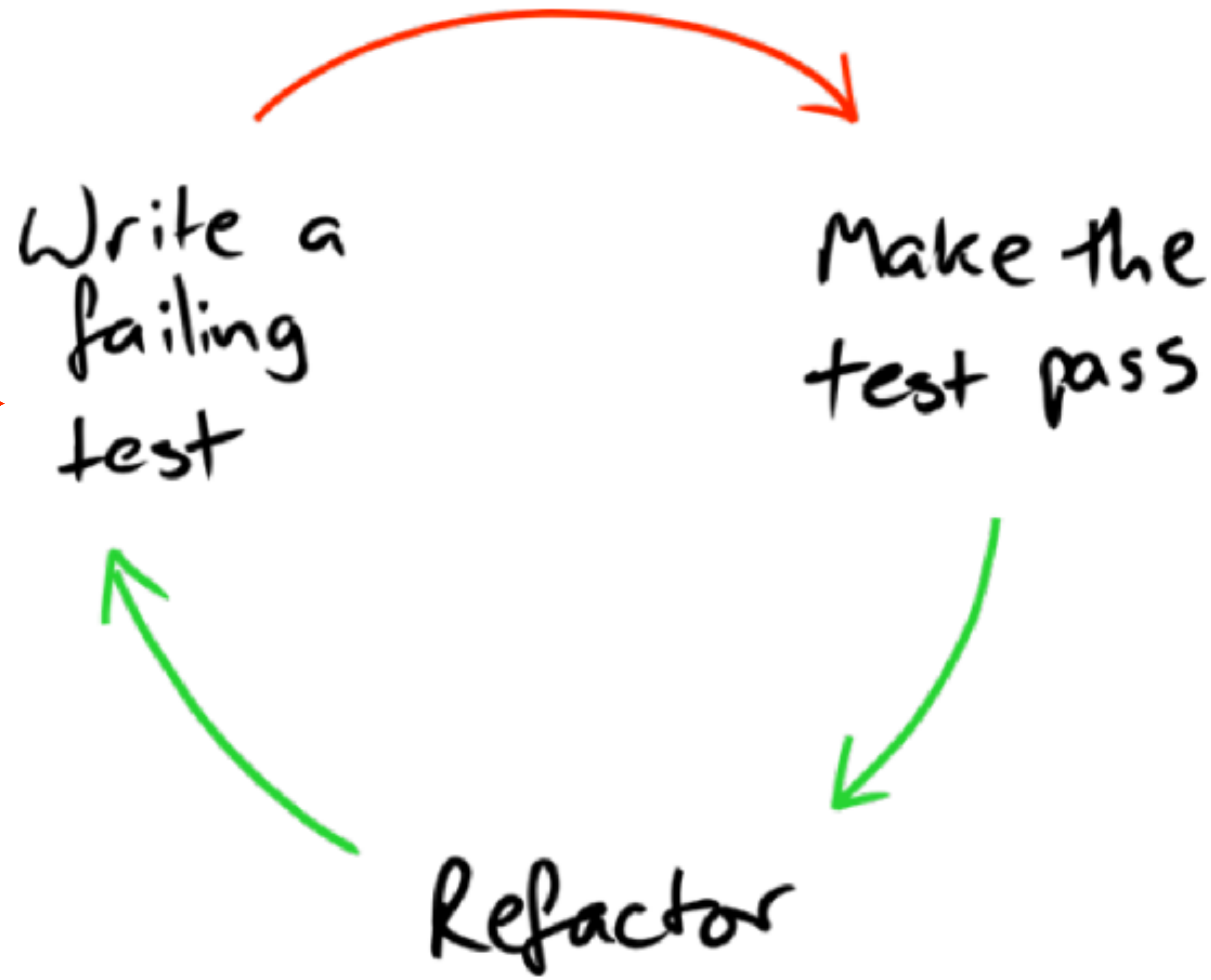


Simple Rule of TDD

1. Write a failing test before write any code
2. Remove duplication code

TDD Cycle

Start with THINKing

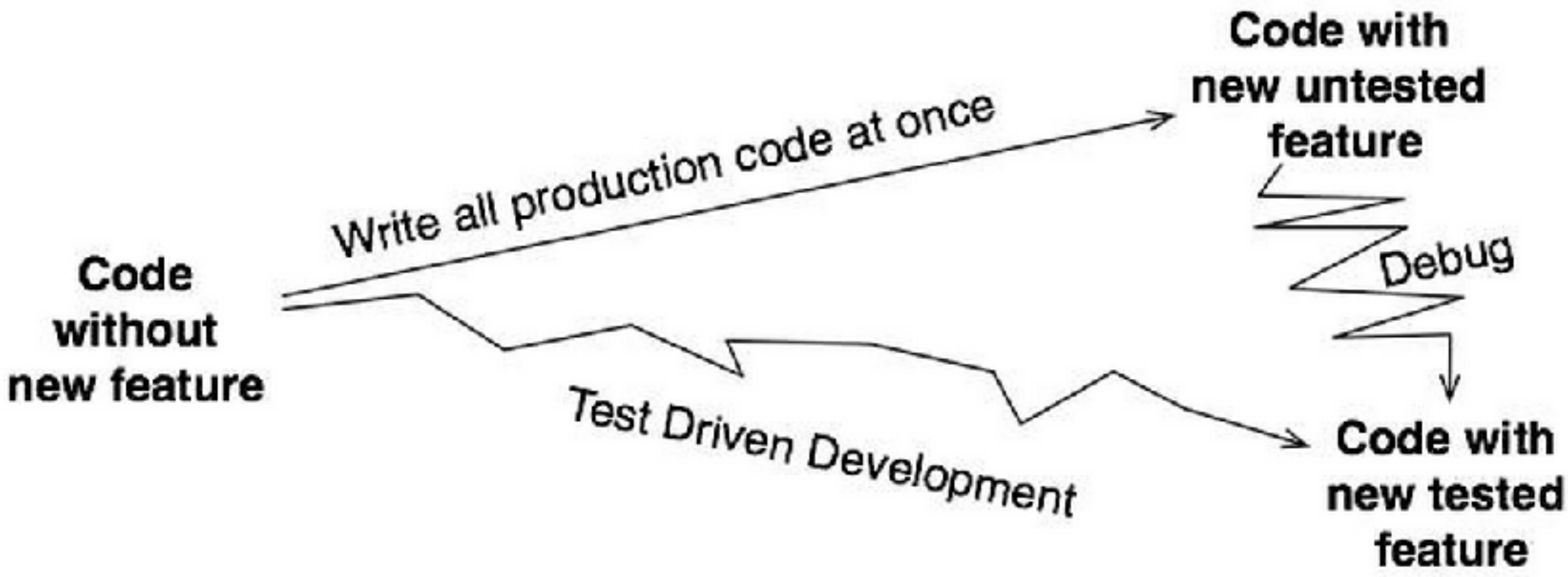


TDD cycle should be

Short
Rhythmic
Incremental
Design-focus
Discipline

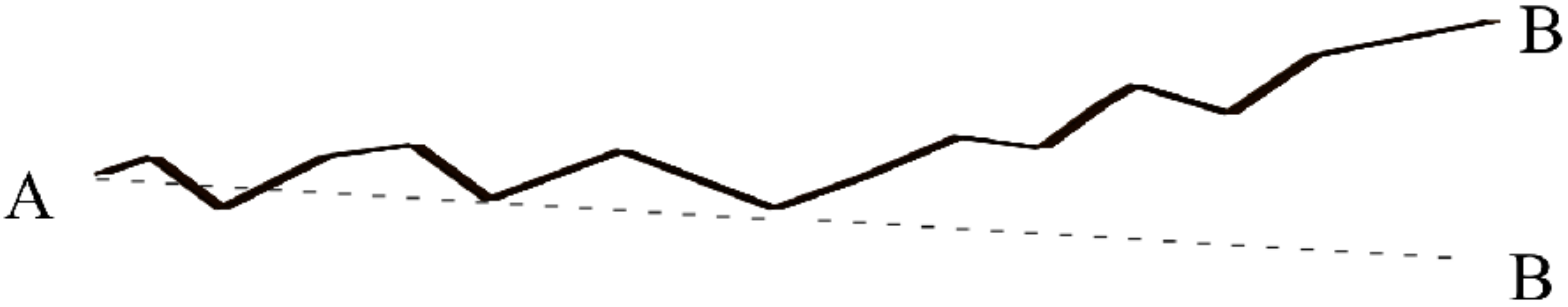
TDD with DLP

Debug Later Programming



https://www.wingman-sw.com/

Let’s start with Small Step



Good Unit Test

Fast
Isolation
Repeatable
Self-validate
Timely

Let’s start to write Unit tests with Java

BUT we need to separate the Test code from Production code !!

We need new **Project Structure**

Maven™

https://maven.apache.org/

POM file (1)

Project **O**bject **M**odel
Represent in XML format
Filename is **pom.xml**

POM file (2)

```
<project xmlns="http://maven.apache.org
  <modelVersion>4.0.0</modelVersion>
  <groupId>demo1</groupId>
  <artifactId>demo1</artifactId>
  <version>0.0.1-SNAPSHOT</version>
</project>
```

POM file (2)

Add new dependency => **junit**

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-inst
    xsi:schemaLocation="http://maven.apache.org/POM/
    <modelVersion>4.0.0</modelVersion>
    <groupId>demo1</groupId>
    <artifactId>demo1</artifactId>
    <version>0.0.1-SNAPSHOT</version>

    <dependencies>
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>4.12</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

</project>
```

https://github.com/junit-team/junit4/wiki/use-with-maven

Write first unit test case

The ratio of time spent in code

Read vs Write is +10:1

Run your test in command line

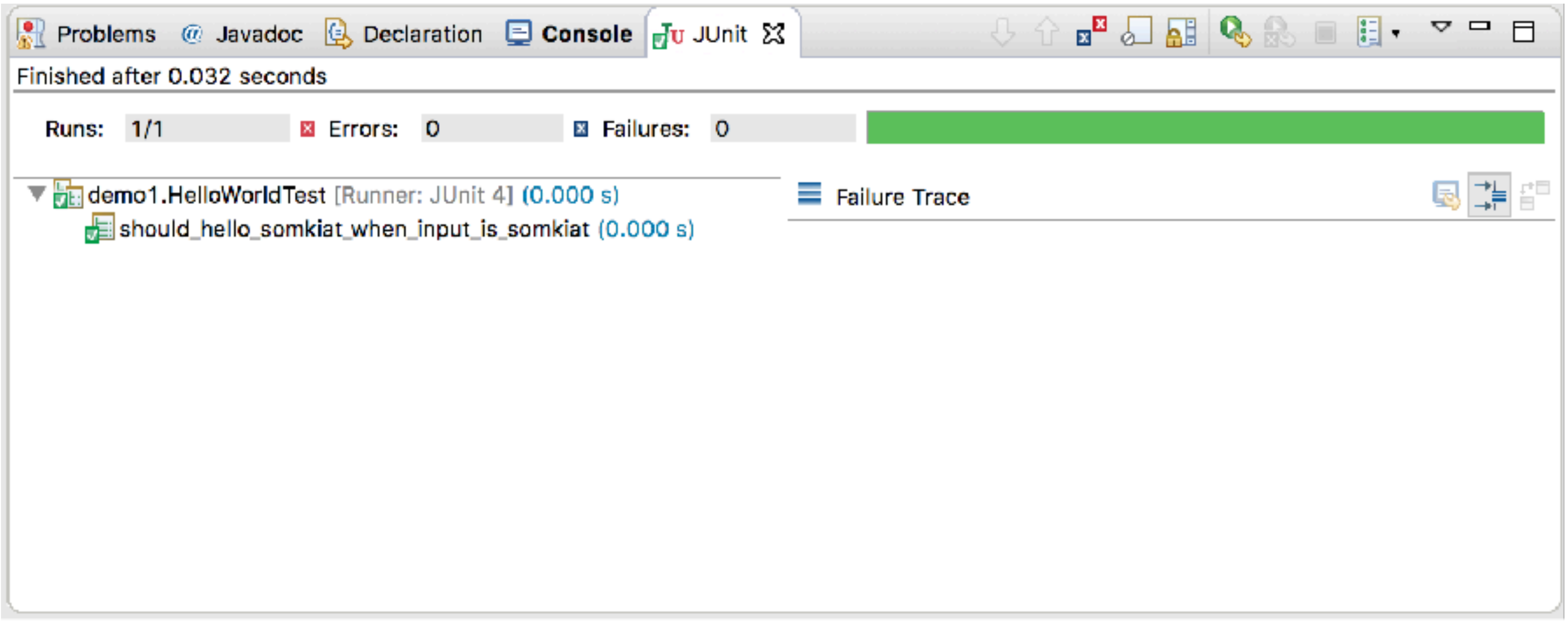
$mvn clean test
$mvn clean package

Make your test pass

```java
public class Reception {

    public String hello(String name) {
        return "Hello " + name;
    }

}
```

Run test again

Result should be pass (Green)

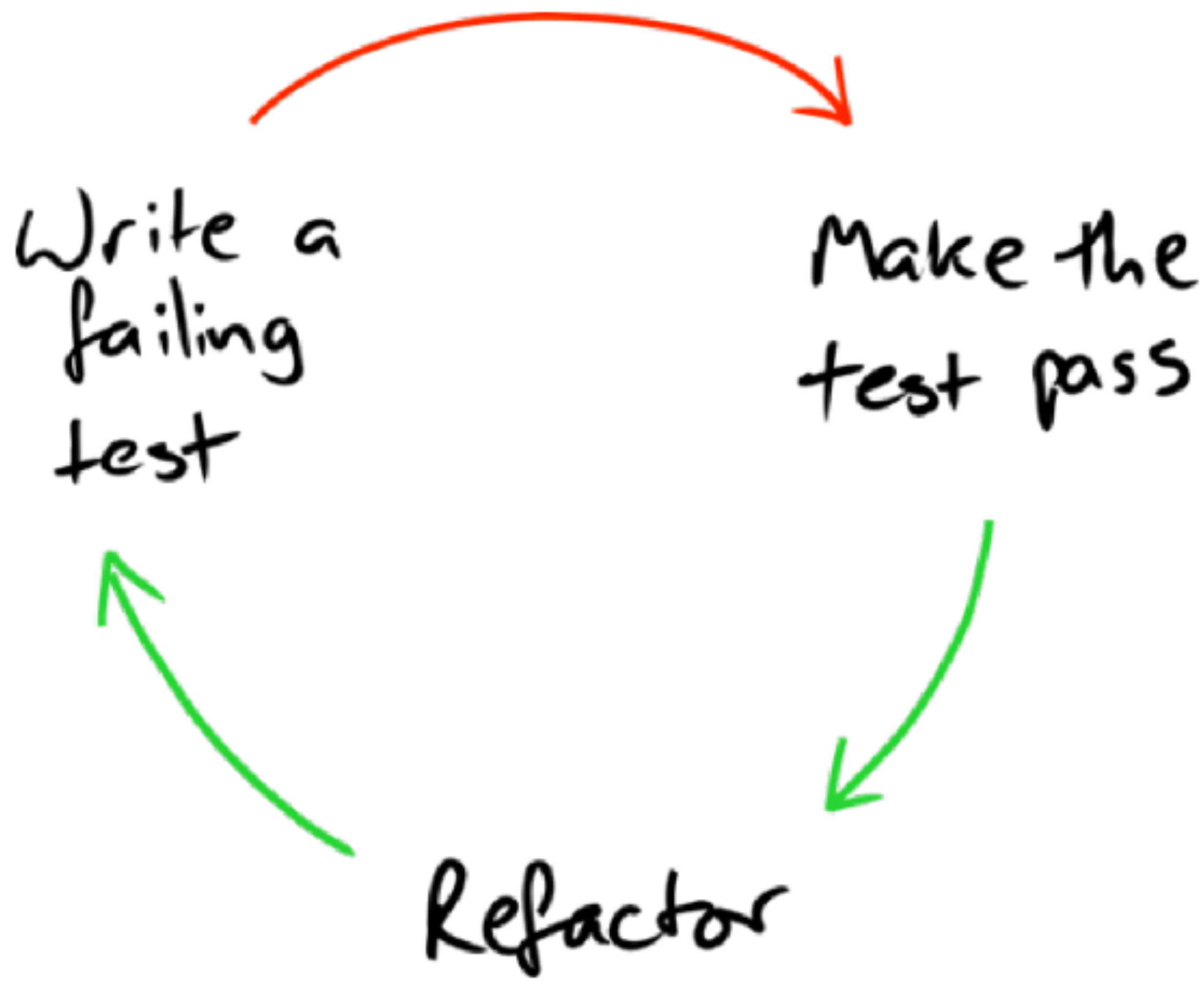


Write next test case …

Simple Rule of TDD (again)

1. Write a failing test before write any code
2. Remove duplication code

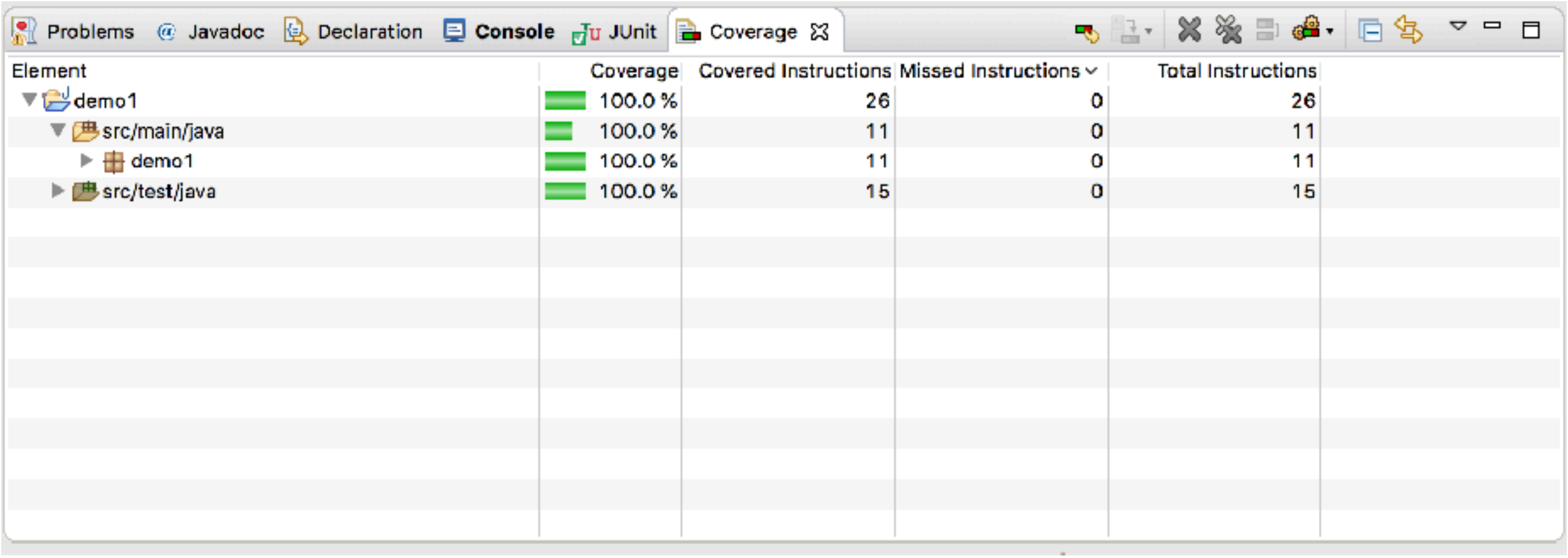
Start with THINKing



Code Coverage

Show result of code coverage

Summary result report



Problems | Javadoc | Declaration | Console | JUnit | Coverage

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
demo1	100.0 %	26	0	26
src/main/java	100.0 %	11	0	11
demo1	100.0 %	11	0	11
src/test/java	100.0 %	15	0	15

Show result of code coverage

See result in each file/class

```java
package demo1;

public class Reception {

    public String hello(String name) {
        return "Hello " + name;
    }

}
```

Workshop

Range of number

Create test case = **RangeTest**

Input	Expected output
[1,5]	1,2,3,4,5
[1,5)	1,2,3,4
(1,5]	2,3,4,5
(1,5)	2,3,4

Thinking before **coding**

Code Smell & Refactoring

https://en.wikipedia.org/wiki/Code_smell

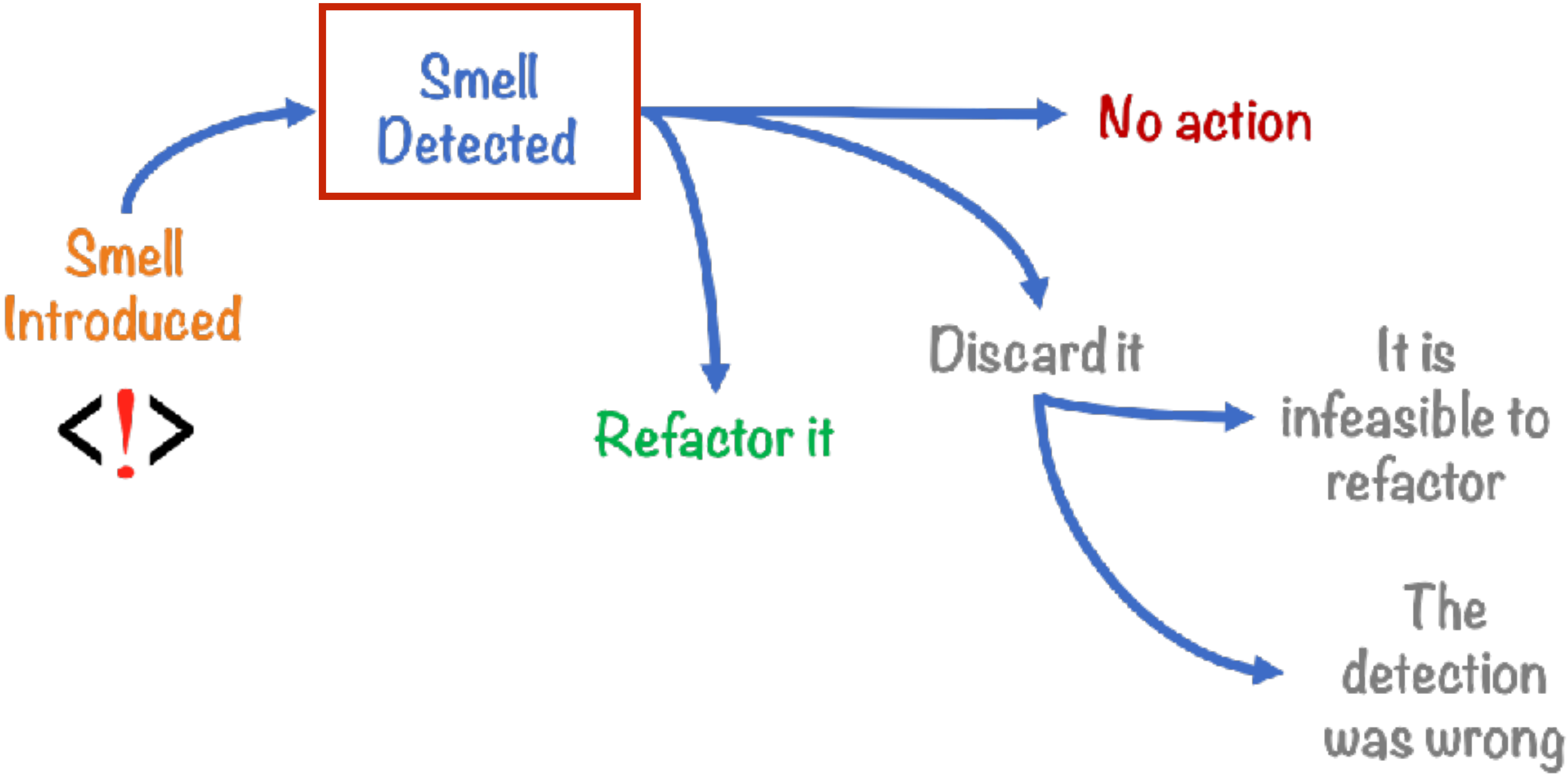
Code Smell

Code Smell


"Bad Code Smells are symptoms of poor design or implementation choices"
[Martin Fowler]

Code Smell make problems

Do you know Code smell ?



Code Smell (example)

Duplication code
Long method
Large class (God class)
Comments
Long parameter list
Switch/If statements
Divergent change (one class)
Shotgun surgery (one class)

https://sourcemaking.com/refactoring/smells

Not taking too much
Show me the code

Code Smell Workshop

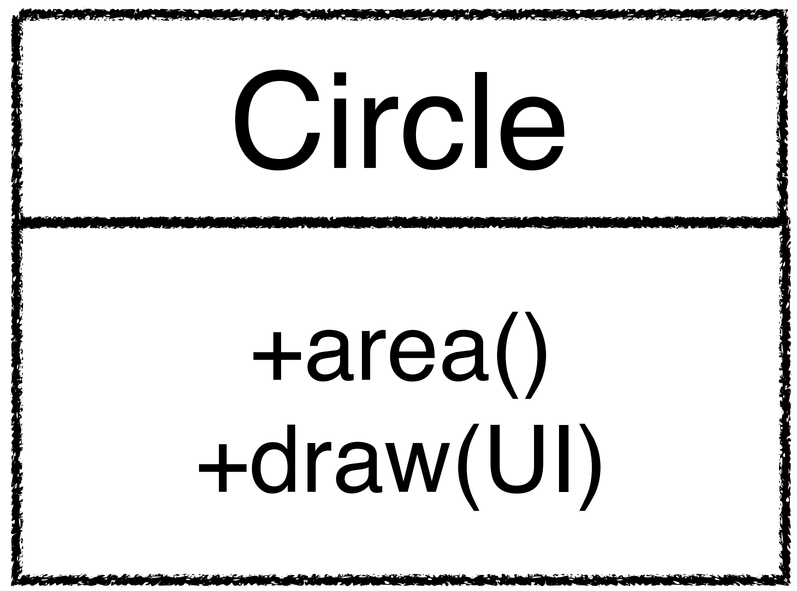
https://github.com/emilybache/Tennis-Refactoring-Kata

SOLID principle

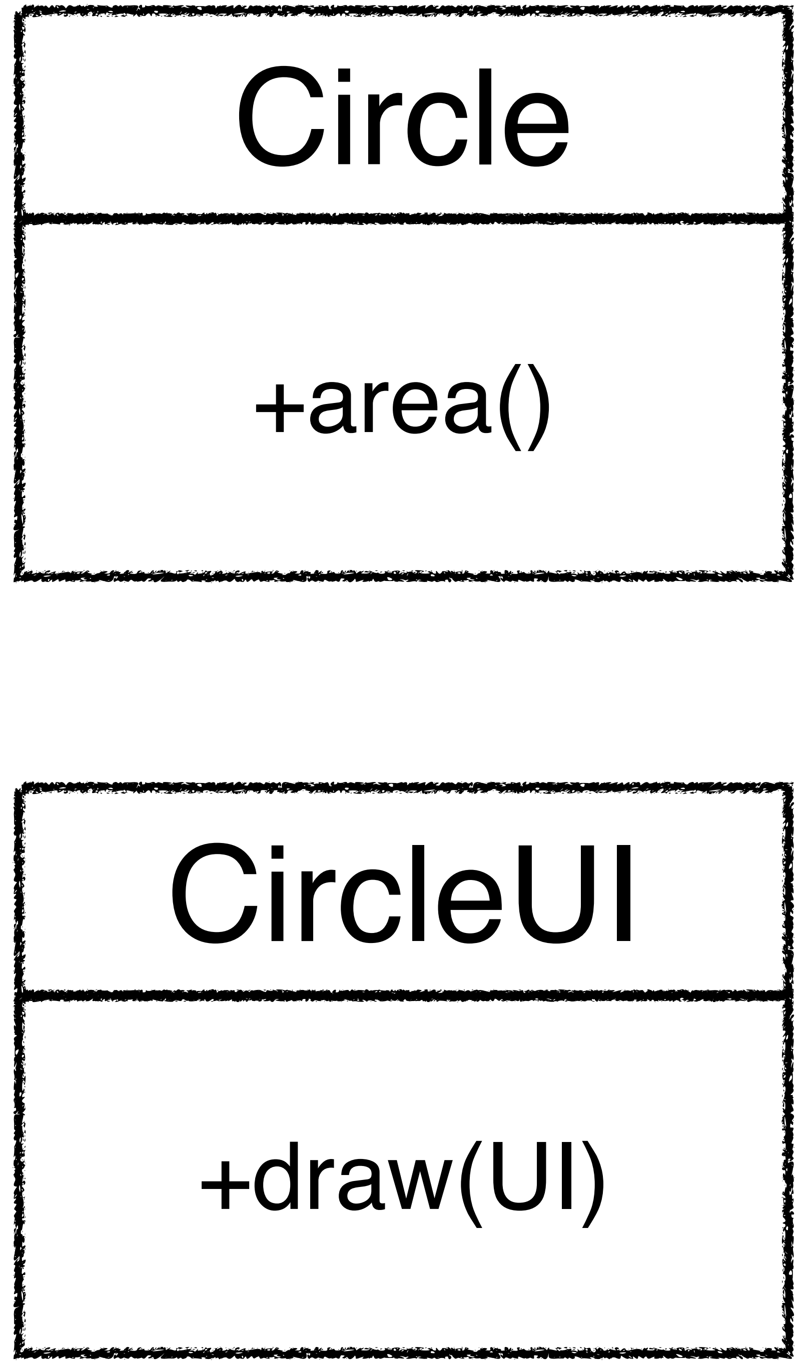
SOLID principle

1. Single Responsibility Principle

Example



Example

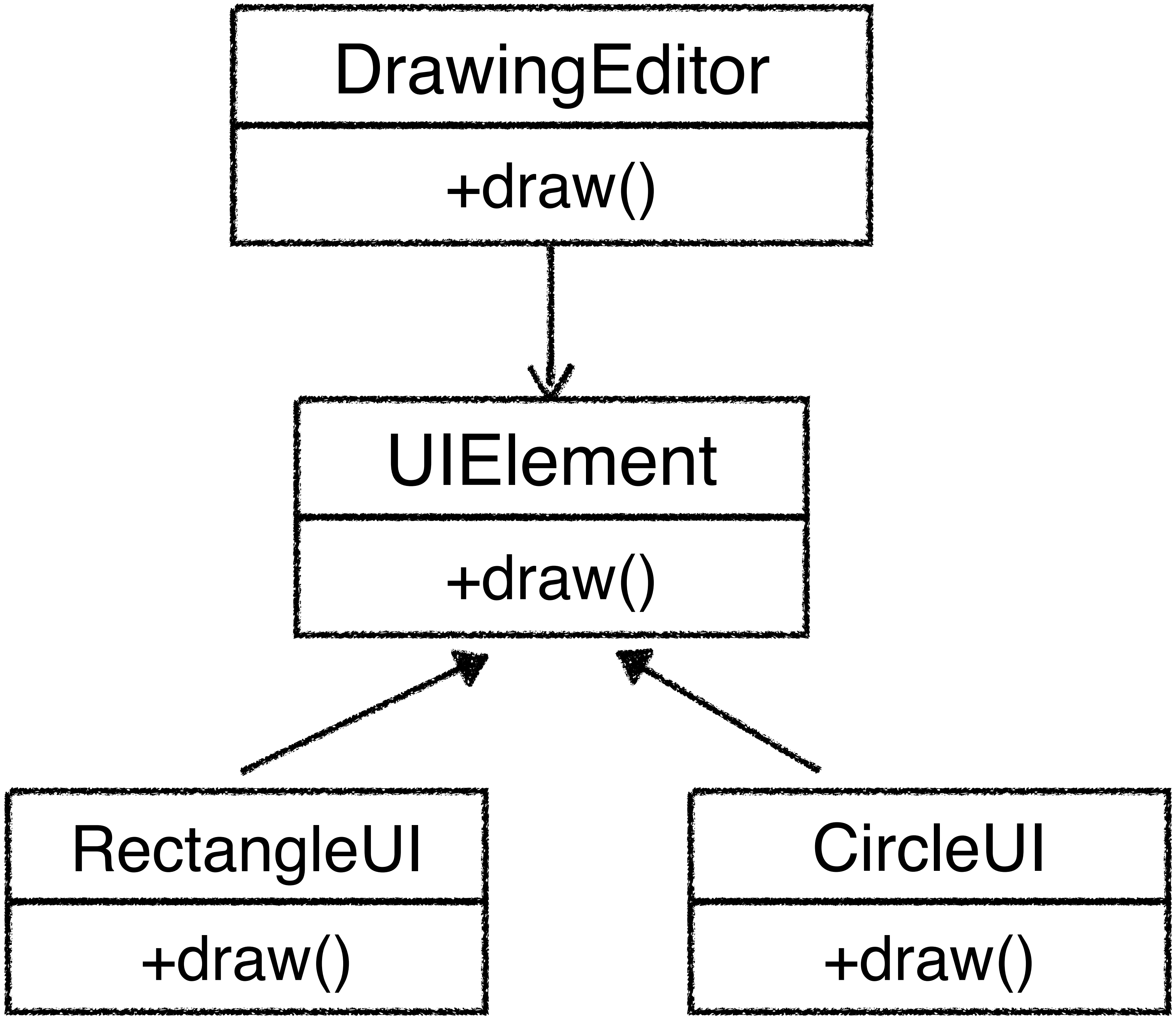


2. Open-Closed Principle

Example

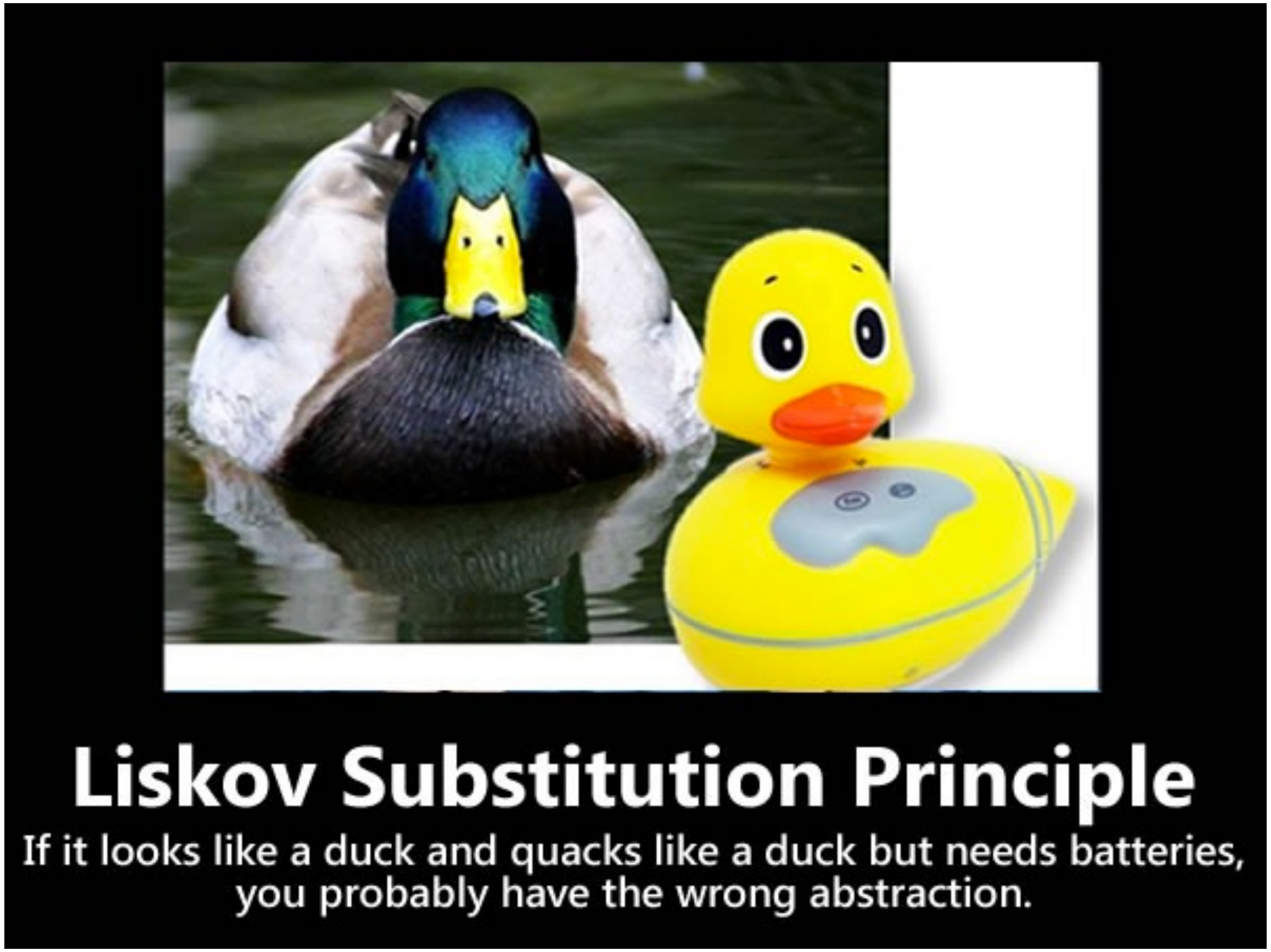
DrawingEditor
+draw() -drawCircle() -drawRectangle()

Example


DrawingEditor
+draw()
UIElement
+draw()
RectangleUI
+draw()
CircleUI
+draw()

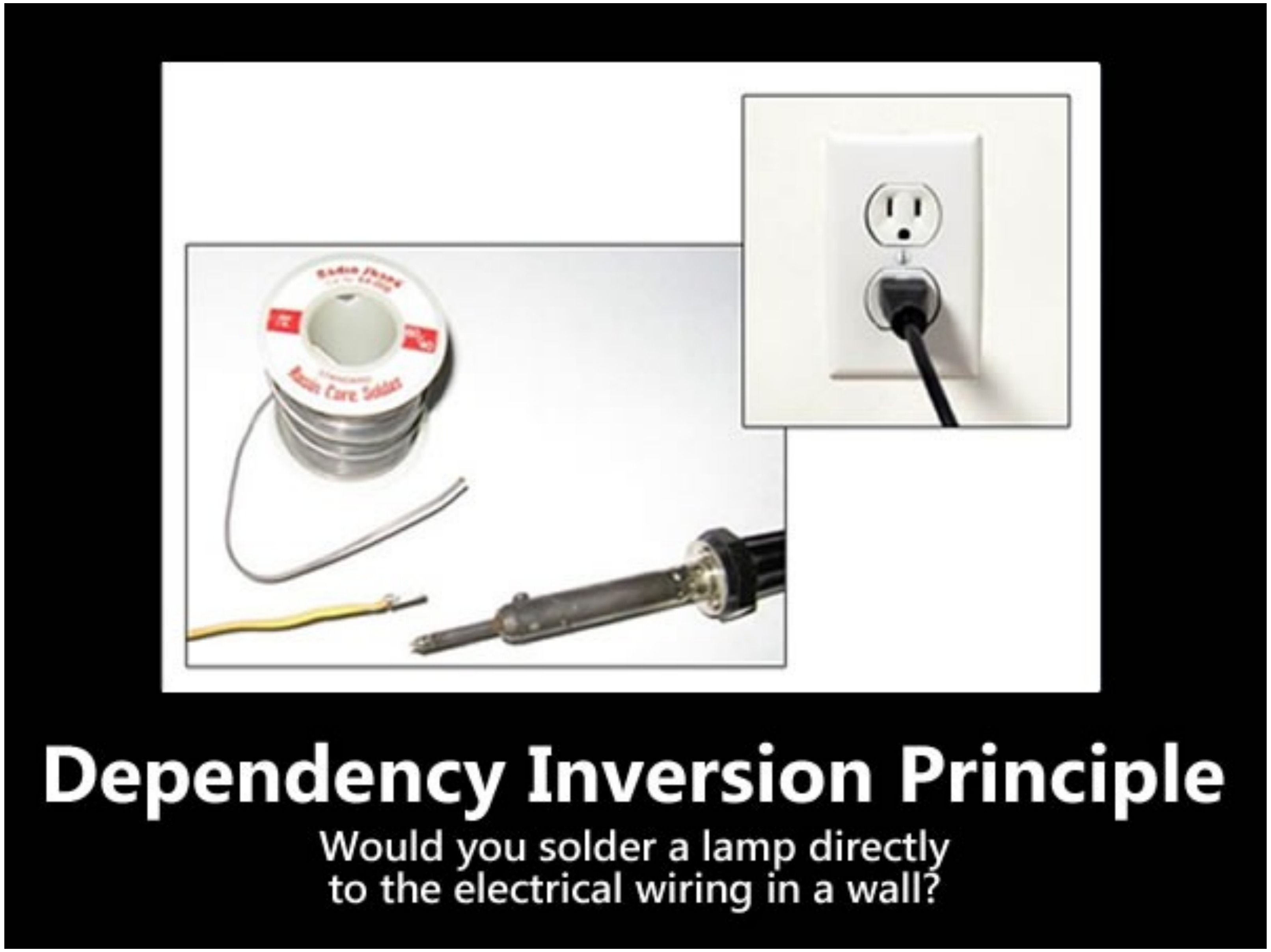
Workshop

3. Liskov Substitution Principle



4. Interface Segregation Principle

5. Dependency Inversion Principle



Workshop

Spring Boot

Agenda

- RESTful API
- Java frameworks
- Building RESTful API with Spring Boot
- Spring Boot with Spring Data (JDBC/JPA)
- Testing with Spring Boot

REST

REpresentation **S**tate **T**ransfer

The style of software architecture behind RESTful services

Defined in 2000 by Roy Fielding

https://www.ics.uci.edu/~fielding/pubs/dissertation/rest_arch_style.htm

Goals

Scalability
Generality of interfaces
Independent deployment of components

RESTful service

REST Request Messages

RESTful request is typically in form of **U**niform **R**esource **I**dentifiers (URI)

REST Request Messages

RESTful request is typically in form of **U**niform **R**esource **I**dentifiers (URI)

Structure of URI depend on specific service

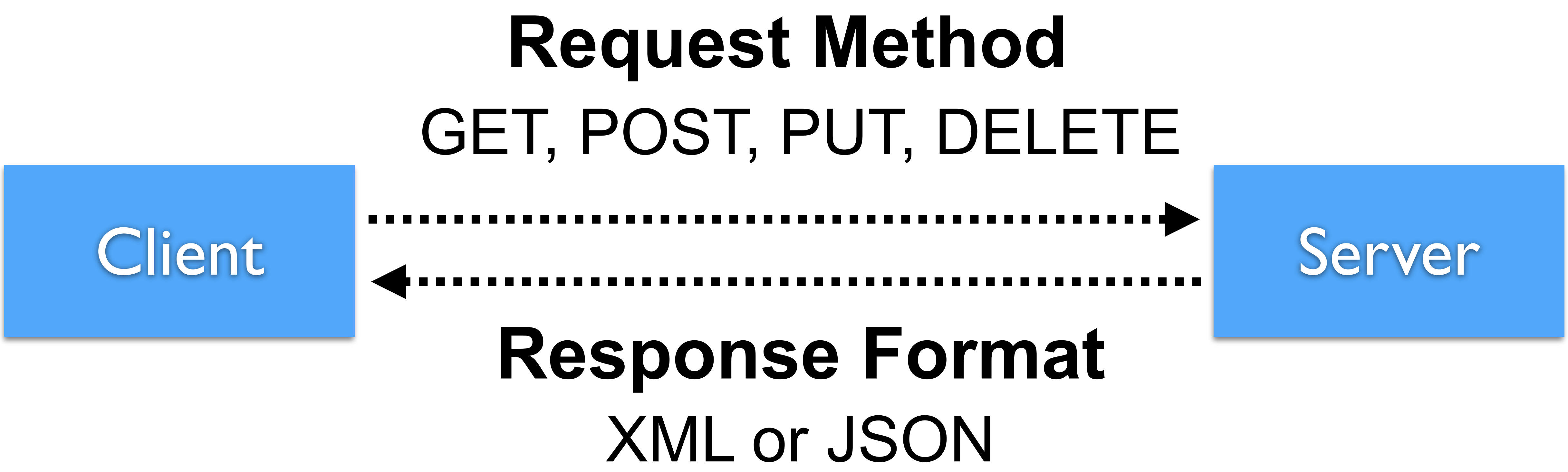
REST Request Messages

RESTful request is typically in form of **U**niform **R**esource **I**dentifiers (URI)

Structure of URI depend on specific service

Request can include parameter and data in body of request as XML, JSON etc.

REST Request & Response



HTTP Methods meaning

Method	Meaning
GET	Read data
POST	Create/Insert new data
PUT/PATCH	Update data or insert if a new id
DELETE	Delete data

Response format ?

Good APIs ?

Java Framework for RESTful

undertow

Jersey
RESTful Web Services in Java

Java
J2EE

WildFly SWARM

JAX-RS

Ratpack
A toolkit for JVM web applications

REST Easy

spring
by Pivotal

Spark

Restlet

akka

Apache CXF

STRUTS

play

NINJA
web framework

unirest
Lightweight HTTP Request Client Libraries

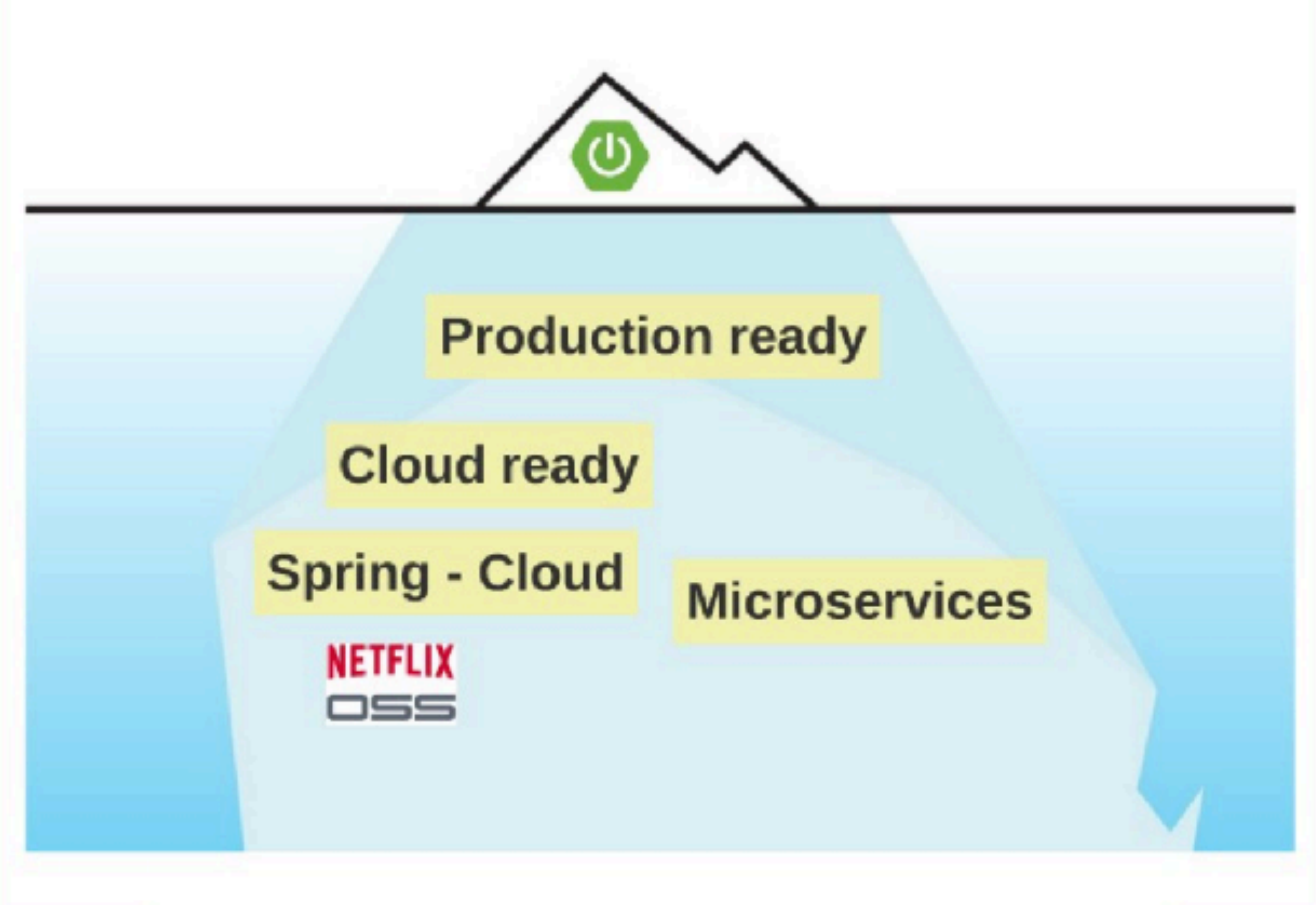
Dropwizard

ACT.framework

Hello Spring Boot 2.x

Why ?

Application skeleton generator
Reduce effort to add new technologies



What ?

Embedded application server
Integration with tools/technologies (starter)
Production tools (monitoring, health check)
Configuration management
Dev tools
No source code generation, no XML

How ?

JAR file

Apache Tomcat

$mvn spring-boot:run

$java -jar demo.jar

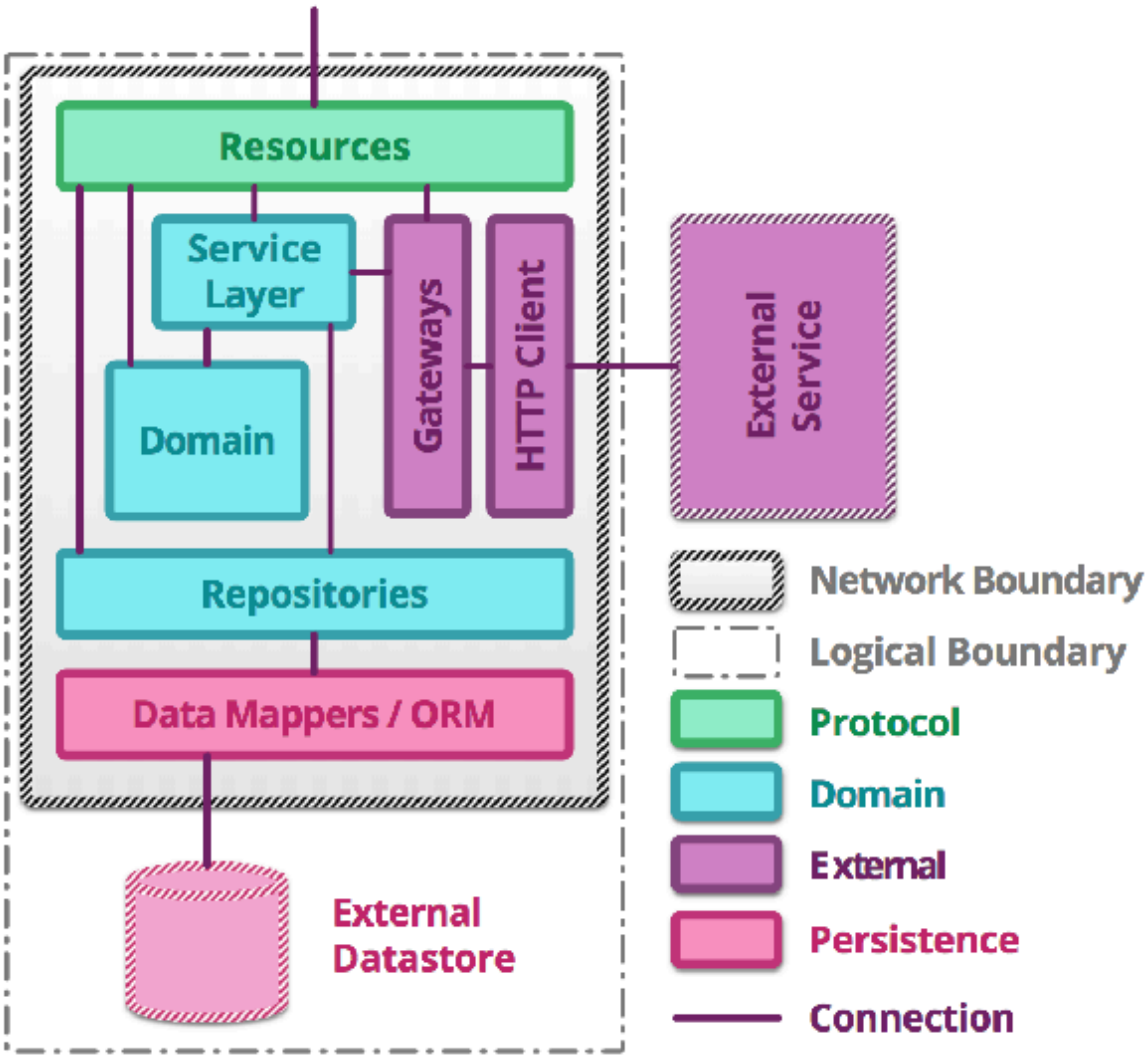
How ?

Apache Tomcat

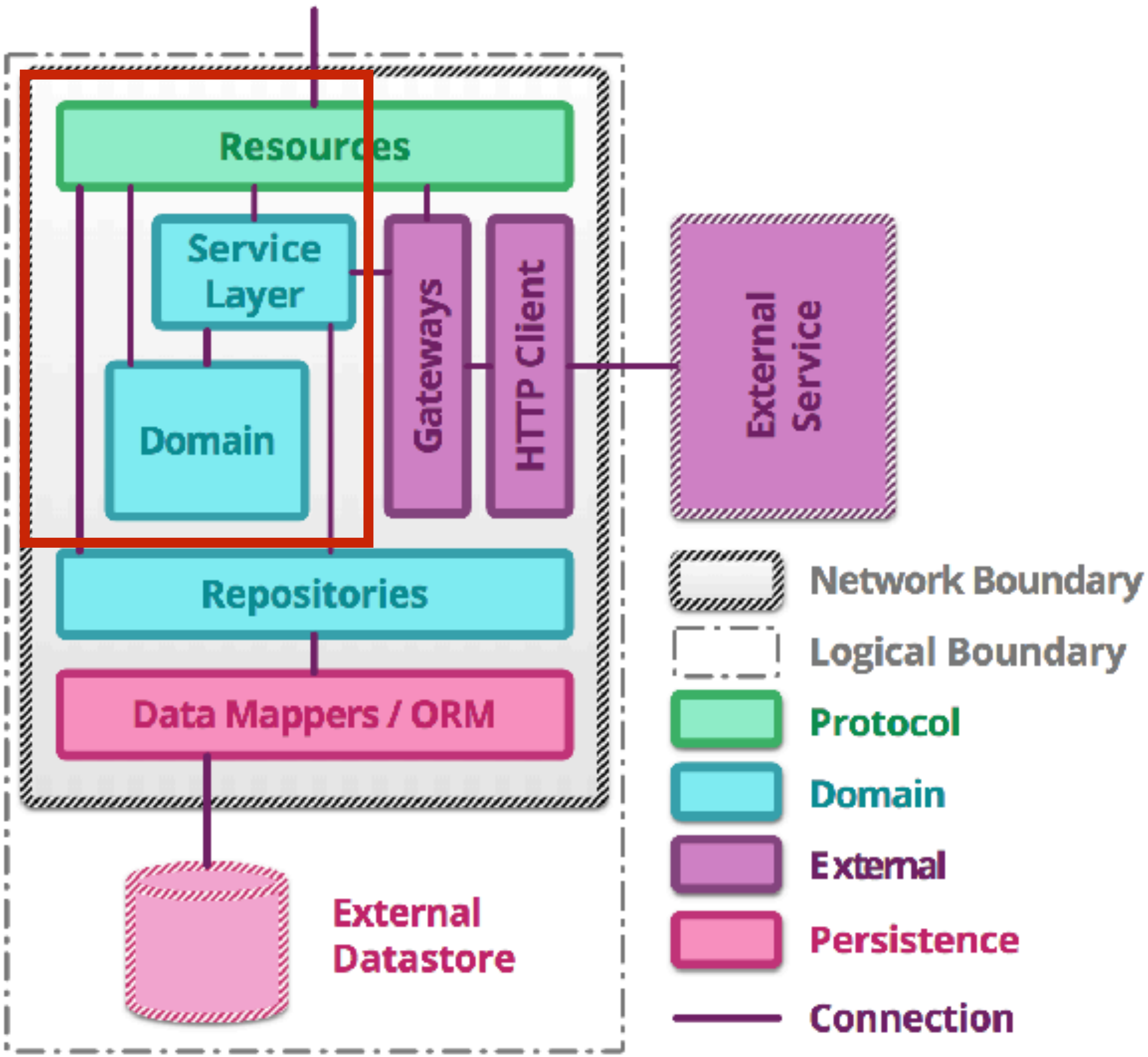
WAR

Building RESTful API with Spring Boot

Service Structure



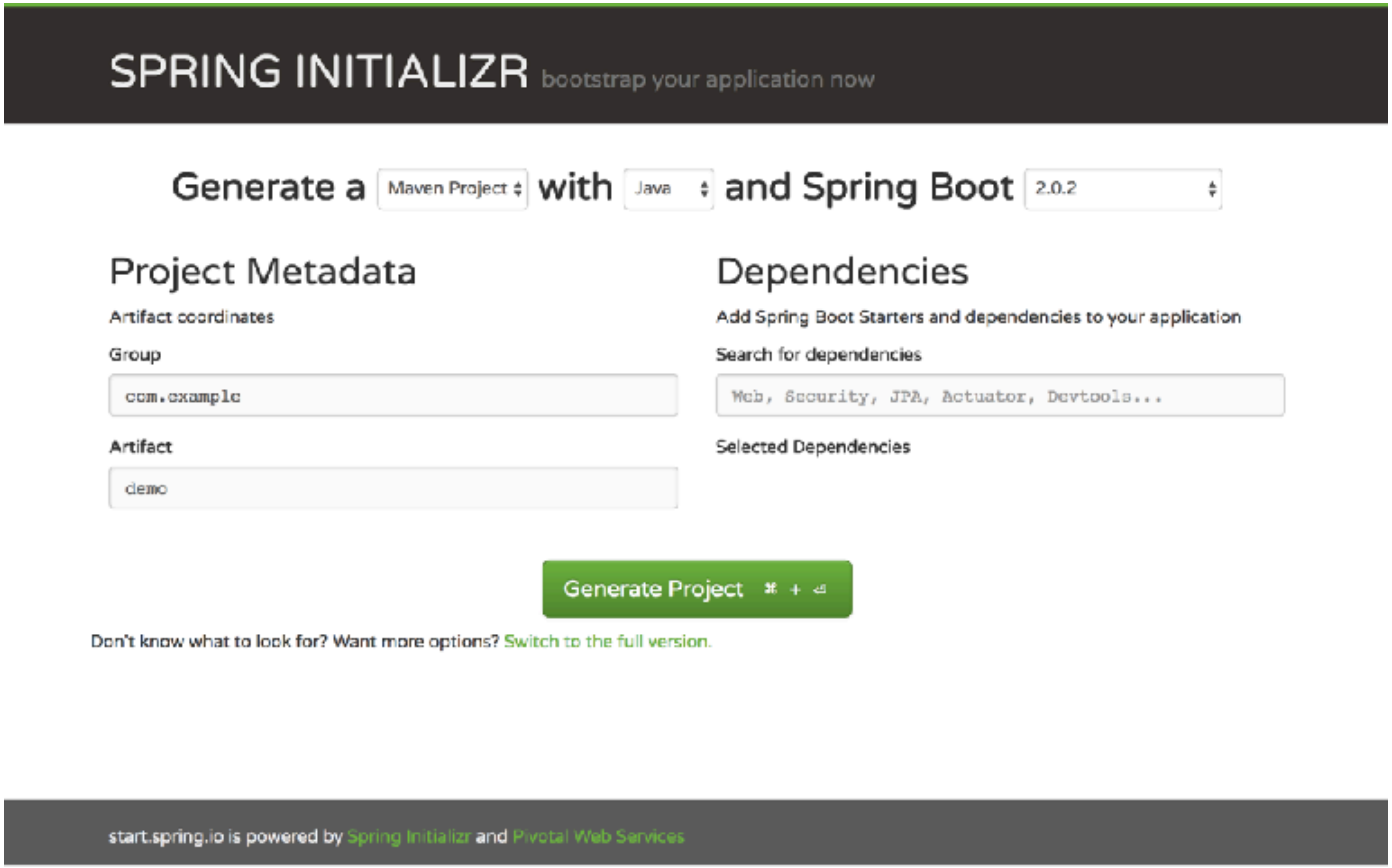
Service Structure



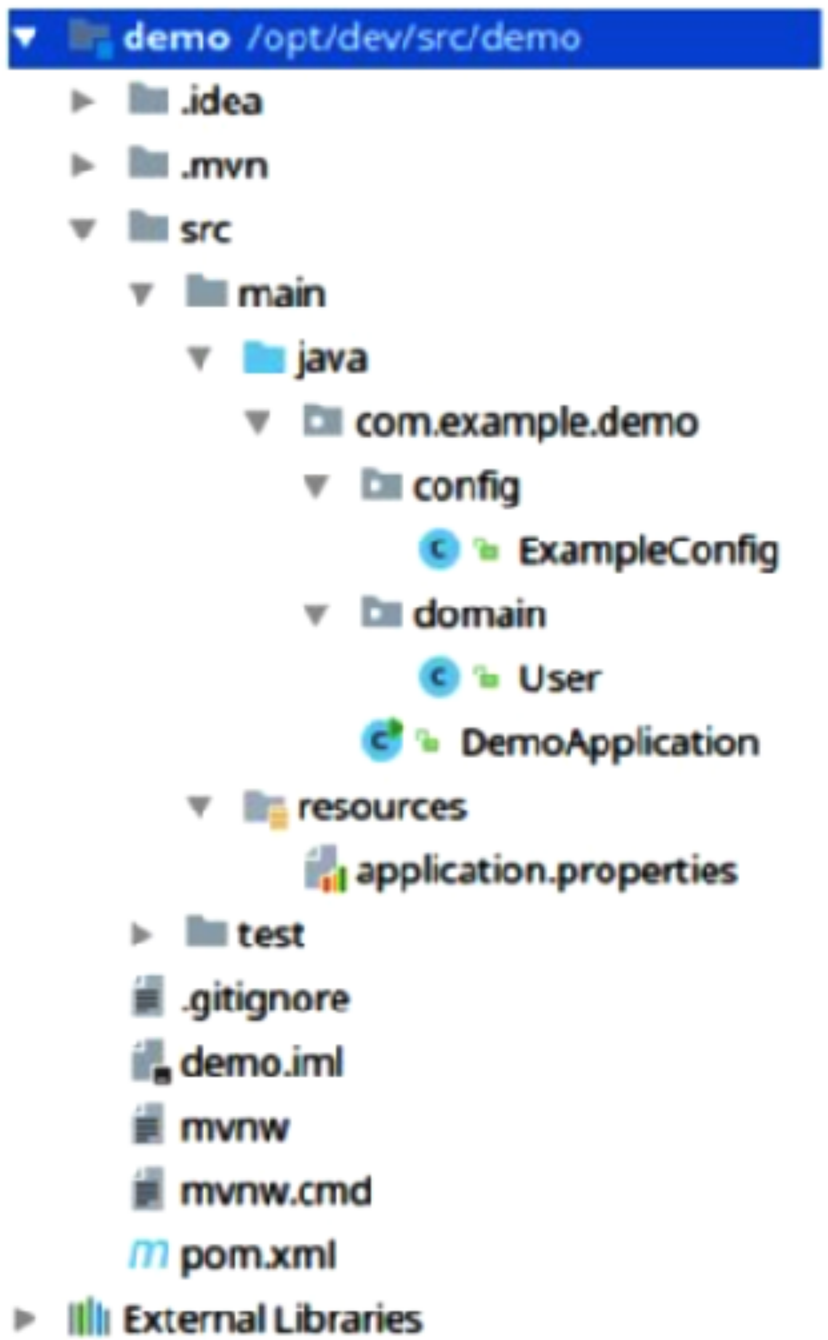
Create project with Spring Initializr

Spring Initializr

https://start.spring.io/



Structure of Spring Boot



Spring Boot main class

```java
package com.example.demo;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class DemoApplication {

    public static void main(String[] args) {
        SpringApplication.run(DemoApplication.class, args);
    }
}
```

Run project

$./mvnw package
$java -jar target/<file name>.jar

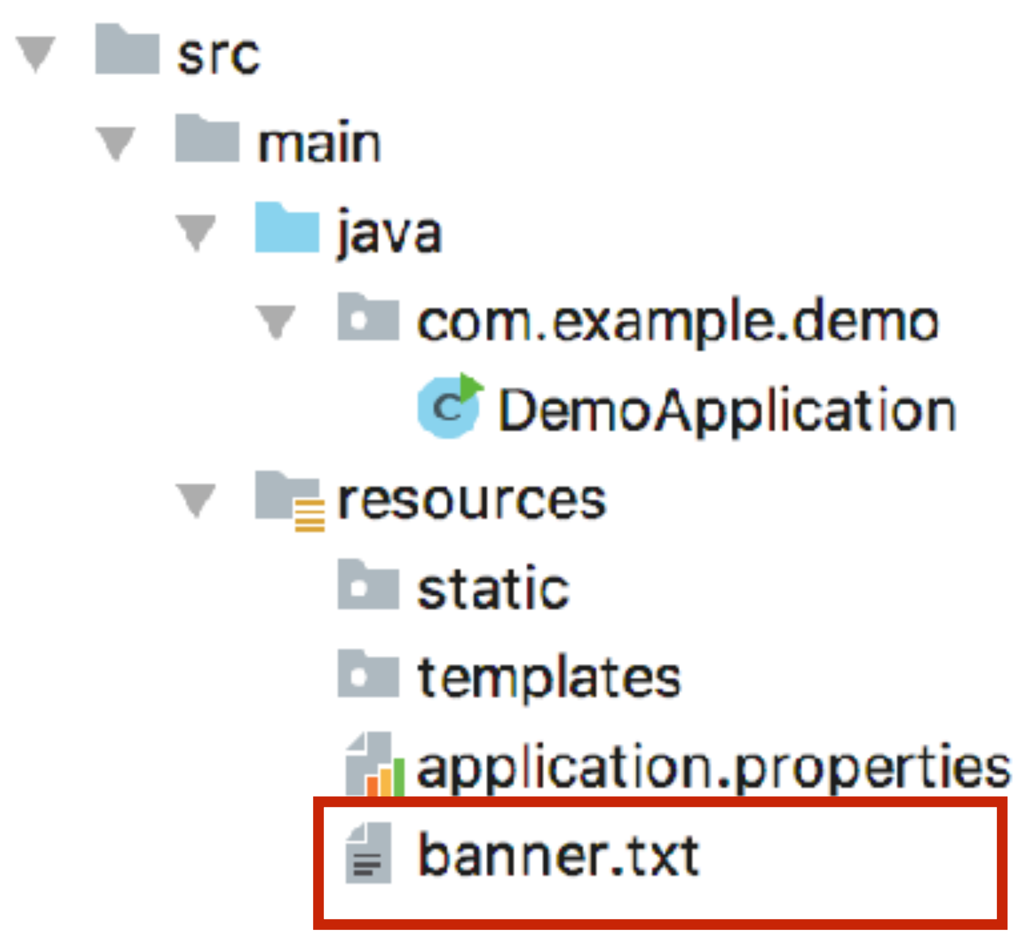
```
  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::        (v2.0.2.RELEASE)

2018-06-07 13:03:30.412  INFO 12828 --- [
oApplication             : Starting DemoApplication on
D 12828 (started by somkiat in /Users/somkiat/Down
2018-06-07 13:03:30.418  INFO 12828 --- [
oApplication             : No active profile set, fall
```

Custom banner

```
  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::        (v2.0.2.RELEASE)

2018-06-07 13:03:30.412  INFO 12828 --- [
oApplication         : Starting DemoApplication on
D 12828 (started by somkiat in /Users/somkiat/Down
2018-06-07 13:03:30.418  INFO 12828 --- [
oApplication         : No active profile set, fall
```

Custom banner (1)

Create file banner.txt or banner.png in resources folder



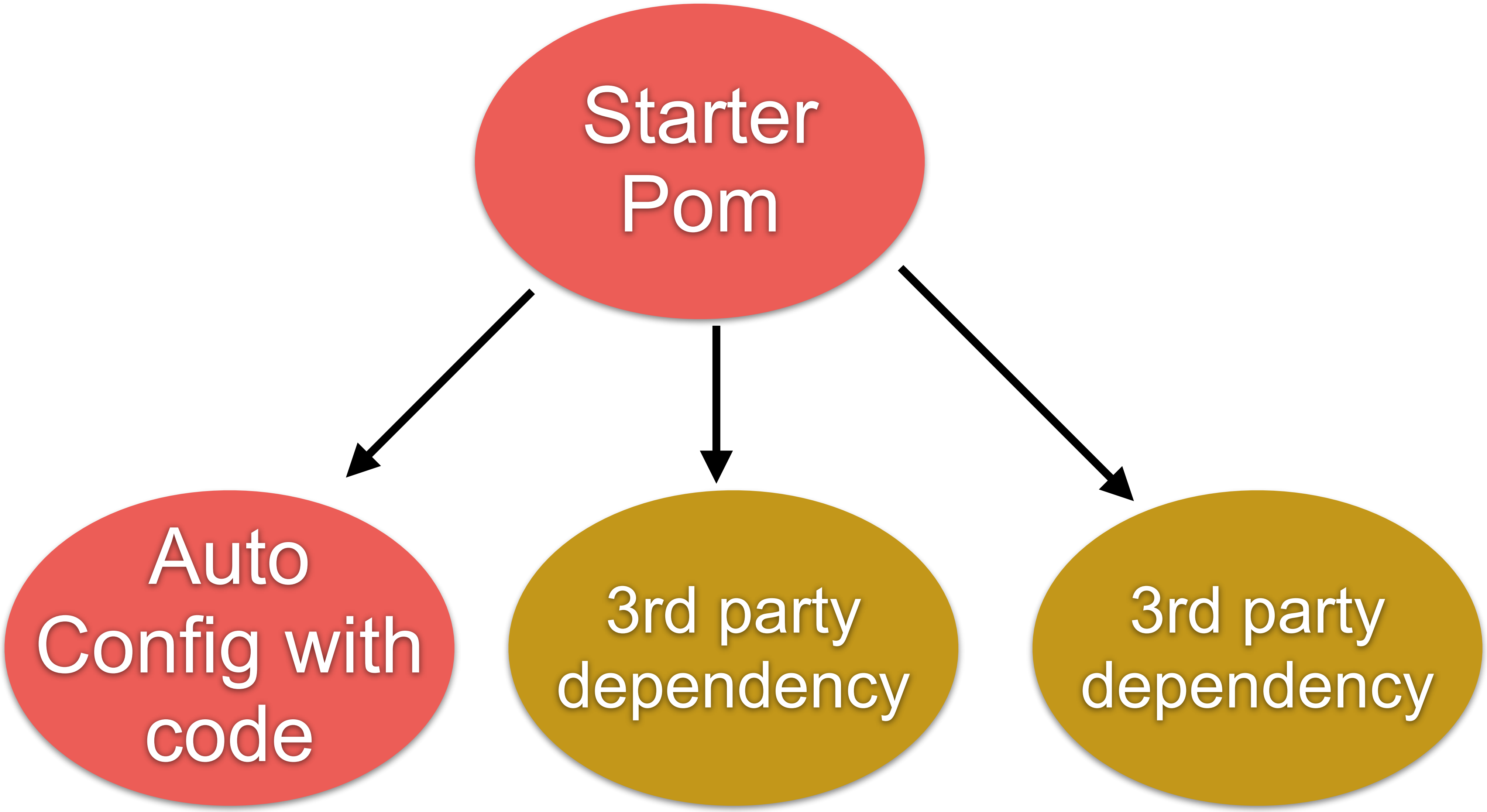
Custom banner (2)

```java
@SpringBootApplication
public class DemoApplication {

    public static void main(String[] args) {
        ImageBanner banner = new ImageBanner(
                new ClassPathResource("try.png"));

        new SpringApplicationBuilder()
                .sources(DemoApplication.class)
                .banner(banner)
                .run();
    }
}
```

Disable banner

```java
@SpringBootApplication
public class DemoApplication {

    public static void main(String[] args) {
        new SpringApplicationBuilder()
                .sources(DemoApplication.class)
                .bannerMode(Banner.Mode.OFF)
                .run(args);
    }
}
```

Anatomy of Starter



Configuration management

Properties/XML/YAML

Config server (App, Spring cloud config, git)

17 ways !!!

https://docs.spring.io/spring-boot/docs/current/reference/html/boot-features-external-config.html

Create project

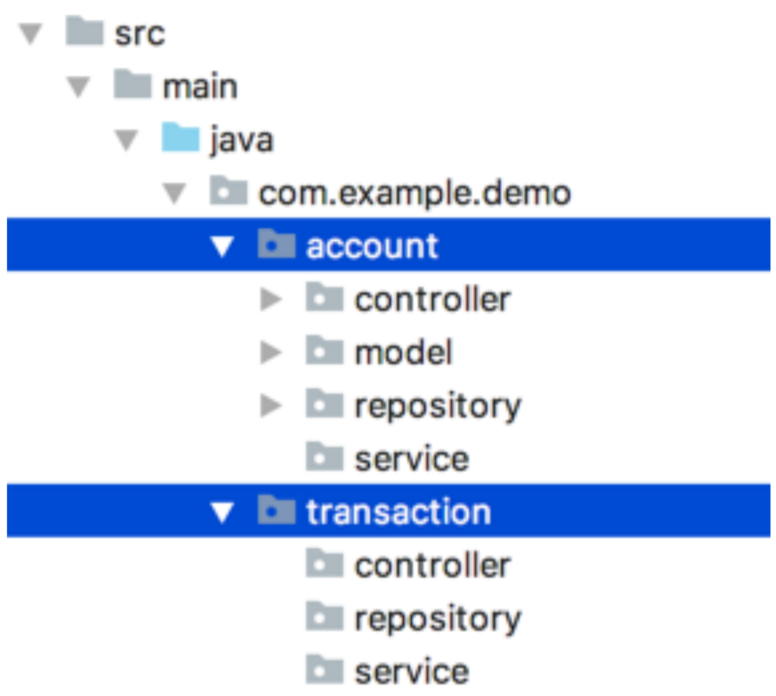
Spring boot application

hello.HelloApplication.java

```
package hello;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class HelloApplication {

    public static void main(String[] args) {
        SpringApplication.run(HelloApplication.class, args);
    }

}
```

Spring Boot Structure

Separate by function/domain



Create model class

hello.model.Hello.java

```java
package hello.domain;

public class Hello {

    private String message;

    public Hello(String message) {
        this.message = message;
    }

    public String getMessage() {
        return message;
    }

    public void setMessage(String message) {
        this.message = message;
    }
}
```

6. Create REST Controller

hello.controller.HelloController.java

```java
package hello.controller;

import hello.domain.Hello;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class HelloController {

    @GetMapping("/hello/{name}")
    public Hello sayHi(@PathVariable String name) {
        return new Hello("Hello " + name);
    }

}
```

7. Compile and Packaging

$mvnw clean package

8. Run

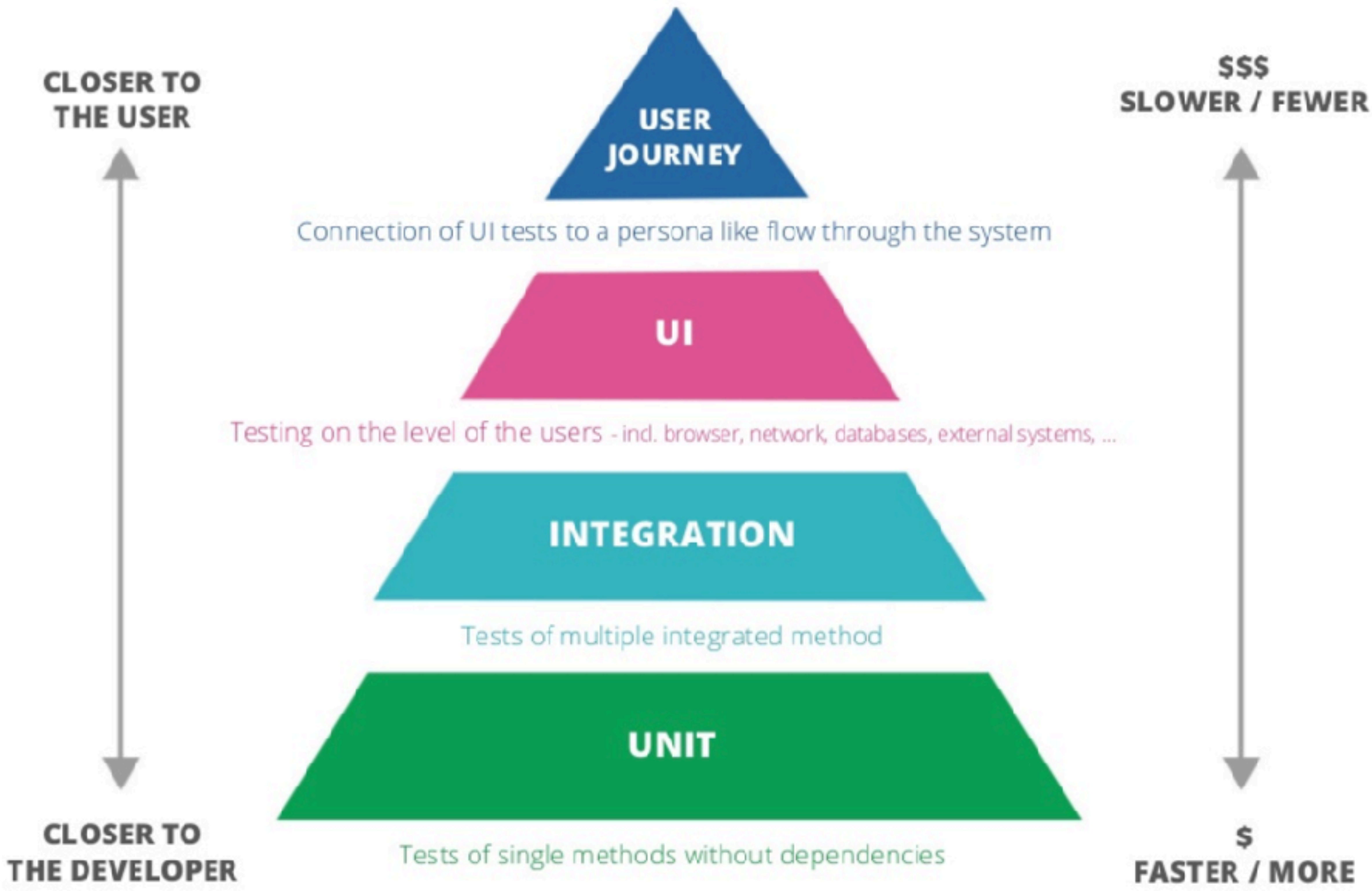
$java -jar target/hello.jar

```
  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::        (v2.0.0.RELEASE)

2018-03-05 23:37:18.018  INFO 30560 --- [           mair
                     : Starting HelloApplication v1.0-SNAPSHC
th PID 30560 (/Users/somkiat/data/slide/microservice/sl
op/course-microservice/slide/4days-workshop/workshop/he
 by somkiat in /Users/somkiat/data/slide/microservice/s
hop/course-microservice/slide/4days-workshop/workshop/he
2018-03-05 23:37:18.023  INFO 30560 --- [           mair
                     : No active profile set, falling back tc
2018-03-05 23:37:18.138  INFO 30560 --- [           mair
```

9. Open in browser

http://localhost:8080/hello/somkiat

How to test the Hello service ?


CLOSER TO
THE USER
$$$
SLOWER / FEWER
USER
JOURNEY
Connection of UI tests to a persona like flow through the system
UI
Testing on the level of the users - ind. browser, network, databases, external systems, ...
INTEGRATION
Tests of multiple integrated method
UNIT
Tests of single methods without dependencies
CLOSER TO
THE DEVELOPER
$
FASTER / MORE

Unit tests

How to use model ?

```
public class HelloTest {

    @Test
    public void success_to_create_model_with_constructor() {
        Hello hello = new Hello("Somkiat");
        assertEquals( expected: "Somkiat", hello.getMessage());
    }

}
```

API/Controller tests

How to use controller ?

Spring boot provides MockMvc to test Controller

```java
package hello.controller;

import hello.domain.Hello;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class HelloController {

    @GetMapping("/hello/{name}")
    public Hello sayHi(@PathVariable String name) {
        return new Hello("Hello " + name);
    }

}
```

API/Controller tests

```
@RunWith(SpringRunner.class)
@WebMvcTest(controllers = HelloController.class)
public class HelloControllerTest {

    @Autowired
    private MockMvc mockMvc;

    @Test
    public void shouldReturnHelloSomkiat() throws Exception {
        mockMvc.perform(get( urlTemplate: "/hello/somkiat"))
                .andExpect(
                        jsonPath( expression: "$.message")
                                .value( expectedValue: "Hello somkiat"))
                .andExpect(status().is2xxSuccessful());
    }

}
```

API/Controller tests

With Spring Boot Testing + TestRestTemplate

```
@RunWith(SpringRunner.class)
@SpringBootTest(
        webEnvironment = WebEnvironment.RANDOM_PORT)
public class AccountControllerTest {

    @Autowired
    private TestRestTemplate testRestTemplate;

    @Autowired
    private AccountRepository accountRepository;

    @Before
    public void initData() {
        accountRepository.save(new Account("01"));
        accountRepository.save(new Account("02"));
    }
```

API/Controller tests

With Spring Boot Testing + TestRestTemplate

```
@Test
public void
เรียกข้อมูลบัญชีของหมายเลข_0868696209_ต้องเจอสองบัญชีนะ() {
    List<AccountResponse> accountResponses
            = testRestTemplate.getForObject(
                    "/account/0868696209", List.class);

    assertEquals(2, accountResponses.size());
}
```

Compile with testing

$mvn clean package

```
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
```

% of Code/Test coverage

Add coverage to pom.xml (1)

```xml
<build>
    <finalName>hello</finalName>
    <plugins>
        <plugin>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-maven-plugin</artifactId>
        </plugin>

        <plugin>
            <artifactId>maven-compiler-plugin</artifactId>
            <version>2.5.1</version>
            <configuration>
                <source>${java.version}</source>
                <target>${java.version}</target>
                <encoding>${project.build.sourceEncoding}</encoding>
            </configuration>
        </plugin>
```

Add coverage to pom.xml (2)

```
<plugin>
    <groupId>org.codehaus.mojo</groupId>
    <artifactId>cobertura-maven-plugin</artifactId>
    <version>2.7</version>
    <configuration>
        <formats>
            <format>html</format>
            <format>xml</format>
        </formats>
    </configuration>
    <executions>
        <execution>
            <phase>package</phase>
            <goals>
                <goal>cobertura</goal>
            </goals>
        </execution>
    </executions>
    <dependencies>
        <dependency>
            <groupId>org.ow2.asm</groupId>
            <artifactId>asm</artifactId>
            <version>5.0.3</version>
        </dependency>
    </dependencies>
</plugin>
```

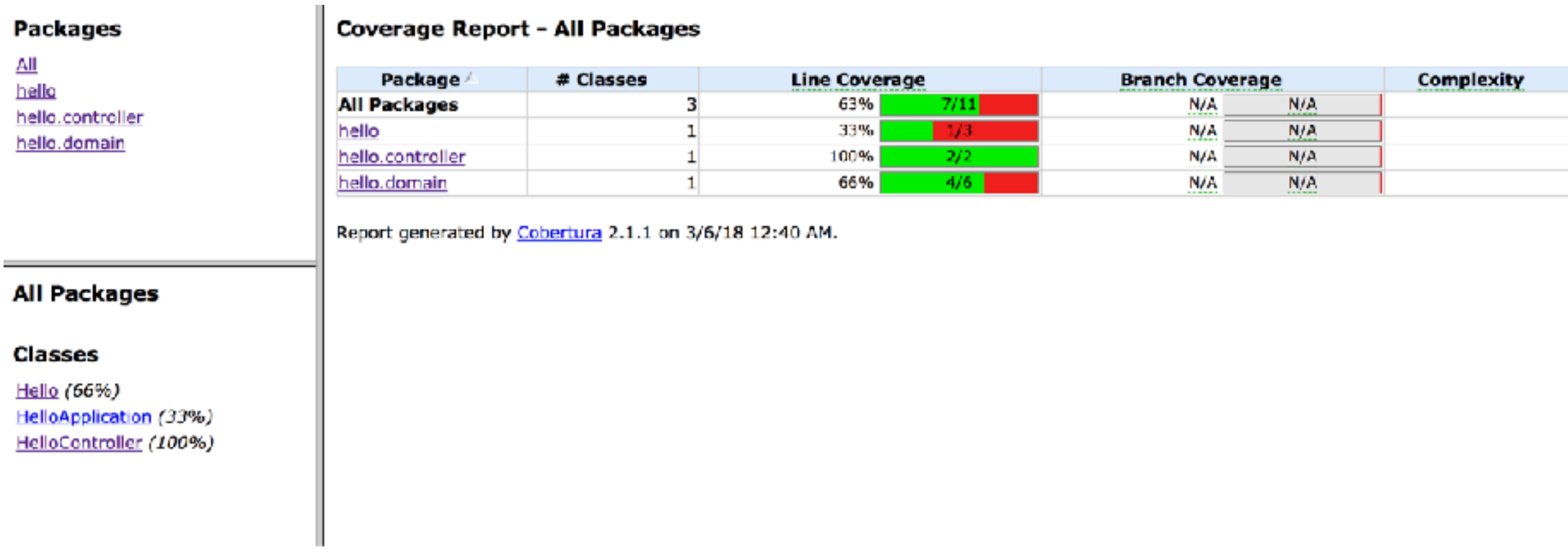
Run test again

$mvn clean package

```
Cobertura Report generation was successful.
Cobertura 2.1.1 - GNU GPL License (NO WARRANTY) - See COPYRIGHT file
Cobertura: Loaded information on 3 classes.
time: 125ms

Cobertura Report generation was successful.
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
```

Coverage report

open target/site/cobertura/index.html



Packages

All
hello
hello.controller
hello.domain

Coverage Report - All Packages

Package	# Classes	Line Coverage		Branch Coverage		Complexity
All Packages	3	63%	7/11	N/A	N/A	1
hello	1	33%	1/3	N/A	N/A	1
hello.controller	1	100%	2/2	N/A	N/A	1
hello.domain	1	66%	4/6	N/A	N/A	1

Report generated by Cobertura 2.1.1 on 3/6/18 12:40 AM.

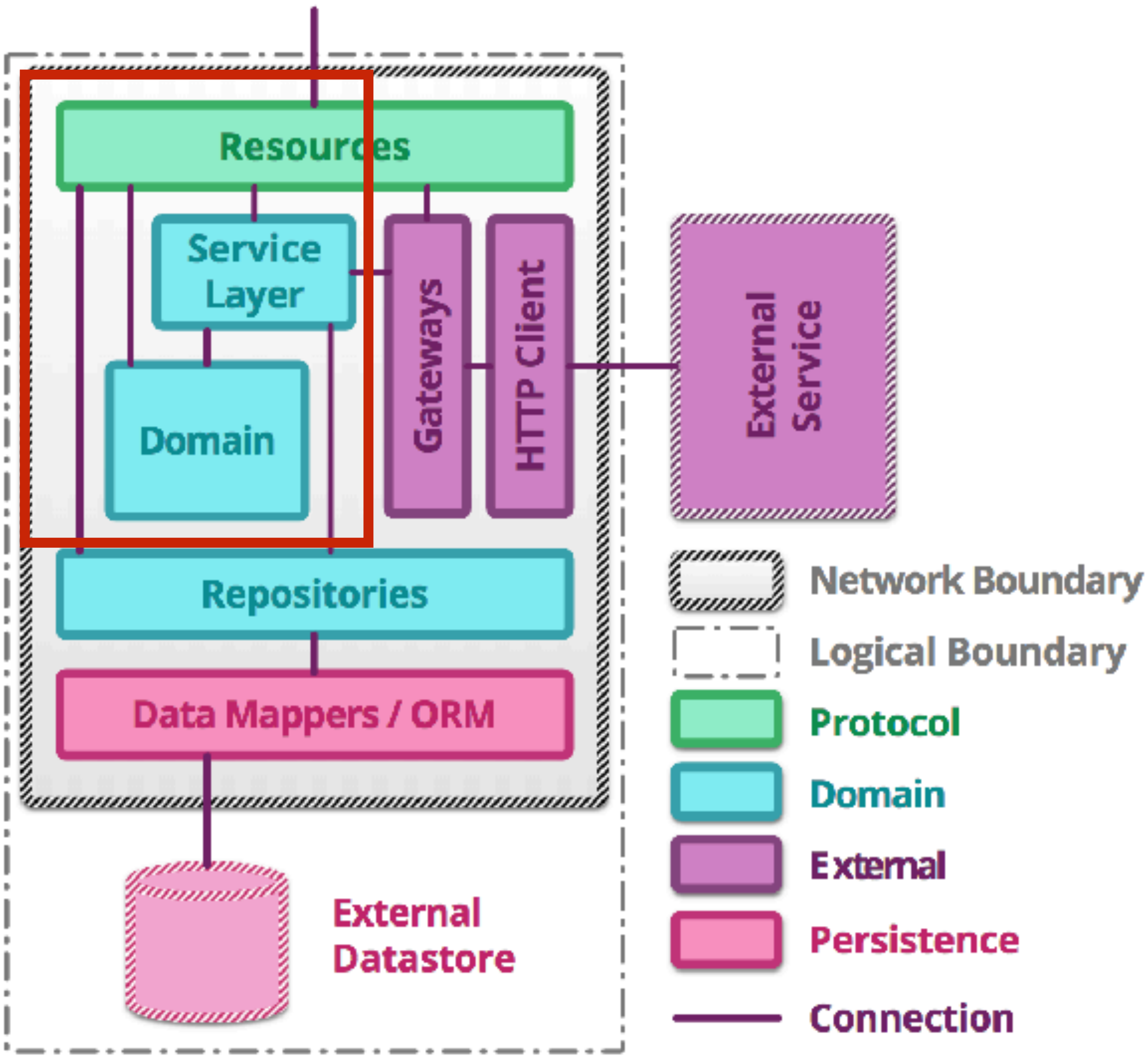
All Packages

Classes

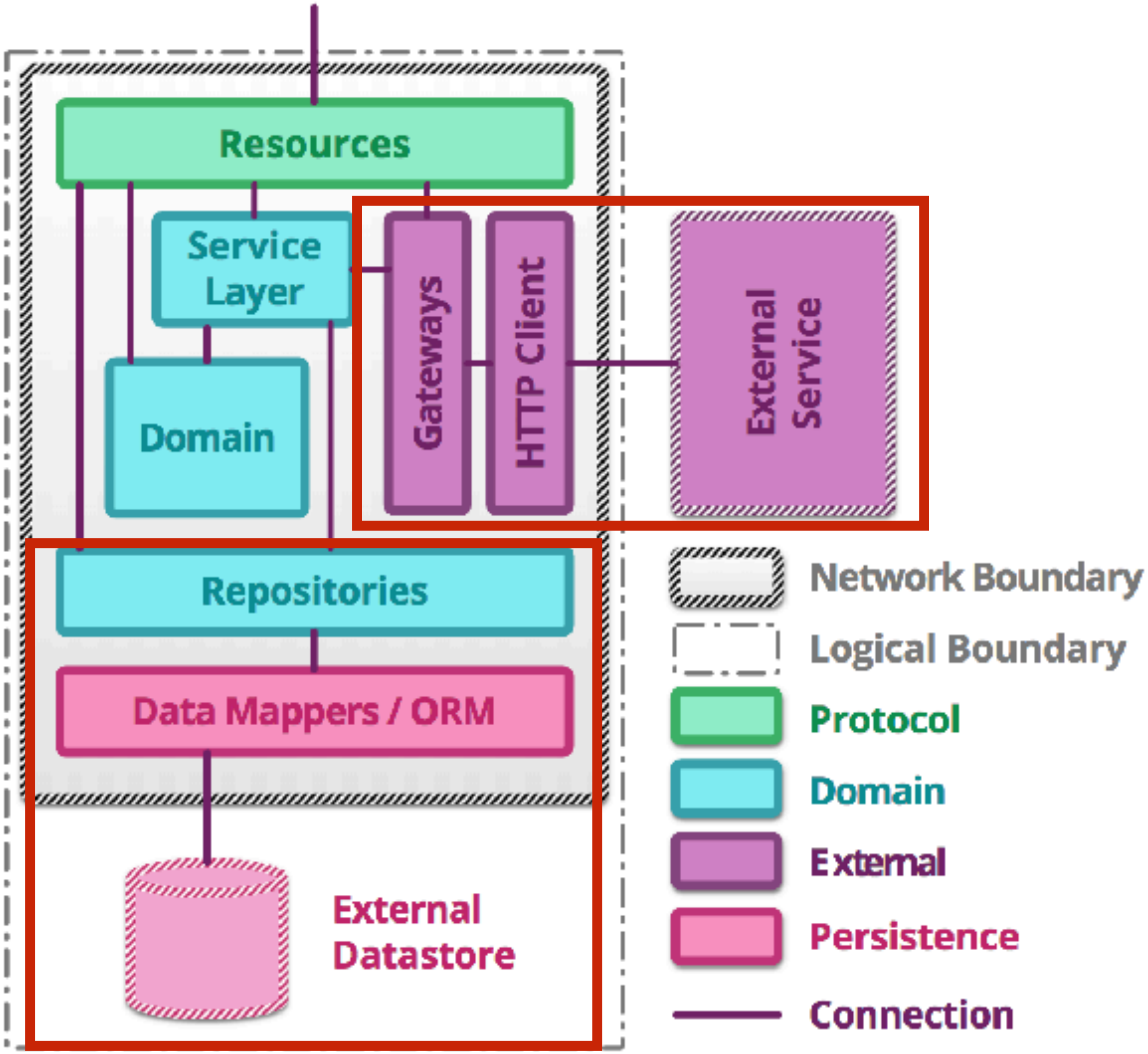
Hello *(66%)*
HelloApplication *(33%)*
HelloController *(100%)*

How to improve % of coverage ?

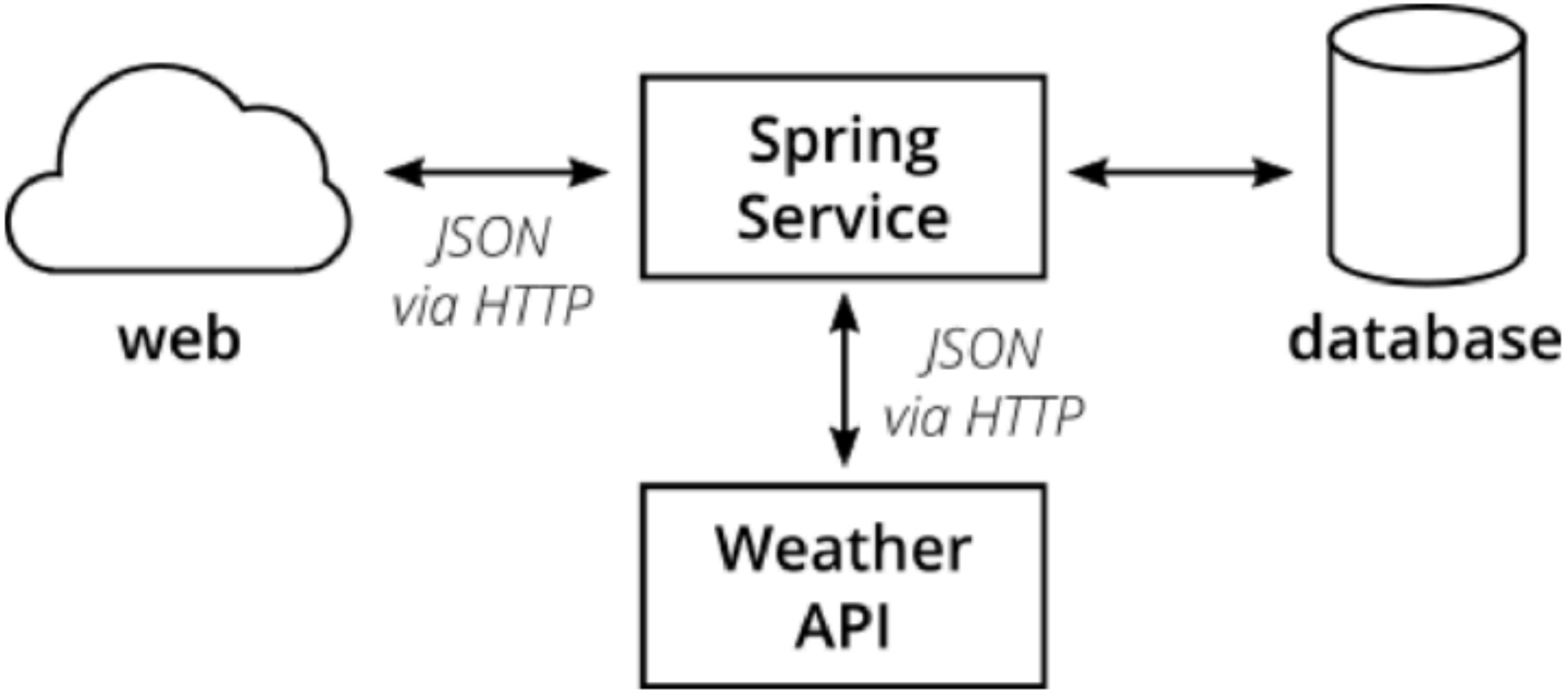
Service Structure



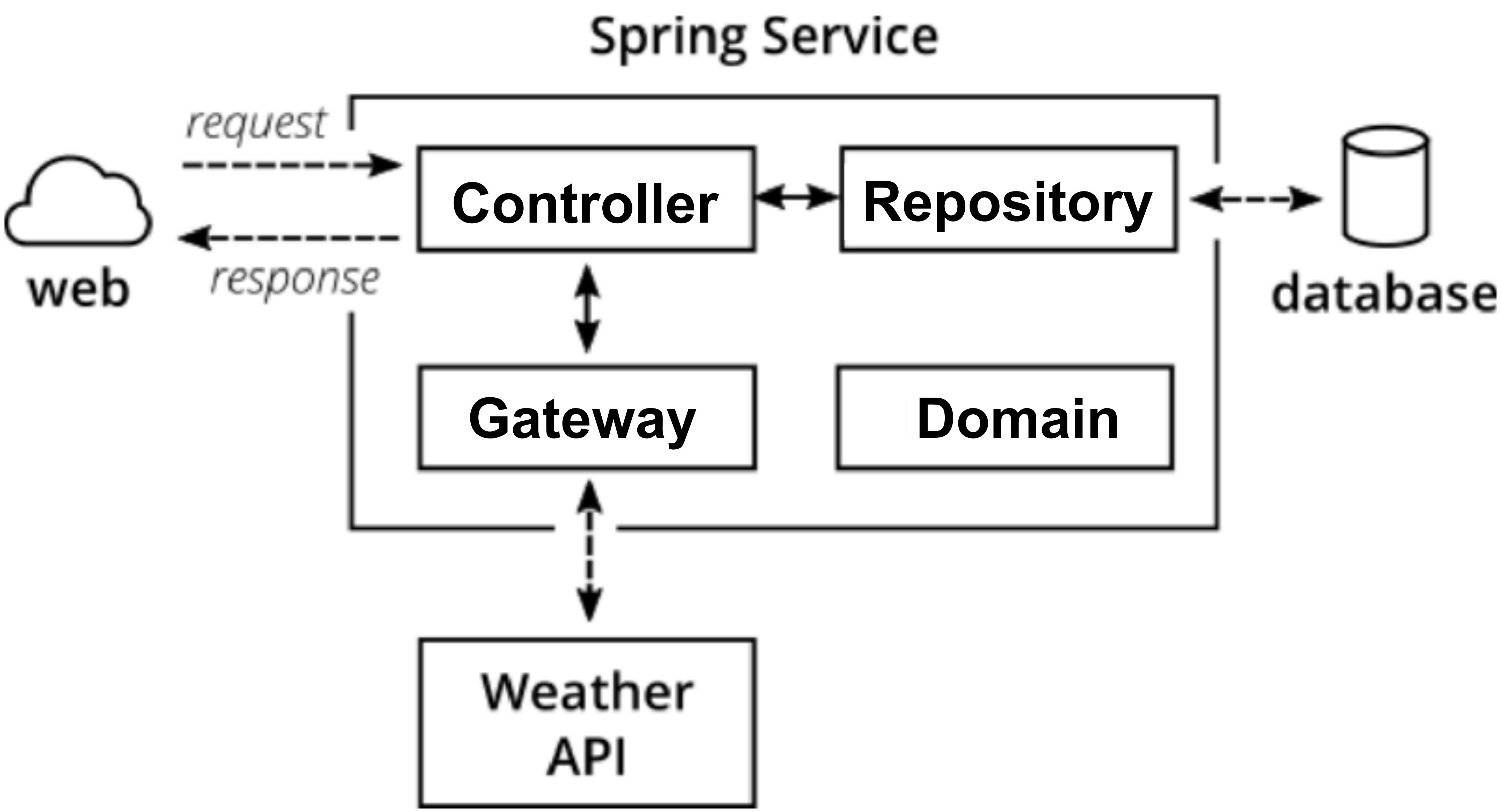
Service Structure



Sample application



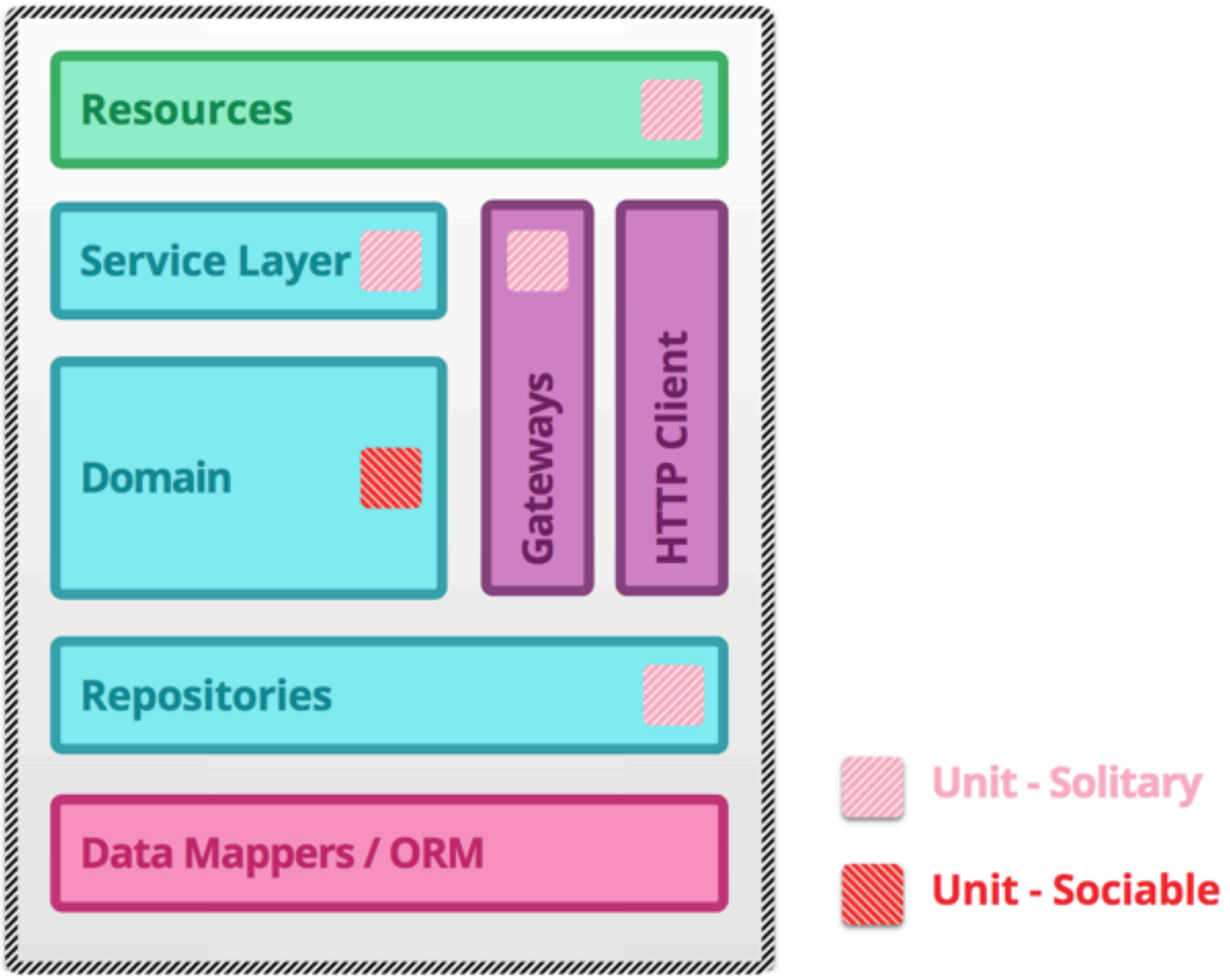
Project structure



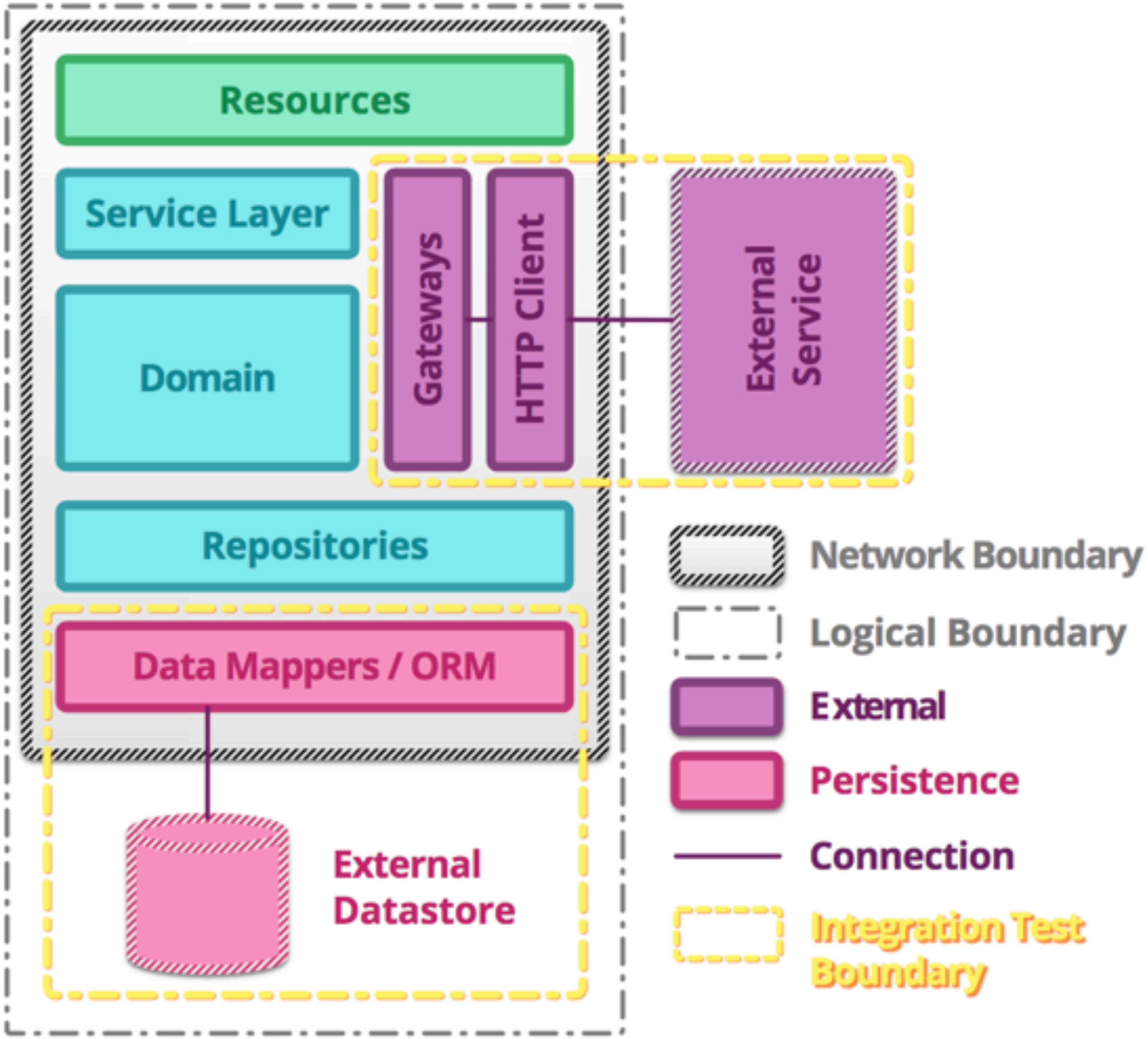
How to test ?


Testing Strategies
Unit Testing
Integration Testing
Component Testing
Contract Testing
End-to-End Testing

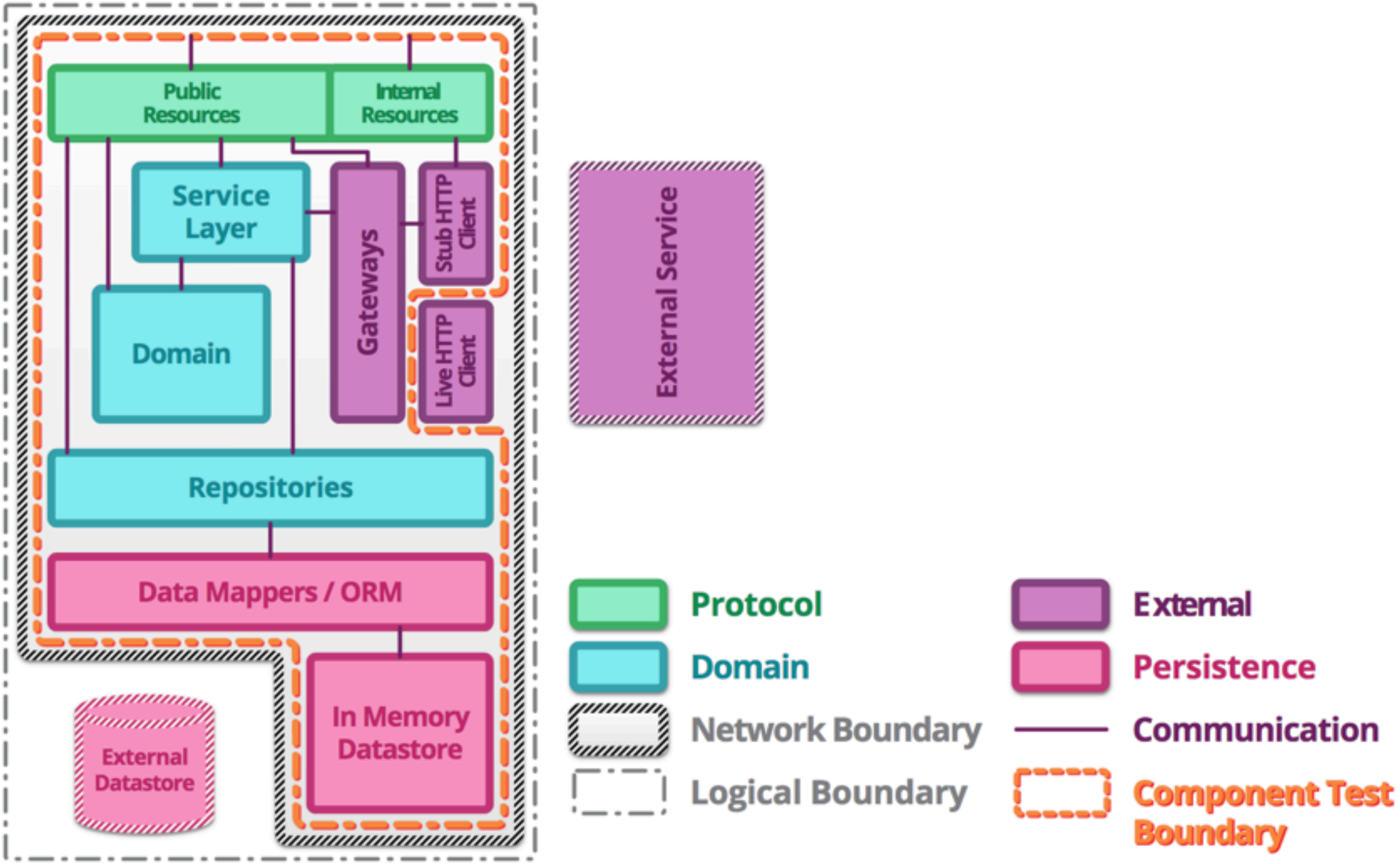
Unit testing



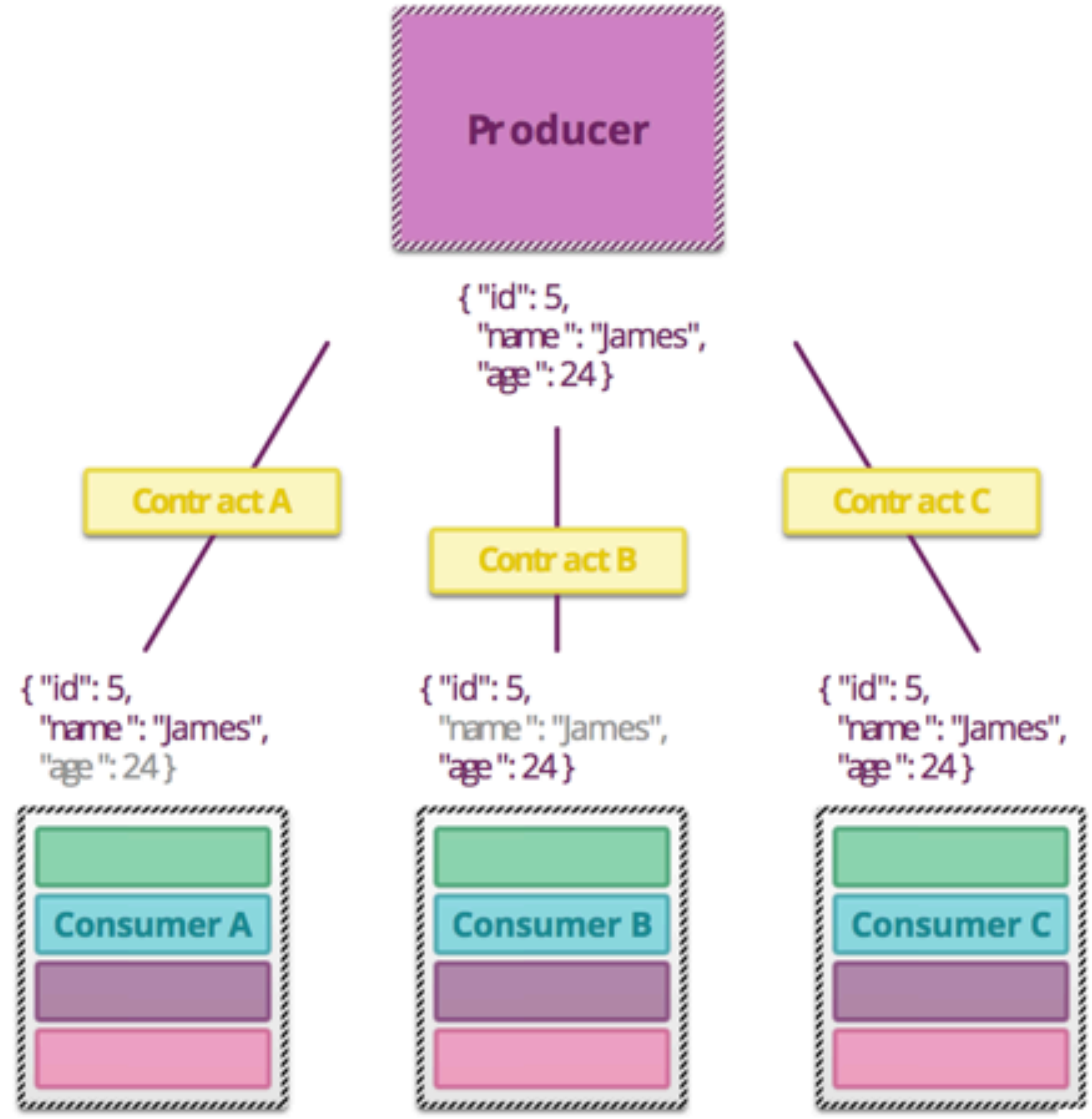
Integration testing



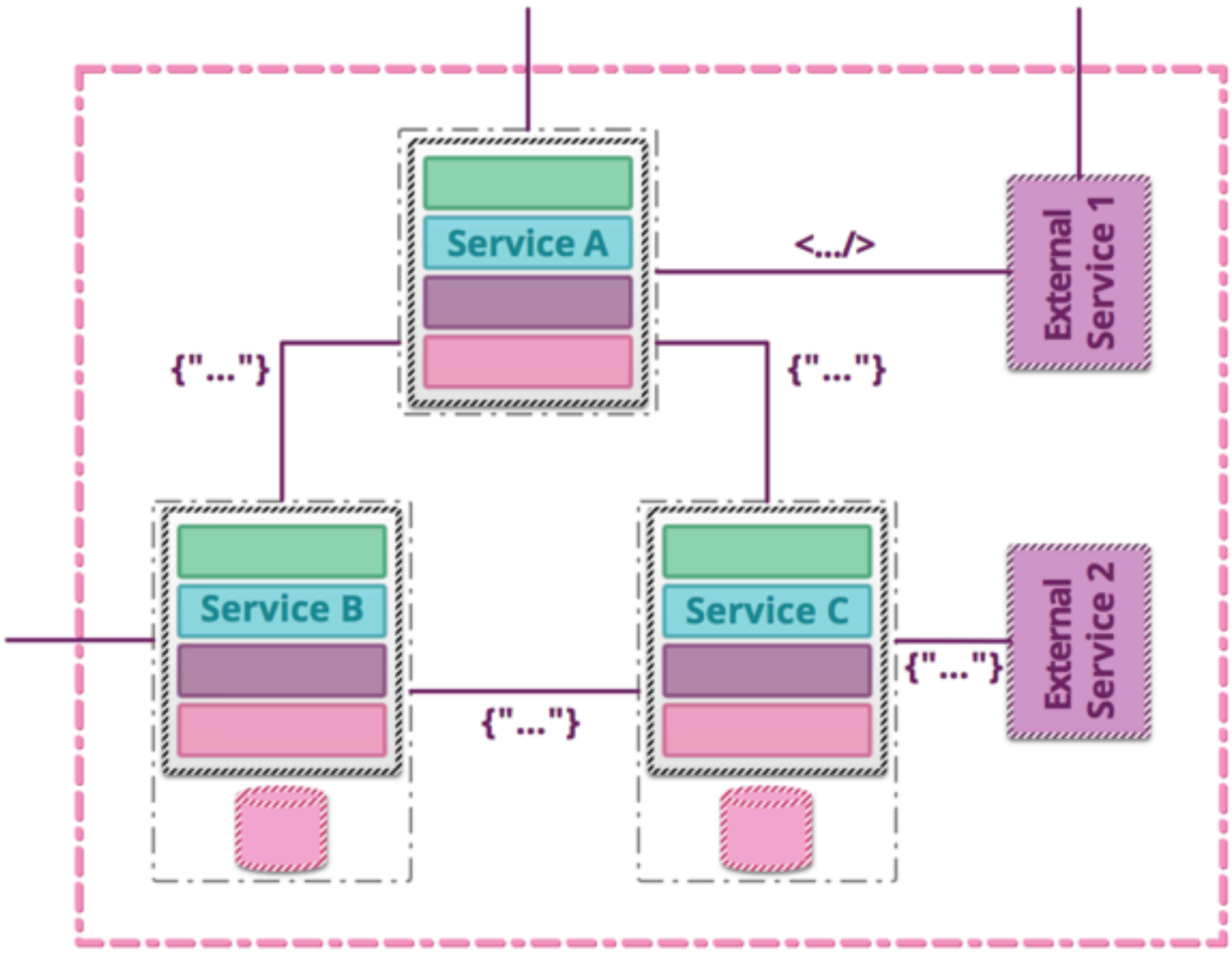
Component testing



Contract testing



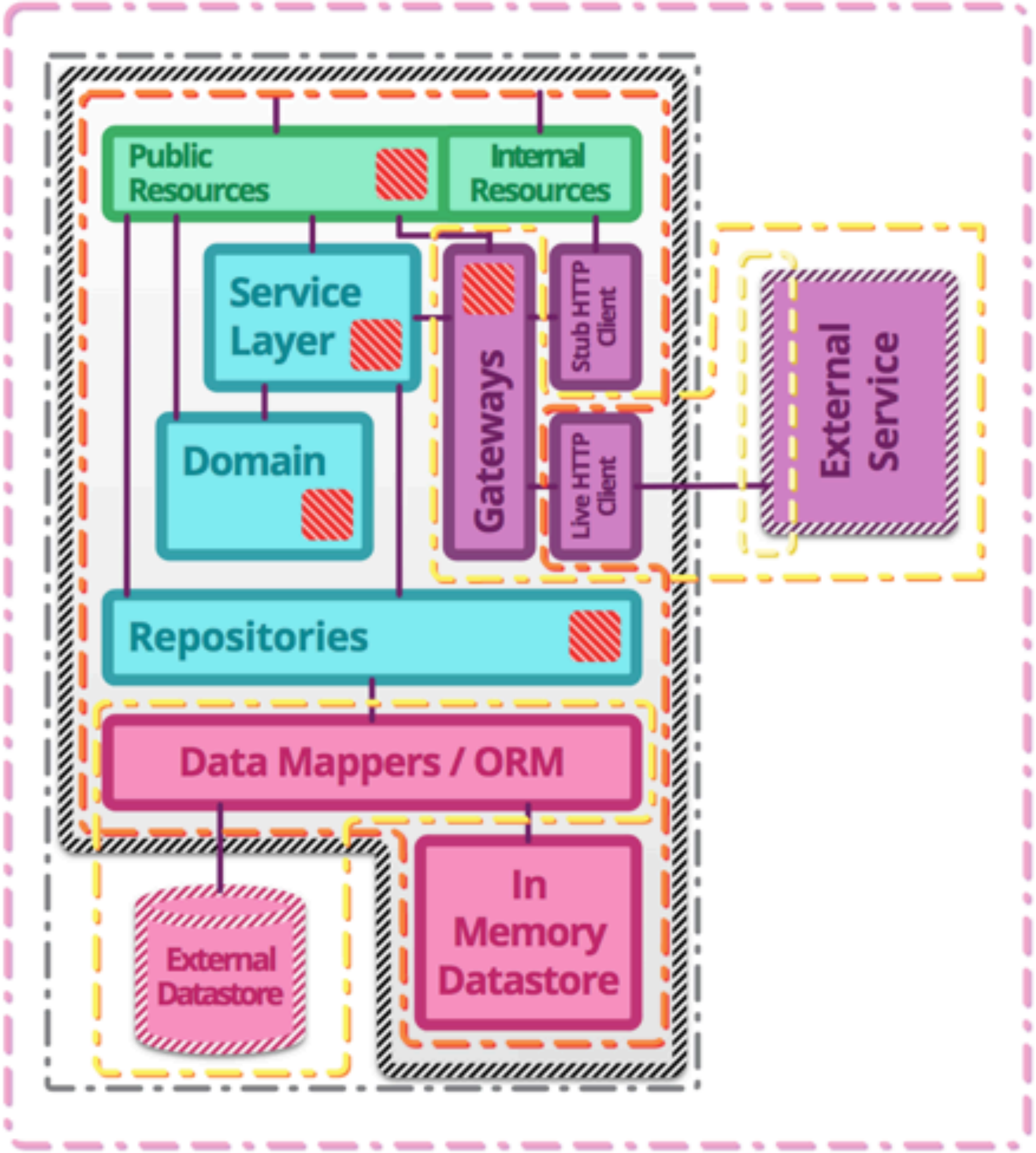
End-to-End testing



Summary

Unit tests : exercise the smallest pieces of testable software in the application to determine whether they behave as expected.

Integration tests : verify the communication paths and interactions between components to detect interface defects.



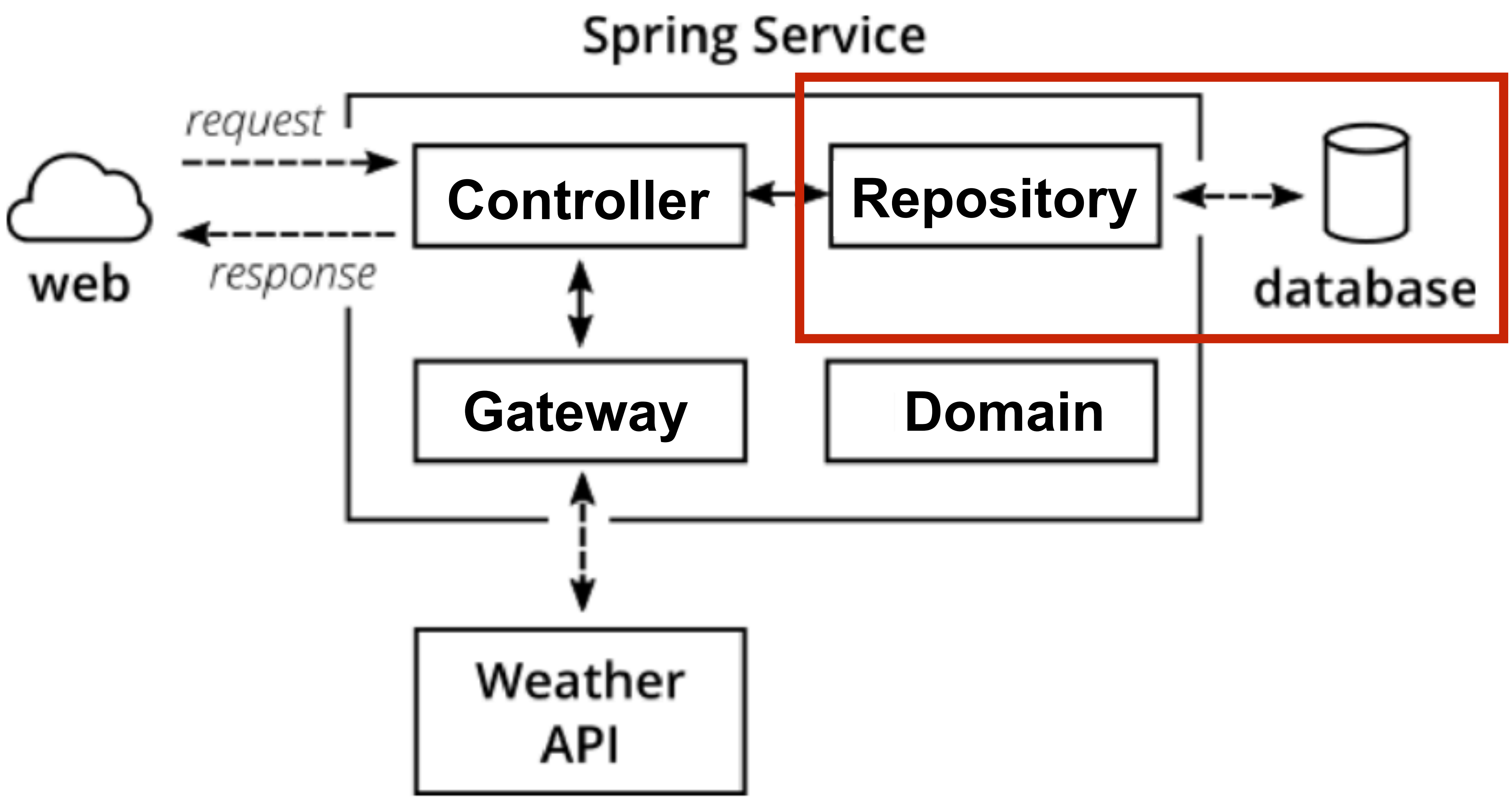
Component tests : limit the scope of the exercised software to a portion of the system under test, manipulating the system through internal code interfaces and using test doubles to isolate the code under test from other components.

Contract tests : verify interactions at the boundary of an external service asserting that it meets the contract expected by a consuming service.

End-to-end tests : verify that a system meets external requirements and achieves its goals, testing the entire system, from end to end.

Let’s workshop with Repository

Working with repository



Working with repository

We’re using Spring Data

Modify pom.xml

Add library of Spring Data

```xml
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>
```

Modify pom.xml

Add library of Persistence/data store

```
<dependency>
    <groupId>org.postgresql</groupId>
    <artifactId>postgresql</artifactId>
    <version>42.1.1</version>
</dependency>
```

Add data store config

In src/main/resources

```
spring.datasource.url= jdbc:postgresql://127.0.0.1:15432/postgres
spring.datasource.username= user
spring.datasource.password= password
spring.datasource.platform= POSTGRESQL

spring.jpa.show-sql= true
spring.jpa.hibernate.ddl-auto= create-drop
spring.jpa.database-platform= org.hibernate.dialect.PostgreSQLDialect
```

Create repository interface

hello.repository.PersonRepository.java

```
public interface PersonRepository
        extends CrudRepository<Person, String> {

    Optional<Person> findByFirstName(String name);

}
```

Create Entity class

hello.repository.Person.java

```
@Entity
public class Person {

    @Id
    @GeneratedValue(strategy = GenerationType.AUTO)
    private int id;
    private String firstName;
    private String lastName;

    public Person() {
    }

    public Person(String firstName, String lastName) {
        this.firstName = firstName;
        this.lastName = lastName;
    }
```

Create new controller

hello.repository.HelloControllerWithRepository.java

```java
public class HelloControllerWithRepository {

    private final PersonRepository personRepository;

    @Autowired
    public HelloControllerWithRepository(PersonRepository personRepository) {
        this.personRepository = personRepository;
    }

    @GetMapping("/hello/data/{name}")
    public Hello sayHi(@PathVariable String name) {
        Optional<Person> foundPerson = personRepository.findByName(name);
        String result = foundPerson
                .map(person -> String.format("Hello %s", person.getFirstName()))
                .orElse( other: "Data not found");
        return new Hello(result);
    }

}
```

Run test and package

$mvn clean package

```
Cobertura Report generation was successful.
Cobertura 2.1.1 - GNU GPL License (NO WARRANTY) - See COPYRIGHT file
Cobertura: Loaded information on 3 classes.
time: 125ms

Cobertura Report generation was successful.
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
```

Run your application

$java -jar target/hello.jar

```
org.postgresql.util.PSQLException: Connection to 127.0.0.1:15432 refused. Check that th
 the postmaster is accepting TCP/IP connections.
        at org.postgresql.core.v3.ConnectionFactoryImpl.openConnectionImpl(ConnectionFa
jar!/:42.1.1]
        at org.postgresql.core.ConnectionFactory.openConnection(ConnectionFactory.java:
        at org.postgresql.jdbc.PgConnection.<init>(PgConnection.java:194) ~[postgresql-
        at org.postgresql.Driver.makeConnection(Driver.java:450) ~[postgresql-42.1.1.ja
        at org.postgresql.Driver.connect(Driver.java:252) ~[postgresql-42.1.1.jar!/:42.
        at com.zaxxer.hikari.util.DriverDataSource.getConnection(DriverDataSource.java:
        at com.zaxxer.hikari.util.DriverDataSource.getConnection(DriverDataSource.java:
        at com.zaxxer.hikari.pool.PoolBase.newConnection(PoolBase.java:365) [HikariCP-2
        at com.zaxxer.hikari.pool.PoolBase.newPoolEntry(PoolBase.java:194) [HikariCP-2.
        at com.zaxxer.hikari.pool.HikariPool.createPoolEntry(HikariPool.java:460) [Hika
        at com.zaxxer.hikari.pool.HikariPool.checkFailFast(HikariPool.java:534) [Hikari
        at com.zaxxer.hikari.pool.HikariPool.<init>(HikariPool.java:115) [HikariCP-2.7.
        at com.zaxxer.hikari.HikariDataSource.getConnection(HikariDataSource.java:112)
```

Coverage report

Packages

- All
- hello
- hello.controller
- hello.domain
- hello.repository

All Packages

Classes

- Hello (65%)
- HelloApplication (33%)
- HelloController (100%)
- HelloControllerWithRepository (0%
- Person (0%)
- PersonRepository (N/A)

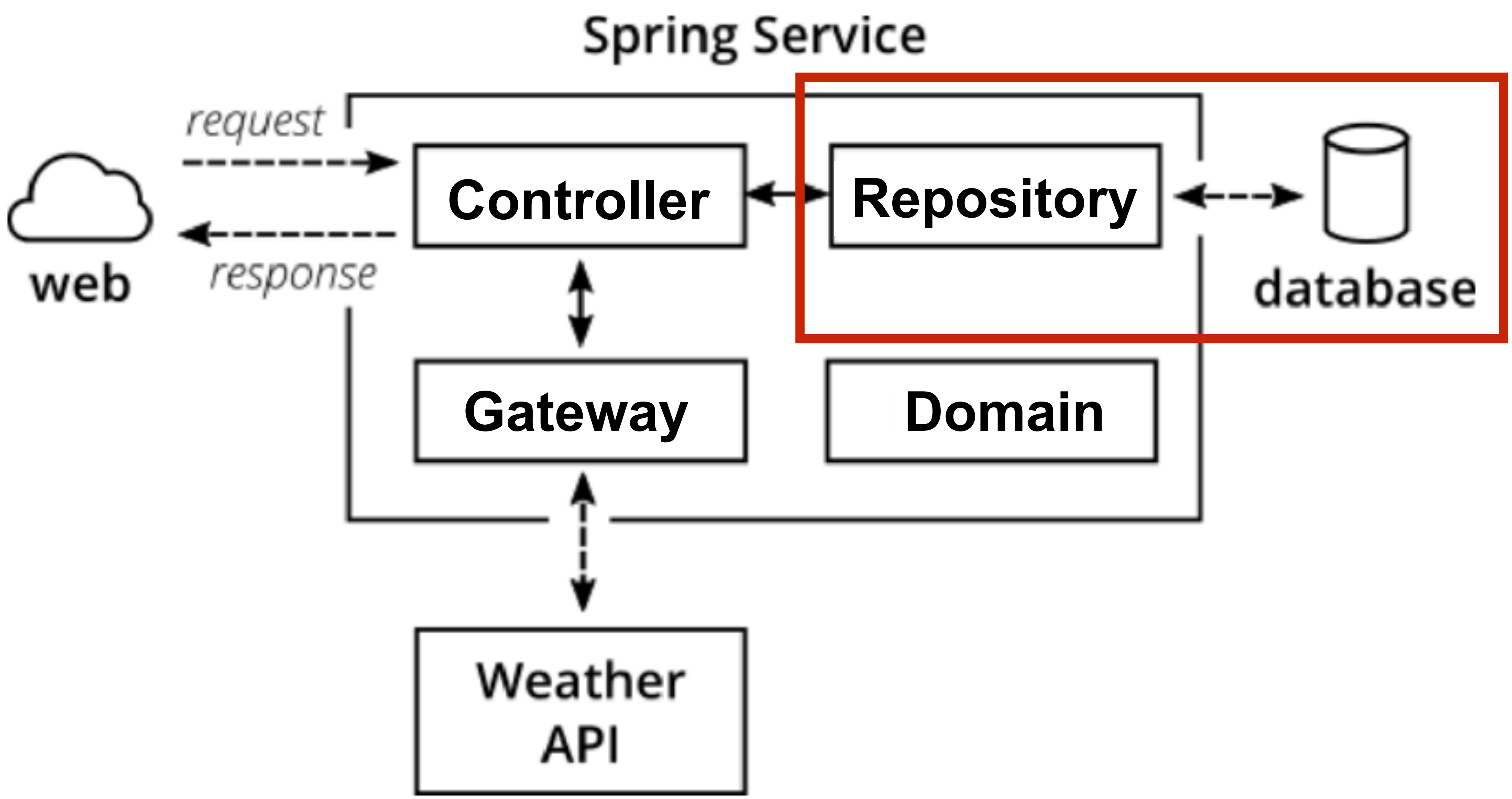
Coverage Report - All Packages

Package	# Classes	Line Coverage		Branch Coverage		Complexity
All Packages	6	25%	7/28	N/A	N/A	1
hello	1	33%	1/3	N/A	N/A	1
hello.controller	2	20%	2/10	N/A	N/A	1
hello.domain	1	66%	4/6	N/A	N/A	1
hello.repository	2	0%	0/9	N/A	N/A	1

Report generated by Cobertura 2.1.1 on 3/6/18 8:52 AM.

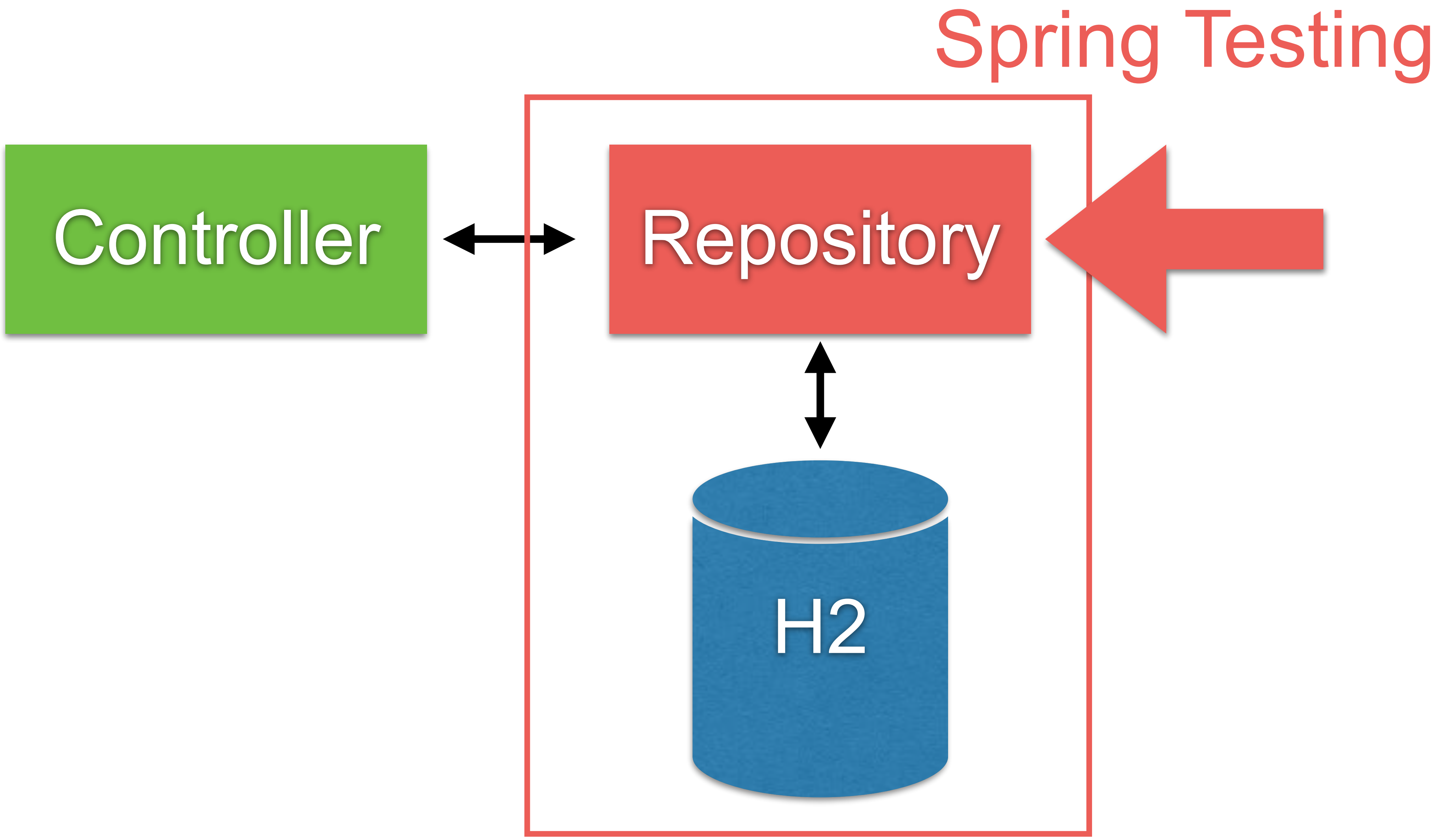
How to test ?

How to test with Repository ?



Repository Testing

Using @DataJpaTest



Spring boot provide DataJpaTest

should be add H2 library to pom.xml

```xml
<dependency>
    <groupId>com.h2database</groupId>
    <artifactId>h2</artifactId>
    <scope>test</scope>
</dependency>
```

Repository Testing (1)

```java
@RunWith(SpringRunner.class)
@DataJpaTest
public class PersonRepositoryTest {

    @Autowired
    private PersonRepository personRepository;

    @After
    public void clearData() {
        personRepository.deleteAll();
    }
```

Repository Testing (2)

Add a test case

```java
@Test
public void shouldSaveAndGetData() throws Exception {
    //Arrange
    Person somkiat = new Person("somkiat", "pui");
    personRepository.save(somkiat);

    Optional<Person> shouldSomkiat
            = personRepository.findByFirstName("somkiat");

    assertEquals( expected: "somkiat",
            shouldSomkiat.get().getFirstName());

}
```

Run test and package

$mvn clean package

```
Cobertura Report generation was successful.
Cobertura 2.1.1 - GNU GPL License (NO WARRANTY) - See COPYRIGHT file
Cobertura: Loaded information on 3 classes.
time: 125ms

Cobertura Report generation was successful.
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
```

Coverage report

Packages

All
hello
hello.controller
hello.domain
hello.repository

All Packages

Classes

Hello (66%)
HelloApplication (33%)
HelloController (100%)
HelloControllerWithRepository (0%
Person (88%)
PersonRepository (N/A)

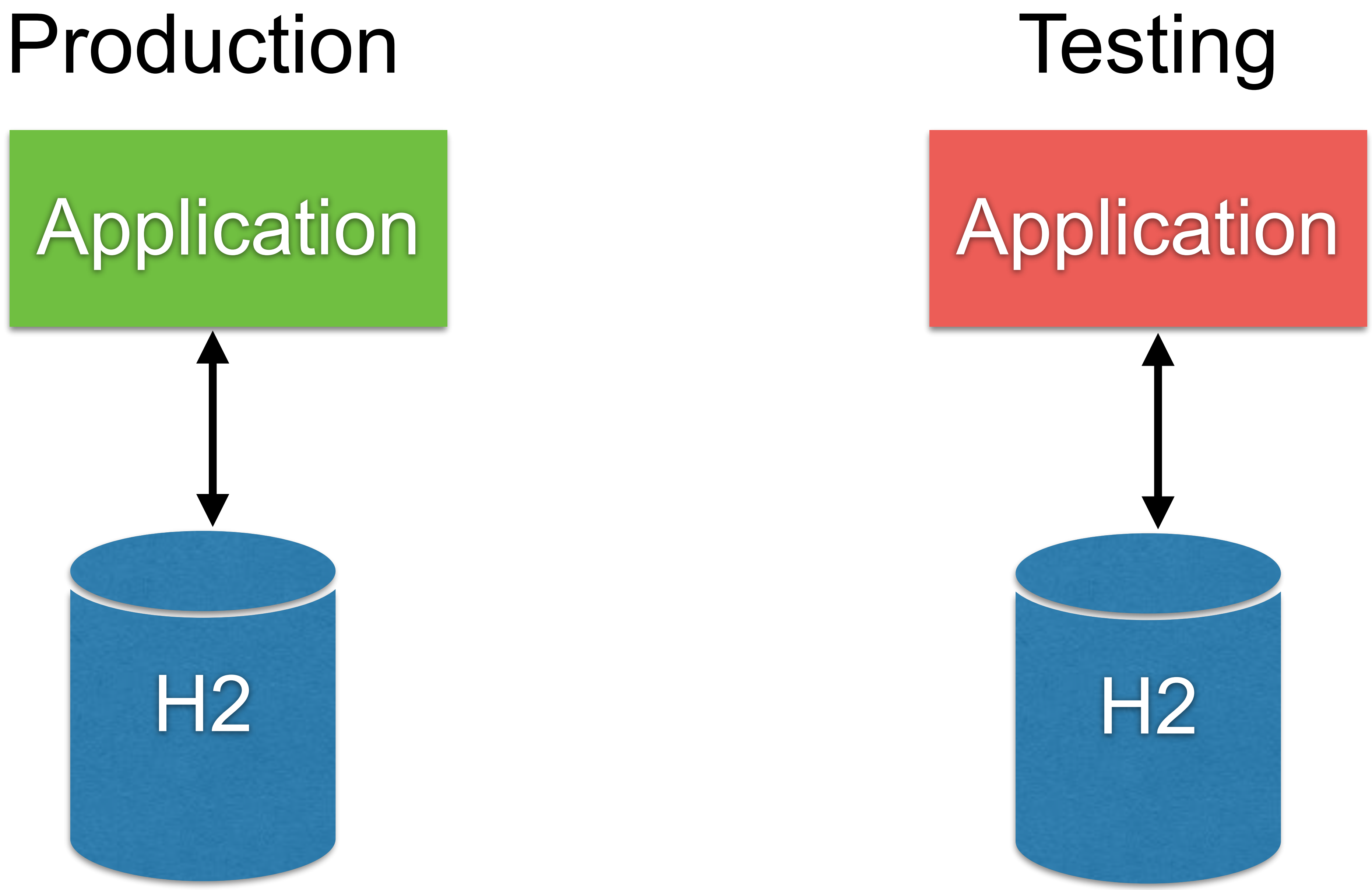
Coverage Report - All Packages

Package	# Classes	Line Coverage		Branch Coverage		Complexity
All Packages	6	53%	15/28	N/A	N/A	1
hello	1	33%	1/3	N/A	N/A	1
hello.controller	2	20%	2/10	N/A	N/A	1
hello.domain	1	66%	4/6	N/A	N/A	1
hello.repository	2	88%	8/9	N/A	N/A	1

Report generated by Cobertura 2.1.1 on 3/6/18 9:32 AM.

Working with Real Database

Working with Database



Initial database 1

Using @PostConstruct

```java
@PostConstruct
public void initData() {
    Account account1 = new Account();
    account1.setAccountId("01");
    accountRepository.save(account1);
    Account account2 = new Account();
    account2.setAccountId("02");
    accountRepository.save(account2);
}
```

Initial database 2

Schema (resources/schema.sql)

Data (resources/data.sql)

Schema.sql

```sql
CREATE TABLE account(
  id BIGINT AUTO_INCREMENT PRIMARY KEY,
  account_Id VARCHAR(16) NOT NULL UNIQUE,
  mobile_No VARCHAR(10),
  name VARCHAR(50),
  account_Type CHAR(2)
);
```

Data.sql

```sql
INSERT INTO account (account_Id) VALUES ('01');
INSERT INTO account (account_Id) VALUES ('02');
```

Initial database 2

Disable auto generate DDL from JPA in file application.yml

```yaml
spring:
  jpa:
    show-sql: true
    hibernate:
        ddl-auto: none
```

Initial database 2

Problem with naming strategy !!

```yaml
spring:
  jpa:
    show-sql: true
    hibernate:
        ddl-auto: none
        naming:
          physical-strategy: 
org.springframework.boot.orm.jpa.hibernate.SpringPhysicalNamingStrategy
          implicit-strategy: 
org.springframework.boot.orm.jpa.hibernate.SpringImplicitNamingStrategy
```

https://github.com/spring-projects/spring-boot/tree/master/spring-boot-project/spring-boot/src/main/java/org/springframework/boot/orm/jpa/hibernate

Run and see from logging

Execute file schema.sql and data.sql

```
o.s.jdbc.datasource.init.ScriptUtils      : Executing SQL script from URL [file:/Users/somkiat/dat
o.s.jdbc.datasource.init.ScriptUtils      : Executed SQL script from URL [file:/Users/somkiat/data
49 ms.
o.s.jdbc.datasource.init.ScriptUtils      : Executing SQL script from URL [file:/Users/somkiat/dat
o.s.jdbc.datasource.init.ScriptUtils      : Executed SQL script from URL [file:/Users/somkiat/data
ms.
```

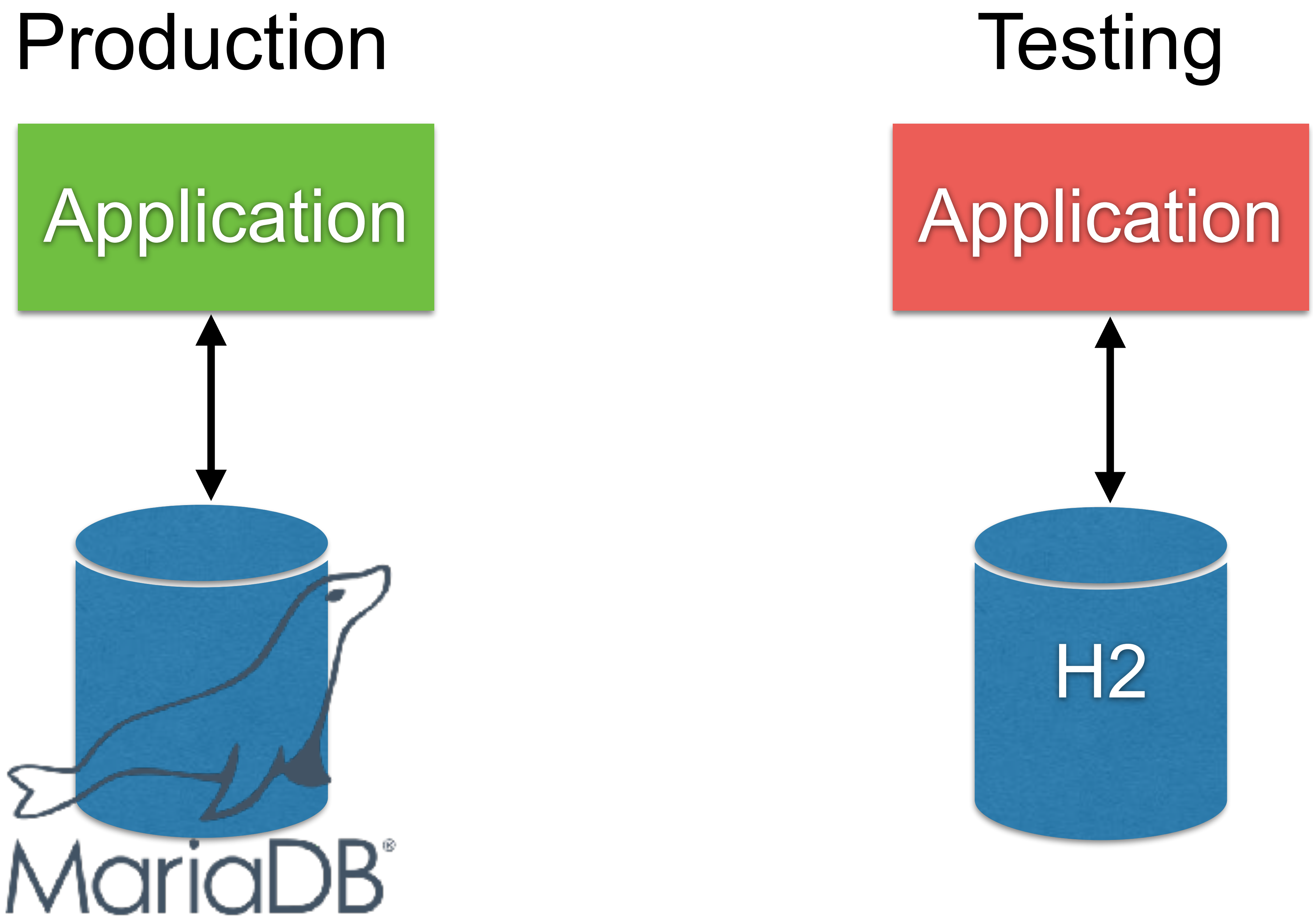
Run service

http://localhost:8080/account/0868696209

localhost:8080/account/0868696209


```
[
  - {
        accountNo: "01",
        mobileNo: null,
        name: "",
        accountType: ""
    },
  - {
        accountNo: "02",
        mobileNo: null,
        name: "",
        accountType: ""
    }
]
```

Working with Database



Add database dependency

In file pom.xml

```xml
<dependency>
  <groupId>org.mariadb.jdbc</groupId>
  <artifactId>mariadb-java-client</artifactId>
  <version>2.2.5</version>
</dependency>
```

https://mariadb.com/kb/en/library/about-mariadb-connector-j/

Config database in application

In file resources/application.yml

```yaml
spring:
  datasource:
    driver-class-name: org.mariadb.jdbc.Driver
    url: jdbc:mariadb://35.198.254.195:3306/account
    username: user01
    password: password

    initialization-mode: always
    testWhileIdle: true
    validationQuery: SELECT 1
  jpa:
    show-sql: true
    properties:
        hibernate:
          dialect: org.hibernate.dialect.MySQL5InnoDBDialect
```

Config database in application

Required config for database

```yaml
spring:
  datasource:
    driver-class-name: org.mariadb.jdbc.Driver
    url: jdbc:mariadb://35.198.254.195:3306/account
    username: user01
    password: password

    initialization-mode: always
    testWhileIdle: true
    validationQuery: SELECT 1
  jpa:
    show-sql: true
    properties:
        hibernate:
          dialect: org.hibernate.dialect.MySQL5InnoDBDialect
```

Config database in application

Required config for database

```yaml
spring:
  datasource:
    driver-class-name: org.mariadb.jdbc.Driver
    url: jdbc:mariadb://35.198.254.195:3306/account
    username: user01
    password: password

    initialization-mode: always
    testWhileIdle: true
    validationQuery: SELECT 1
  jpa:
    show-sql: true
    properties:
        hibernate:
          dialect: org.hibernate.dialect.MySQL5InnoDBDialect
```

By default not run

Run service

http://localhost:8080/account/0868696209

localhost:8080/account/0868696209


```
[
  - {
        accountNo: "01",
        mobileNo: null,
        name: "",
        accountType: ""
    },
  - {
        accountNo: "02",
        mobileNo: null,
        name: "",
        accountType: ""
    }
]
```

Check your test ?

$mvnw clean test

Step to test

Stop the real database before run test

See result

Error ?

Spring boot try to connect to the real database ?

Fix Error

Create file /resources/application.yml in test folder

config of H2 database

```
spring:
  datasource:
    driver-class-name: org.h2.Driver
    url: jdbc:h2:mem:db;DB_CLOSE_DELAY=-1
    username: sa
    password: sa
  jpa:
    show-sql: true
    properties:
        hibernate:
          dialect: org.hibernate.dialect.H2Dialect
```

Question ?

Dependency Scopes in file pom.xml

runtime ?

compile

provide

runtime

test

system

import

https://maven.apache.org/guides/introduction/introduction-to-dependency-mechanism.html#Dependency_Scope

Question ?

show_sql=true not working ?

```yaml
jpa:
  show_sql: true
  hibernate:
    ddl-auto: none
  properties:
    hibernate:
      format_sql: true
      dialect: org.hibernate.dialect.MySQL5InnoDBDialect
```

Beautiful sql format

```yaml
logging:
  level:
    org.hibernate.SQL: DEBUG
```

Default log level = INFO

http://www.baeldung.com/sql-logging-spring-boot

Result of logging message

```
 13:18:36.336  INFO 79616 --- [nio-8080-exec-2] o.s.web.servlet.Dispatc
FrameworkServlet 'dispatcherServlet': initialization completed in 37 ms
 13:18:36.864  INFO 79616 --- [nio-8080-exec-2] o.h.h.i.QueryTranslator
HHH000397: Using ASTQueryTranslatorFactory
 13:18:37.138 DEBUG 79616 --- [nio-8080-exec-2] org.hibernate.SQL
select account0_.id as id1_0_, account0_.account_id as account_2_0_, ac
 as account_3_0_, account0_.mobile_no as mobile_n4_0_, account0_.name a
unt account0_
```

Log Levels

Level	Color
FATAL	Red
ERROR	Red
WARN	Yellow
INFO (default)	Green
DEBUG	Green
TRACE	Green

https://docs.spring.io/spring-boot/docs/current/reference/html/boot-features-logging.html

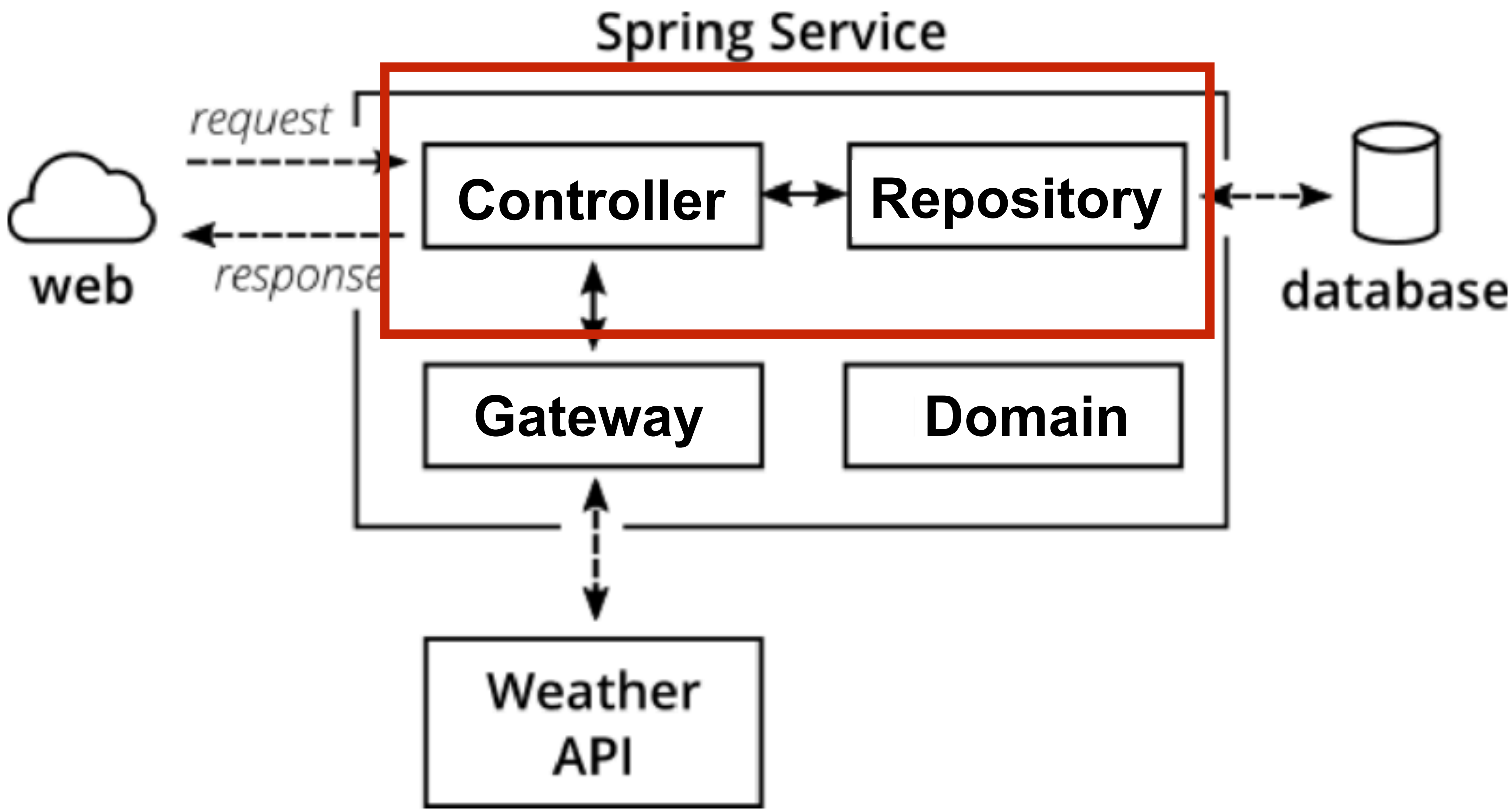
Testing with Mocking

Controller/Service Testing

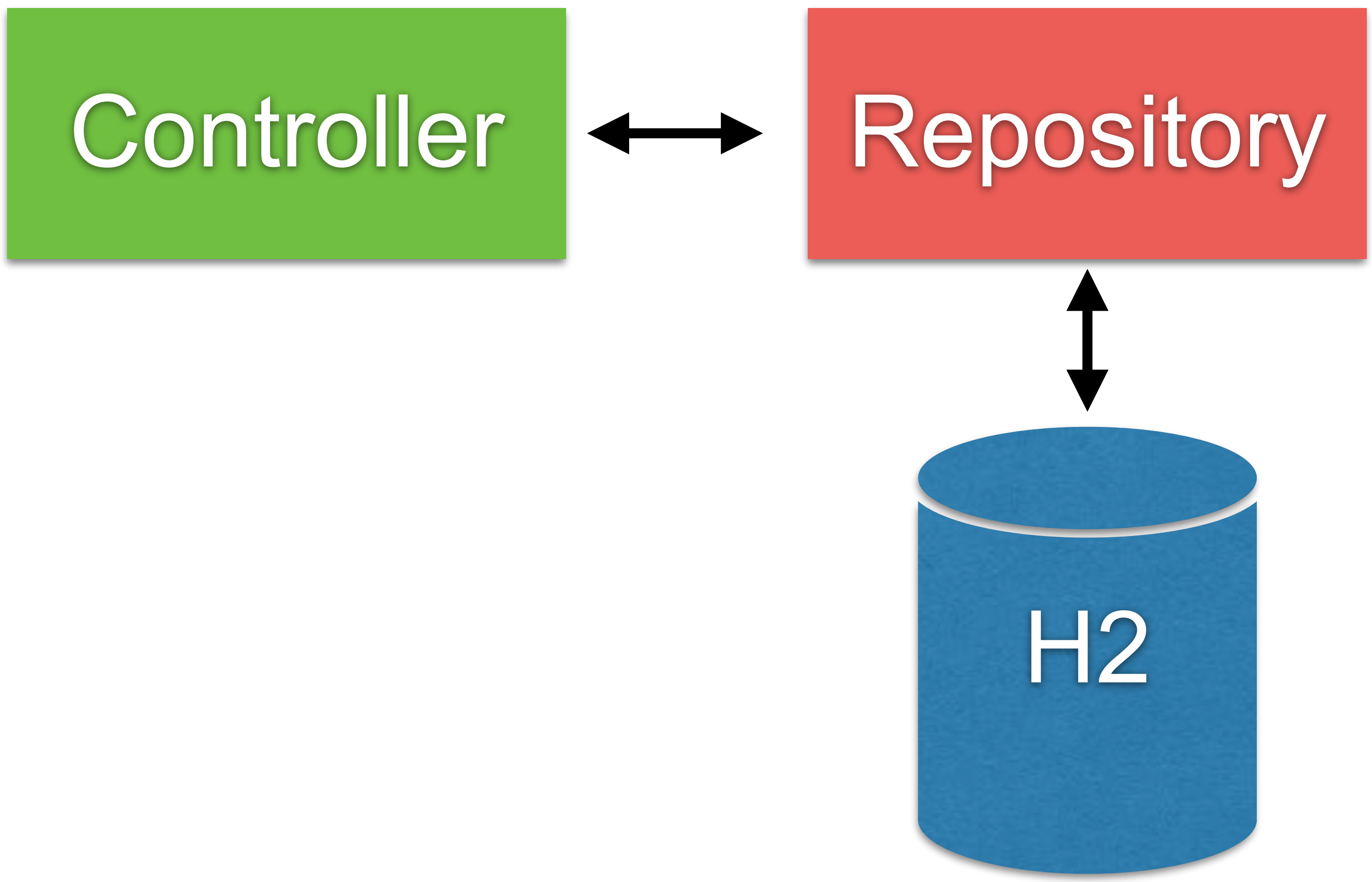
Unit testing ?

Spring Unit testing with MockMvc ?

Controller Testing with Unit test

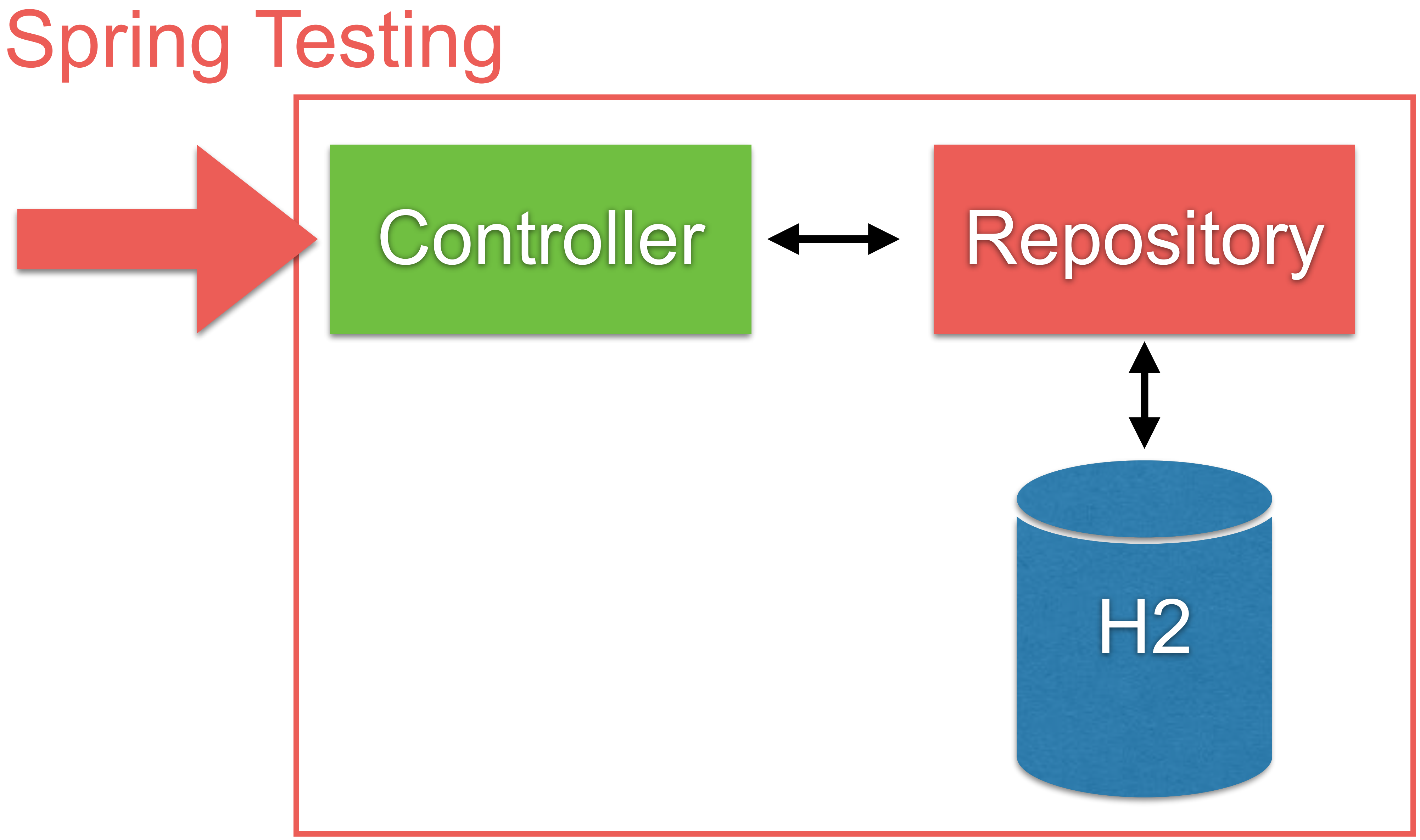


Controller Testing



Controller Testing

Using @SpringBootTest



Unit test

Use Test Double

In java, use Mockito library

http://site.mockito.org/

Unit test with Mockito (1)

```
public class HelloControllerWithRepositoryTest {

    private HelloControllerWithRepository controllerWithRepository;

    @Mock
    private PersonRepository personRepository;

    @Before
    public void init() {
        initMocks( testClass: this);
        controllerWithRepository
                = new HelloControllerWithRepository(personRepository);
    }
```

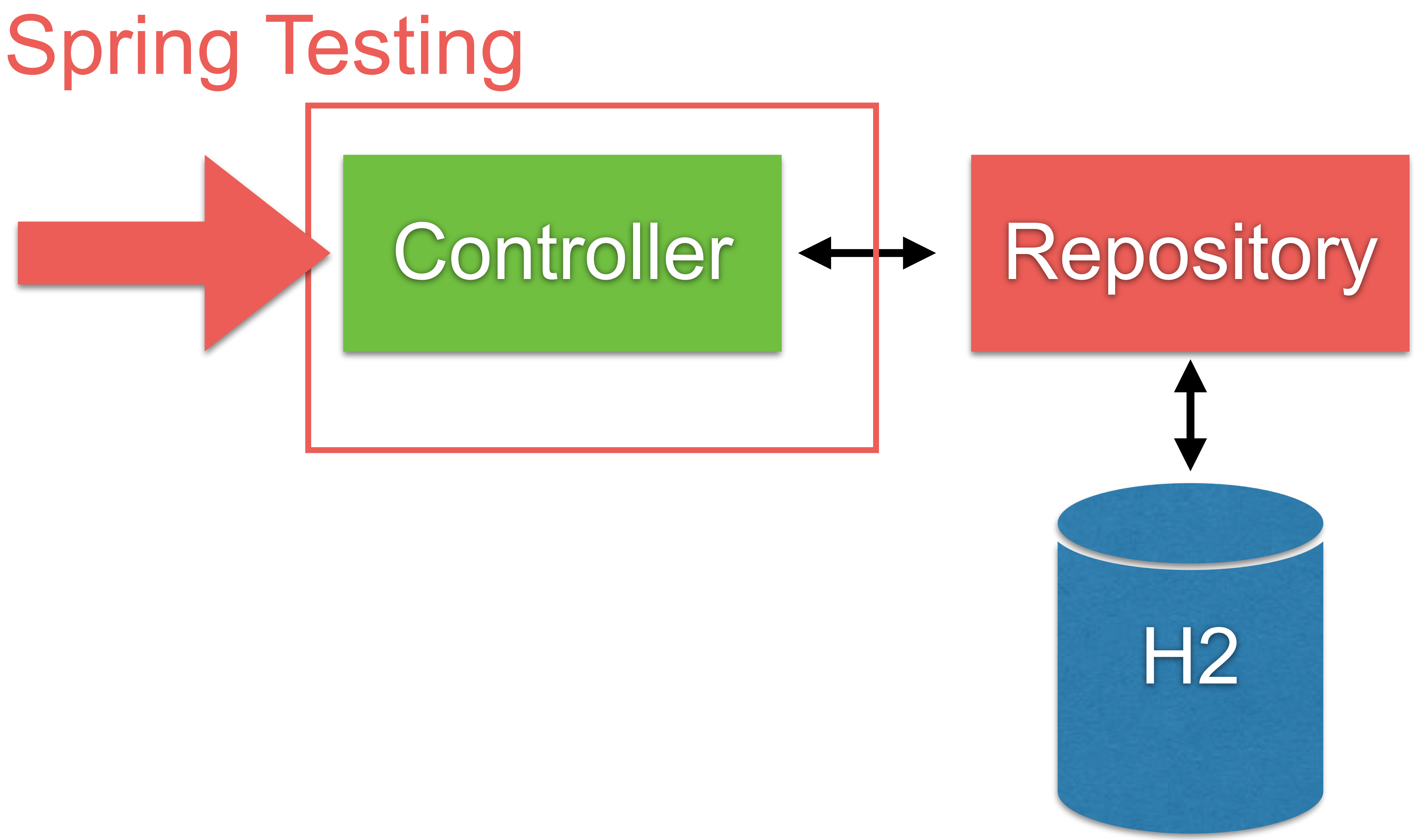
Unit test with Mockito (2)

```java
@Test
public void shouldReturnHelloSomkiat() {
    //Arrange
    Person somkiat = new Person("somkiat", "pui");
    given(personRepository.findByFirstName("somkiat"))
            .willReturn(Optional.of(somkiat));

    // Action
    Hello hello = controllerWithRepository.sayHi( name: "somkiat");

    // Assert
    assertEquals( expected: "Hello somkiat", hello.getMessage());
}
```

Try by yourself

Controller Testing with mock

Using @SpringBootTest



Controller Testing with mock (1)

```java
@RunWith(SpringRunner.class)
@SpringBootTest(
        webEnvironment = WebEnvironment.RANDOM_PORT)
public class AccountControllerMockTest {

    @Autowired
    private TestRestTemplate testRestTemplate;

    @MockBean
    private AccountRepository accountRepository;
```

Controller Testing with mock (2)

Setup datafrom repository

```
@Test
public void
เรียกข้อมูลบัญชีของหมายเลข_0868696209_ต้องเจอสองบัญชีนะ() {
    // Arrange
    List<Account> accountIterable = new ArrayList<>();
    accountIterable.add(new Account("01"));
    accountIterable.add(new Account("02"));
    given(accountRepository.findAll()).willReturn(accountIterable);

    // Act
    List<AccountResponse> accountResponses
            = testRestTemplate.getForObject(
                    "/account/0868696209", List.class);

    // Assert
    assertEquals(2, accountResponses.size());
}
```

Controller Testing with mock (3)

```java
@Test
public void
เรียกข้อมูลบัญชีของหมายเลข_0868696209_ต้องเจอสองบัญชีนะ() {
    // Arrange
    List<Account> accountIterable = new ArrayList<>();
    accountIterable.add(new Account("01"));
    accountIterable.add(new Account("02"));
    given(accountRepository.findAll()).willReturn(accountIterable);

    // Act
    List<AccountResponse> accountResponses
            = testRestTemplate.getForObject(
                    "/account/0868696209", List.class);

    // Assert
    assertEquals(2, accountResponses.size());
}
```

Call service with rest template

Coverage report

Packages

All
toystore
toystore.controller
toystore.domain
toystore.repository

All Packages

Classes

Hello *(100%)*
HelloController *(83%)*
HelloWithRepositoryController *(100*
Person *(100%)*
PersonRepository *(N/A)*
ToyStoreApplication *(33%)*

Coverage Report - All Packages

Package	# Classes	Line Coverage		Branch Coverage		Complexity
All Packages	6	90%	29/32	50%	1/2	1.1
toystore	1	33%	1/3	N/A	N/A	1
toystore.controller	2	93%	15/16	50%	1/2	2
toystore.domain	1	100%	4/4	N/A	N/A	1
toystore.repository	2	100%	9/9	N/A	N/A	1

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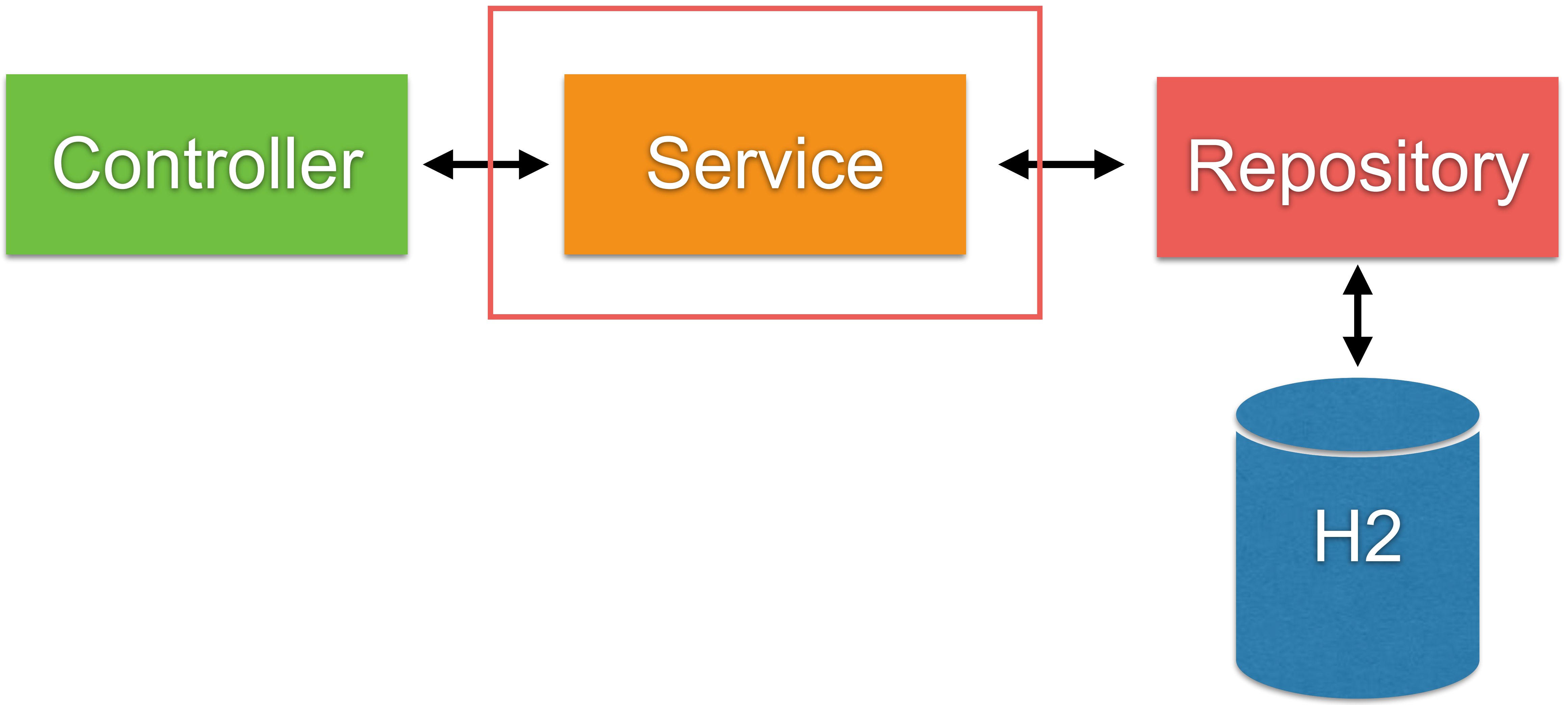
Controller Testing with Unit test



Q/A

More topics

Concurrency and Threading

Key points

Performance
Responsiveness
Throughput